Contents

1 Introduction

1.1 Sets

1.1.1 Equivalence between sets

Lemma 1.1.1

Suppose $f : X \to Y, g : Y \to X$, then there are decompositions

$$X = A \cup \tilde{A}, \quad Y = B \cup \tilde{B}$$

such that $A \cap \tilde{A} = \varnothing, B \cap \tilde{B} = \varnothing$ and $f(A) = B, g\left(\tilde{B}\right) = \tilde{A}$.

Note.
我们从随便一个 A 出发，先让一些简单的条件得到满足，比如 $B = f(A), \tilde{B} = Y \backslash B, \tilde{A} = g(Y \setminus f(E))$，就它们确定地写出来. 剩下 $A \cap \tilde{A} = \varnothing$ 和 $A \cup \tilde{A} = X$ 需要满足. 直接找到特定的目标一开始是困难的，于是我们先把满足 $A \cap \tilde{A} = \varnothing$ 的“好东西”都收集起来，即 $\mathcal{S}$. 再在 $\mathcal{S}$ 中寻找一个满足 $A \cup \tilde{A} = X$ 的候选. “极大元”就是一个看起来非常可疑的对象 (如果它也落在 $\mathcal{S}$，事实上也的确如此).

Proof. Let

$$\mathcal{S} := \{E \subseteq X : E \cap g(Y \setminus f(E)) = \varnothing\}.$$

We define

$$A := \bigcup_{E \in \mathcal{S}} E$$

Let

$$B := f(A), \quad \tilde{B} := Y \setminus f(A), \quad \tilde{A} := g\left(\tilde{B}\right) = g(Y \setminus f(A))$$

It is obvious that $B \cap \tilde{B} = \varnothing, B \cup \tilde{B} = Y$. We should show $A \cap \tilde{A} = \varnothing$, which is equivalent to $A \in \mathcal{S}$, and $A \cup \tilde{A} = X$. First , take any $E \in \mathcal{S}$, we have

$$E \cap g(Y \setminus f(A)) \subseteq E \cap g(Y \setminus f(E)) = \varnothing.$$

Then we concludes that

$$A \cap g(Y \setminus f(A)) = \varnothing \implies A \in \mathcal{S}$$

To show that $A \cup \tilde{A} = X$, we suppose conversely that there is $x \in X$ such that

$$x \notin A \cup \tilde{A}$$

Let $A' := A \cup \{x\}$, then

$$\begin{aligned} A' \cap g(Y \setminus f(A')) &= (A \cup \{x\}) \cap g(Y \setminus (f(A) \cup \{x\})) \\ &\subseteq (A \cup \{x\}) \cap \tilde{A} \\ &= \left(A \cap \tilde{A}\right) \cup \left(\{x\} \cap \tilde{A}\right) \\ &= \varnothing \end{aligned}$$

Thus $A' \in \mathcal{S}$, which is a contradiction to the definition of A. □

Theorem 1.1.2 ▶ Cantro-Bernstein

If $X \supseteq A$, $Y \supseteq B$, and if $X \sim B$, $Y \sim A$, then $X \sim Y$.

Proof. We know that there are injection $f : X \to Y$ and $g : Y \to X$. There are decompositions

$$X = A \cup \tilde{A}, \quad Y = B \cup \tilde{B}$$

such that $A \cap \tilde{A} = \varnothing, B \cap \tilde{B} = \varnothing$ and $f(A) = B$, $g\left(\tilde{B}\right) = \tilde{A}$. Then we define $F : X \to Y$

$$F(x) := \begin{cases} f(x), & x \in A \\ \left(g|_{\tilde{B}}\right)^{-1}(x), & x \in \tilde{A} \end{cases}$$

is a bijection. □

1.1.2 Countable sets and the Cardinality of the Continuum

Example 1.1.3

The following sets are countable:

1. $\{I_\alpha\}_{\alpha \in \Lambda}$, where $I_\alpha \subseteq \mathbb{R}^1$ is an interval and $I_\alpha \cap I_\beta = \varnothing$ whenever $\alpha \neq \beta$.
2. Let $f : \mathbb{R}^1 \to \mathbb{R}^1$ be a monotone function, then

$$A := \{x : f(x) \text{ is not continuous at } x\}$$

is a countable set.

3. Let $f : \mathbb{R} \to \mathbb{R}$, define

$$E := \{x \in \mathbb{R} : f(x) \text{ is not continuous on } x, \text{but the right limit } f(x+0) \text{ exists}\}$$

Proof. 1. We take a rational number r_α in each I_α.

2. We take a rational number r_x in $(f(x-0), f(x+0))$ for each x that f is not continuous on it.

□

Theorem 1.1.4

Let A be a infinite set and $|A| = \alpha$. If B is countable, then

$$|A \cup B| = \alpha$$

Proof. Since $B' := B \setminus A$ is countable, and $A \cup B' = A \cup (B \setminus A) = A$, we may assume that $A \cup B = \varnothing$. There exists countable set $A_1 \subseteq A$. Then $A_1 \cup B$ is countable. We have $A_1 \cup B \sim A_1$. There exists bijection $f : A_1 \cup B \to A_1$. Since

$$A = A_1 \cup (A \setminus A_1), \quad A \cup B = (A_1 \cup B) \cup (A \setminus A_1)$$

Define $F : A \cup B \to A$

$$F(x) = \begin{cases} f(x), & x \in A_1 \cup B \\ x, & x \in A \setminus A_1 \end{cases}$$

Then F is a bijection. Thus $|A \cup B| = |A| = \alpha$. □

Theorem 1.1.5

$$A \text{ is a infinite set} \iff \exists A_1 \text{ is a proper subset of } A, s.t. |A_1| = |A|$$

Theorem 1.1.6

$[0,1] = \{x : 0 \leq x \leq 1\}$ is a uncoutable set.

Proof sketch.
试图建立 $(0,1] \sim \mathscr{H}$，其中 $\mathscr{H}$ 表示自然数列全体. 将 $[0,1]$ 上的数按二进制展开 (只采用具有无穷多个 1 的表示). 将展开式从低到高位次, 将两个数码 1 之间

的间隔 (位置差) 组成一个自然数列. 从而建立了 $(0,1] \sim \mathscr{H}$.

Definition 1.1.7

Define
$$c := |(0,1]|$$
called the ***cardinality of the continuum***.

Theorem 1.1.8

Let $\{A_k\}$ be a suquence of sets. If
$$|A_k| = c, \quad \forall k \in \mathbb{N}$$
Then
$$\left|\bigcup_{k=1}^{\infty} A_k\right| = c$$

Proof sketch.
$A_k \sim [k, k+1)$, 故 $\left|\bigcup_{k=1}^{\infty} A_k\right| \leq \left|\coprod_{k=1}^{\infty} A_k\right| \leq |[1,\infty)| = c$. 而 $A_k \subseteq \bigcup_{k=1}^{\infty} A_i \implies \left|\bigcup_{k=1}^{\infty} A_k\right| \geq c$. 因此由 CSB 定理, $\left|\bigcup_{k=1}^{\infty} A_k\right| = c$.

1.2 Arithmetic in [0,infty]

1.3 Metric Space

Lemma 1.3.1

X is a metric space, $x \in X$, B_1, B_2 are twos balls in X. If $x \in B_1 \cap B_2$, then x is the center of an open ball $B \subseteq B_1 \cap B_2$.

Proof. We may assume that B_1, B_2 have different center. Let $y_1 \neq y_2$,
$$B_1 = \{x : d(x, y_1) < r_1\}, \quad B_2 = \{x : d(x, y_2) < r_2\}$$
Take $x \in B_1 \cap B_2$, we claim that there exists $r_0 > 0$ such that $B := \{y : d(x, y) < r_0\} \subseteq B_1 \cap B_2$. Otherwise , for all $r > 0$, there exists a point y_r such that the following two hold at the same time

1. $d(x, y_r) < r$
2. $d(y_r, y_1) \geq r_1$ or $d(y_r, y_2) \geq r_2$.

Thus

$$d(x, y_1) \geq d(y_r, y_1) - d(x, y_r) > r_1 - r, \quad \text{or} \quad d(x, y_2) > r_2 - r$$

The above shows that

$$r > \min\{r_1 - d(x, y_1), r_2 - d(x, y_2)\}, \quad \forall r > 0$$

which is a contradiction. □

1.4 Limits of Set Sequences

Definition 1.4.1 ▶ Limit Inferior and Limit Superior

Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of sets.

1. By the ***Limit Inferior of the sequence***, we mean

$$\liminf_{n\to\infty} A_n = \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} A_n = \{x : \exists N_0 \in \mathbb{N}, \forall n > N_0, x \in A_n\}$$

2. By the ***Limit Superior of the sequence***,we mean

$$\limsup_{n\to\infty} A_n = \bigcap_{N=1}^{\infty} \bigcup_{n=N}^{\infty} A_n = \{x : \forall N \in \mathbb{N}, \exists n \geq N, x \in A_n\}$$

Note.

1. Limit Inferior 包含那些“最终稳定下来”的元素，即从某个点之后就永远属于序列中所有后续集合的元素。
2. Limit Superior 包含那些“反复出现”的元素，即在无限多个集合中出现的元素。

Proposition 1.4.2

For any sequence of sets $\{A_n\}$, it always holds that:

$$\liminf_{n\to\infty} A_n \subseteq \limsup_{n\to\infty} A_n$$

Proof sketch.
直觉上是显然的，因为“最终稳定下来”的元素一定会“反复出现”.

Proof. It is obvious by the $\{x : x \in P\}$ form representation of the Limit Inferior and Superior. □

Definition 1.4.3 ▶ Limit of a Set Sequence

If the limit inferior and limit superior of a set sequence $\{A_n\}_{n=1}^{\infty}$ are equal, i.e., $\liminf_{n\to\infty} A_n = \limsup_{n\to\infty} A_n$, then we say the ***limit*** of the sequence exists, and it is defined as:

$$\lim_{n\to\infty} A_n = \liminf_{n\to\infty} A_n = \limsup_{n\to\infty} A_n$$

Proposition 1.4.4

Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of sets.

1. If $\{A_n\}$ is an increasing sequence, then the limit of $\{A_n\}$ exists and

$$\lim_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} A_n$$

2. If $\{A_n\}$ is a decreasing sequence, then the limit of $\{A_n\}$ exists and

$$\lim_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

Proof. 1. If $\{A_n\}$ is increasing , then

$$\bigcap_{n=N}^{\infty} A_n = A_N$$

and

$$\bigcup_{n=N}^{\infty} A_n = \bigcup_{n=1}^{\infty} A_n, \forall N \in \mathbb{N}.$$

Thus

$$\liminf_{n\to\infty} A_n = \bigcup_{N=1}^{\infty}\bigcap_{n=N}^{\infty} A_n = \bigcup_{N=1}^{\infty} A_N$$

and

$$\limsup_{n\to\infty} A_n = \bigcap_{N=1}^{\infty}\bigcup_{n=N}^{\infty} A_n = \bigcap_{N=1}^{\infty}\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} A_n$$

which are the same.

2. If $\{A_n\}$ is decreasing, then

$$\bigcap_{n=N}^{\infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

and

$$\bigcup_{n=N}^{\infty} A_n = A_N$$

Thus

$$\liminf_{n\to\infty} A_n = \bigcup_{N=1}^{\infty}\bigcap_{n=N}^{\infty} A_n = \bigcup_{N=1}^{\infty}\bigcap_{n=1}^{\infty} A_n = \bigcap_{n=1}^{\infty} A_n$$

and

$$\limsup_{n\to\infty} A_n = \bigcap_{N=1}^{\infty}\bigcup_{n=N}^{\infty} A_n = \bigcap_{N=1}^{\infty} A_N$$

which are the same.

□

Proposition 1.4.5

For a sequence of sets $\{A_n\}_{n=1}^{\infty}$ and their corresponding sequence of indicator functions $\{\chi_{A_n}(x)\}_{n=1}^{\infty}$:

- $\chi_{\liminf_{n\to\infty} A_n}(x) = \liminf_{n\to\infty} \chi_{A_n}(x)$
- $\chi_{\limsup_{n\to\infty} A_n}(x) = \limsup_{n\to\infty} \chi_{A_n}(x)$
- $\lim_{n\to\infty} A_n$ exists, then $\chi_{\lim_{n\to\infty} A_n}(x) = \lim_{n\to\infty} \chi_{A_n}(x)$

Proof sketch.

- 若 x 在 limit inferior 里面, 则 x 是"最终稳定"的, $\chi_{A_n}(x)$ 是关于 n "最终"恒为 1 的.
- 若 x 在 limit superior 里面, 则 x 是"反复出现"的, 即相当于 N 多大, 总会出现之后的某个 n 使得 $\chi_{A_n}(x) = 1$.
- 当极限存在时, 函数列 $\left\{\chi_{A_n}(x)\right\}_{n=1}^{\infty}$ 的 limit inferior 和 supperior 根据上两条相等, 等于极限集合的 χ.

1.5 Ordinal and Transfinite Induction

Definition 1.5.1 ▶ Ordinal

A set α is said to be an ***ordinal***, if the following two hold:

1. Transitive Set: If $x \in \alpha$, then $x \subseteq \alpha$. (Equivalently, if $z \in y$, $y \in \alpha$, then $z \in \alpha$).
2. Well-ordered by $\in$: For any $x, y \in \alpha$, either $x \in y$ or $y \in x$. And there is a $\in$-smallest element in any nonempty subset of α.

Lemma 1.5.2

If α is an ordinal, then $\alpha \notin \alpha$.

Proof. If $\alpha \in \alpha$, then $\{\alpha\}$ is a subset of α. But there is no smallest ordinal in $\{\alpha\}$, which is a contradiction. □

Definition 1.5.3 ▶ Classification of Ordinal

1. ***Zero Ordinal***: Only $0 = \varnothing$.
2. ***Successor Ordinal***: Ordinal numbers of the form $\alpha + 1 = \alpha \cup \{\alpha\}$
3. ***Limit Ordinal***: Ordinal that is neither zero ordinal nor successor ordinal.

Example 1.5.4 ▶ Ordinal

1. $$\omega = \{0, 1, 2, 3, \cdots\}$$
   is the smallest infinite ordinal, which is a limit ordinal.
2. $$\omega \cdot 2 := \{0, 1, 2, \cdots, \omega, \omega + 1, \omega + 2, \cdots\}$$

is a limit ordinal.

3. $$\omega_1 := \{\alpha : \alpha \text{ is a ountable ordinal}\}$$

is the smallest uncountable ordinal.

(a) ***Unountable***: If ω_1 is countable, then $\omega_1 \in \omega_1$ by definition, which is a contradiction.

(b) ***Smallest***: If there is a uncountable ordinal β such that $\beta < \omega_1$. Then $\beta \in \omega_1$, which is a contradiction to the uncountability of β.

Theorem 1.5.5 ▸ Transfinite Induction

For any ordinal α, let $P(\alpha)$ be a property of α. If the following holds :

- For any ordinal α, if $P(\beta)$ holds for any $\beta < \alpha$, then $P(\alpha)$ holds.

Then

- For any ordinal α, $P(\alpha)$ is true.

Proof. Let S be the collection of α such that $P(\alpha)$ is false. Then if the transfinite indunction is not true, S is nonempty. Since the class of ordinal is well-ordered, there is a smallest ordinal γ in S. Then for any $\beta < \gamma$, $\beta \notin S$, that is $P(\beta)$ is true. But it contradicts to what we supposed at first. □

1.6 Meager Sets in Topology

Definition 1.6.1 ▸ DEFINITIONS

- A subset E of a metric space X is said to be ***dense in an open set*** U if $\overline{E} \supset U$.
- A set E is defined to be ***nowhere dense*** if it is not dense in any open subset U of X. Alternatively, we could say that E is nowhere dense if $\overline{E}$ does not contain any open set.
- A set E is said to be of ***first category*** in X if it is the union of a countable collection of nowhere dense sets.
- A set that is not of the first category is said to be of the ***second category*** in X.

Theorem 1.6.2 ▸ Baire category theorem

A ***complete metric space*** X is not the union of a ***countable collection of nowhere dense sets***. That is, a ***complete metric space*** is of the ***second category***.

1.7 Lemma on Metric Space

Theorem 1.7.1

If $x_0 \in \mathbb{R}^n$, $F \subseteq \mathbb{R}^n$ is a nonempty closed set, then

$$\exists y_0 \in F, s.t. \left|x_0 - y_0\right| = d\left(x_0, F\right)$$

Lemma 1.7.2

Let $E \subseteq \mathbb{R}^n$ is a nonempty set. If

$$\forall x \notin E, \exists y \in E, s.t. \left|x - y\right| = d\left(x, E\right)$$

Then E is a closed set.

Corollary 1.7.3

If $E \subseteq \mathbb{R}^n$ is nonempty, the following statements equivalent

1. E is a closed set;
2. $\forall x \in \mathbb{R}^n, \exists y \in E$, s.t. $|x - y| = d(x, E)$
3. $\forall x \notin E, \exists y \in E$, s.t. $|x - y| = d(x, E)$

Theorem 1.7.4

If $E \subseteq \mathbb{R}^n$ is nonempty, then the function $x \mapsto d(x, E)$ is uniformly continuous on $\mathbb{R}^n$.

Corollary 1.7.5

If $F_1, F_2 \subseteq \mathbb{R}^n$ are nonempty closed set, and one of them is bounded, then

$$\exists x_1 \in F_1, x_2 \in F_2, s.t. \left|x_1 - x_2\right| = d\left(F_1, F_2\right)$$

Lemma 1.7.6 ▸ Uryson

If $F_1, F_2 \subseteq \mathbb{R}^n$ are nonempty closed sets, and if

$$F_1 \cap F_2 = \varnothing$$

Then there exists $f \in C(\mathbb{R}^n)$ such that

1. $0 \leq f(x) \leq 1, (x \in \mathbb{R}^n)$;
2. $$F_1 = \{x : f(x) = 1\}, \quad F_2 = \{x : f(x) = 0\}$$

Theorem 1.7.7

Let (X, d) be a metric space, $A \subseteq X$ is a closde set.

1. If $f : A \to [-1, 1]$ is continuous, then there exists continuous function $F : X \to [a, b]$ such that $F(x) = f(x)$ for all $x \in A$.
2. If $f : A \to \mathbb{R}$ is a contiuous function, then there exists continuous function $F : X \to \mathbb{R}$ such that $F(x) = f(x)$ for all $x \in A$.

2

Abstract Integration

2.1 The Concept of Measurbility

Definition 2.1.1

1. A collection $\mathfrak{M}$ of subsets of a set X is said to be a *σ-**algebra*** in X if $\mathfrak{M}$ has the following properties:
   (a) $X \in \mathfrak{M}$.
   (b) If $A \in \mathfrak{M}$, then $A^c \in \mathfrak{M}$.
   (c) If $A = \bigcup_{n=1}^{\infty} A_n$ and $A_n \in \mathfrak{M}$ for $n = 1, 2, 3, \cdots$, then $A \in \mathfrak{M}$.
2. If $\mathfrak{M}$ is a σ-algebra in X, then X is called a ***measurable*** provided that $f^{-1}(V)$ is measurable set in X for every open set V in Y.
3. If X is a measurable space, Y is a topological space, and f is a mapping of X into Y, then f is said to be ***measurable*** provided that $f^{-1}(V)$ is a measurable set in X for every open set V in Y.

Note.
对比 topological space 的定义, σ-algebra 的定义是对称的, 即可测集的补集仍是可测的, 而对于拓扑空间, 开集的补集不是开集, 而是被定义为了闭集这样的对象.

Lemma 2.1.2

1. Let X be a measurable space , Y, Z be topological spaces. If $f : X \to Y$ is measurable , and if $g : Y \to Z$ is continuous, then $g \circ f : X \to Z$ is measurable.
2. Let u, v be real measurable functions on a measurable space X, let Φ be a continuous mapping of the plane into a topological space Y, and define
$$h(x) = \Phi(u(x), v(x))$$
for $x \in X$. Then $h : X \to Y$ is measurable.

Proof. 1. Take any open subset $V \subseteq Z$. Then $g^{-1}(V)$ is an open subset of

Y . Since f is measurable,

$$(g \circ f)^{-1}(V) = f^{-1}\left(g^{-1}(V)\right)$$

is a mearuable set.

2. Let $V \subseteq Y$ be a open subset. Then $\Phi^{-1}(V)$ is a open subset of $[-\infty, \infty]^2$. There exists countably many open sets of $[-\infty, \infty]$:$U_i, V_i, i \in \mathbb{N}$, such that

$$\Phi^{-1}(V) = \bigcup_i U_i \times V_i$$

Thus

$$\begin{aligned} h^{-1}(V) &= (u, v)^{-1}\left(\bigcup_i U_i \times V_i\right) \\ &= \bigcup_i (u, v)^{-1}(U_i \times V_i) \\ &= \bigcup_i \left(u^{-1}(U_i) \cap v^{-1}(V_i)\right) \end{aligned}$$

which is a measurable set.

□

Corollary 2.1.3

Let X be a measurable space. The following propositions hold

1. If $f = u + iv$, where u and v are real measurable functions on X, then f is a complex measurable function on X.
2. If $f = u + iv$ is a complex measurable function on X, then u, v, and $|f|$ are real measurable functions on X.
3. If f and g are complex measurable functions on X, then so are $f + g$ and fg.
4. If E is a measurable set in X and if

$$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$$

then χ_E is a measurable function.

5. If f is a complex measurable function on X , there is a complex measurable function α on X such that $|\alpha| = 1$ and $f = \alpha |f|$.

Theorem 2.1.4

If $\mathscr{F}$ is any collection of subsets of X , there exists a smallest σ-algebra $\mathfrak{M}^*$ such that $\mathscr{F} \subseteq \mathfrak{M}^*$.

Definition 2.1.5 ▸ Borel

Let X be a topological space.

1. By ***Borel σ-algebra*** , we mean the smallest σ-algebra $\mathscr{B}$ in X such that every open set in X belongs to $\mathscr{B}$. The members of $\mathscr{B}$ are called the ***Borel sets*** of X.
2. All countable unions of closed sets and all countable intersections of open sets are Borel sets, which we called ***F_σ's and G_δ's***, respectively.
3. By a ***Borel function***, we mean a measurable function on the measurable space $(X, \mathscr{B})$.

Remark.

1. The letters F and G were used for colsed and opensets,respectively, and σ refers to union, δ to intersection.
2. A continuous function is Borel-measurable, since the preimage of any open set is open and therefore a Borel set.

Theorem 2.1.6

Suppose $\mathfrak{M}$ is a σ-algbera in X ,and Y is a topological space. Let f map X into Y.

1. If Ω is the collection of all sets $E \subseteq Y$ such that $f^{-1}(E) \in \mathfrak{M}$, then Ω is a σ- algebra in Y.
2. If f is measurable and E is a Borel set in Y , then $f^{-1}(E) \subseteq \mathfrak{M}$.
3. If $Y = [-\infty, \infty]$ and $f^{-1}((\alpha, \infty]) \subseteq \mathfrak{M}$ for every real α, then f is measurable .
4. If f is measurable, if Z is topological space, if $g : Y \to Z$ is a Borel mapping, and if $h = g \circ f$, then $h : X \to Z$ is measurable.

Theorem 2.1.7

If $f_n : X \to [-\infty, \infty]$ is measurable,for $\neq 1, 2, 3, \cdots$, and

$$g = \sup_{n \geq 1} f_n, \quad h = \lim_{n \to \infty} \sup f_n,$$

then g and h are measurable.

Corollary 2.1.8

1. The limit of every pointwise convergent sequence of complex measurable functions is measurable.
2. If f and g are measurable(with range in $[-\infty, \infty]$), then so are $\max\{f, g\}$ and $\min\{f, g\}$. In particular , this is true of the functions

$$f^+ = \max\{f, 0\}, \quad f^- = -\min\{f, 0\}.$$

which are called the ***positive part*** and ***negative part*** of f , respectively.

Remark.
1. There are the standrd representation

$$|f| = f^+ + f^-, \quad f = f^+ - f^-$$

2. And easy but (may) useful observation is : If $f = g-h, g \geq 0, h \geq 0$, then $f^+ \leq g$ and $f^- \leq h$.

2.2 Simple Functions

Definition 2.2.1

A complex function s on a measurable space X whose range consists of only finitely many points will be called a ***simple function***. Among these are the nonnegative simple functions, whose range is a finite subset of $[0, \infty)$.
Specifically, if $\alpha_1, \cdots, \alpha_n$ are distinct values of a simple function s, and if we set $A_i = \{x : s(x) = \alpha_i\}$, then clearly

$$s = \sum_{i=1}^{n} \alpha_i \chi_{A_i}.$$

Where χ_{A_i} is the characteristic function of A_i.

Remark.
- Here, we explicity exclude ∞ from the values of a simple function.
- It is clear that s is mearuable if and only if each of the sets A_i is measurable.

Theorem 2.2.2

Let $f : X \to [0, \infty]$ be measurable space. There exists simple measurable functions s_n on X such that
1. $0 \le s_1 \le s_2 \le \cdots \le f$.
2. $s_n(x) \to f(x)$ as $n \to \infty$, for every $x \in X$.

Proof sketch.
We construst a sequence of Borel simple functions $\{\varphi_n(x)\}$ to act as an ***identify***. 当 n 增大的同时，我们同时让 $\varphi_n(x)$ 的单位逼近精度和单位逼近范围随着 n 提升. 并且在舍弃误差时，总是向下取整，使得该单位逼近是自下而上的.

Proof. For every $x \in [0, \infty]$, and for every $n \in \mathbb{N}$, there exists a unique integer $k_n(x)$, such that

$$k_n(x) 2^{-n} \le x < (k_n(x) + 1) 2^{-n}$$

For every $n \in \mathbb{N}$, we define

$$\varphi_n(x) = \begin{cases} k_n(x) 2^{-n}, & 0 \le x \le n \\ n, & x \ge n \end{cases}$$

Each $\varphi_n(x)$ is then a Borel simple function. It is not hard to show that $0 \le \varphi_1 \le \varphi_2 \cdots \le \cdots \le \mathrm{Id}$. For each n, we define

$$s_n(x) := (\varphi_n \circ f)(x)$$

Then $\{s_n\}$ is a suquence of simple measurable functions such that $0 \le s_1 \le s_2 \le \cdots \le f$. Since $\lim_{n\to\infty} \varphi_n(x) = x$, then $\lim_{n\to\infty} s_n(x) = f(x)$. □

2.3 Measure

Definition 2.3.1 ▶ Measure and Measure Space

(a) A ***positive measure*** is a function μ, defined on a σ-algebra $\mathfrak{M}$, whose range is in $[0, \infty]$ and which is ***countably additive***. This means that if $\{A_i\}$ is a disjoint countable collection of members of $\mathfrak{M}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu\left(A_i\right).$$

To avoid trivialities, we shall also assume that $\mu(A) < \infty$ for at least one $A \in \mathfrak{M}$.

(b) A ***measure space*** is a measurable space which has a positive measure defined on the σ-algebra of its measurable sets.

(c) A ***complex measure*** is a complex-valued countably additive function defined on a σ-algebra.

Theorem 2.3.2

Let μ be a positive measure on a σ-algebra $\mathfrak{M}$. Then

1. $\mu(\varnothing) = 0$.
2. $\mu(A_1 \cup \cdots \cup A_n) = \mu(A_1) + \cdots + \mu(A_n)$ if $A_1, \dots, A_n$ are pairwise disjoint members of $\mathfrak{M}$.
3. $A \subset B$ implies $\mu(A) \le \mu(B)$ if $A \in \mathfrak{M}, B \in \mathfrak{M}$.
4. $\mu(A_n) \to \mu(A)$ as $n \to \infty$ if $A = \bigcup_{n=1}^{\infty} A_n, A_n \in \mathfrak{M}$, and
$$A_1 \subset A_2 \subset A_3 \subset \dots$$
5. $\mu(A_n) \to \mu(A)$ as $n \to \infty$ if $A = \bigcap_{n=1}^{\infty} A_n, A_n \in \mathfrak{M}$,
$$A_1 \supset A_2 \supset A_3 \supset \dots,$$
and $\mu(A_1)$ is finite.

Proof. 1. Take $A \in \mathfrak{M}$ such that $\mu(A) < \infty$. [1] And let $A_2 = A_3 = \cdots = \varnothing$, then $\mu(\varnothing) > 0$ leads to a contradition to the countably additive.

2. Take $A_{n+1} = A_{n+2} = \cdots = \varnothing$.

[1]That is what we supposed at the definition of measure.

3. Note that $B = (B \setminus A) \cup A$, then by additivity

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$$

4. Let $A_0 = \varnothing$, and let $B_n = A_n \setminus A_{n-1}$ for all $n \in \mathbb{N}$. Then $B_1, \cdots, B_n$ are pairwise disjoint members of $\mathfrak{M}$ such that $A = \bigcup_{n=1}^{\infty} B_n$. We have

$$\mu(A) = \sum_{n=1}^{\infty} \mu(B_n) = \sum_{n=1}^{\infty} \mu(A_n \setminus A_{n-1})$$

If one of the $\mu(A_n)$ is ∞, then $\lim_{n\to\infty} \mu(A_n)$ and $\mu(A)$ both are ∞ . Otherwise, we have

$$\mu(A_n \setminus A_{n-1}) = \mu(A_n) - \mu(A_{n-1}), \quad \forall n \mathbb{N}$$

Thus

$$\mu(A) = \sum_{n=1}^{\infty} \mu(A_n \setminus A_{n-1}) = \lim_{n\to\infty} \mu(A_n)$$

5. Let $B_n = A_n \setminus A_{n+1}$ for all $n \in \mathbb{N}$. $B_1, \cdots, B_n$ are pariwise disjoint members of $\mathfrak{M}$ with finite measure, such that

$$A_1 \setminus A = A_1 \setminus \left(\bigcap_{n=1}^{\infty} A_n \right) = \bigcup_{n=1}^{\infty} (A_1 \setminus A_n) = \bigcup_{n=1}^{\infty} \left(\bigcup_{k=1}^{n} B_k \right) = \bigcup_{n=1}^{\infty} B_n$$

. We have

$$\mu(A_1 \setminus A) = \mu\left(\bigcup_{n=1}^{\infty} B_n \right),$$

where the RHS is $\mu(A_1) - \mu(A)$, and the LHS is

$$\sum_{n=1}^{\infty} (\mu(A_n) - \mu(A_{n+1})) = \mu(A_1) - \lim_{n\to\infty} \mu(A_{n+1})$$

Since $\mu(A_1) < \infty$, we have

$$\mu(A) = \lim_{n\to\infty} \mu(A_n)$$

□

Example 2.3.3

1. For any $E \subset X$, where X is any set, define $\mu(E) = \infty$ if E is an infinite set, and let $\mu(E)$ be the number of points in E if E is finite. This μ is called the ***counting measure*** on X.
2. Fix $x_0 \in X$, define $\mu(E) = 1$ if $x_0 \in E$ and $\mu(E) = 0$ if $x_0 \notin E$, for any $E \subset X$. This μ may be called the ***unit mass concentrated at*** x_0.
3. Let μ be the counting measure on the set $\{1, 2, 3, \dots\}$, let $A_n = \{n, n+1, n+2, \dots\}$. Then $\bigcap A_n = \varnothing$ but $\mu(A_n) = \infty$ for $n = 1, 2, 3, \dots$. This shows that the hypothesis

$$\mu(A_1) < \infty$$

is not superfluous in Theorem 2.3.2(5).

2.4 Integration of Positive Funtions

Definition 2.4.1 ▶ Integration of Positive Funtions

1. If $s : X \to [0, \infty)$ is a measurable simple function, of the form

$$s = \sum_{i=1}^{n} \alpha_i \chi_{A_i},$$

where $\alpha_1, \dots, \alpha_n$ are the distinct values of s, and if $E \in \mathfrak{M}$, we define

$$\int_E s \, d\mu = \sum_{i=1}^{n} \alpha_i \mu(A_i \cap E).$$

The convention $0 \cdot \infty = 0$ is used here; it may happen that $\alpha_i = 0$ for some i and that $\mu(A_i \cap E) = \infty$.
2. If $f : X \to [0, \infty]$ is measurable, and $E \in \mathfrak{M}$, we define

$$\int_E f \, d\mu = \sup \int_E s \, d\mu,$$

the supremum being taken over all simple measurable functions s such that $0 \le s \le f$. The left member above is called the ***Lebesgue integral*** of f over E, with respect to the measure μ. It is a number in $[0, \infty]$.

Remark.
We apparently have two definitions for $\int_E f \, \mathrm{d}\mu$ if f is simple, they are the same.

Corollary 2.4.2

(a) If $0 \le f \le g$, then $\int_E f \, d\mu \le \int_E g \, d\mu$.
(b) If $A \subset B$ and $f \ge 0$, then $\int_A f \, d\mu \le \int_B f \, d\mu$.
(c) If $f \ge 0$ and c is a constant, $0 \le c < \infty$, then

$$\int_E cf \, d\mu = c \int_E f \, d\mu.$$

(d) If $f(x) = 0$ for all $x \in E$, then $\int_E f \, d\mu = 0$, even if $\mu(E) = \infty$.
(e) If $\mu(E) = 0$, then $\int_E f \, d\mu = 0$, even if $f(x) = \infty$ for every $x \in E$.
(f) If $f \ge 0$, then $\int_E f \, d\mu = \int_X \chi_E f \, d\mu$.

Remark.
We can also regard the $\int_E f \, \mathrm{d}\mu$ as the restricted function $f|_E$ integrates on the induced measure space E.

Proposition 2.4.3

Let s and t be nonnegative measurable simple functions on X. For $E \in \mathfrak{M}$, define

$$\varphi(E) = \int_E s \, d\mu.$$

Then φ is a measure on $\mathfrak{M}$. Also

$$\int_X (s+t) \, d\mu = \int_X s \, d\mu + \int_X t \, d\mu.$$

Note.
$\varphi(E) = \int_E s \, \mathrm{d}\mu$ 就是依据 s 对测度 μ 进行一个有限的算数的加权. 自然地也就成为一个测度.

Proof. If $E_1, E_2, \cdots$ are disjoint members of $\mathfrak{M}$ whose union is E, the countable additivity of μ shows that

$$\begin{aligned}
\varphi(E) &= \sum_{i=1}^{n} \alpha_i \mu(A_i \cap E) \\
&= \sum_{i=1}^{n} \sum_{j=0}^{\infty} \alpha_i \mu\left(A_i \cap E_j\right) \\
&= \sum_{j=0}^{\infty} \sum_{i=1}^{n} \alpha_i \mu\left(A_i \cap E_j\right) \\
&= \sum_{j=0}^{\infty} \varphi\left(E_j\right)
\end{aligned}$$

Also, $\varphi(\varnothing) \neq \infty$, μ is not identically ∞.

Next, let $\beta_1, \cdots, \beta_m$ be distinct values of t. And let $B_j := \left\{x : t(x) = \beta_j\right\}$. Then

$$\begin{aligned}
\int_X (s+t)\, \mathrm{d}\mu &= \sum_{i,j} \int_{A_i \cap B_j} (s+t)\, \mathrm{d}\mu \\
&= \sum_{i,j} \alpha_i \mu\left(A_i \cap B_j\right) + \sum_{i,j} \beta_j \mu\left(A_i \cap B_j\right) \\
&= \sum_{i} \alpha_i \mu(A_i) + \sum_{j} \beta_j \mu\left(B_j\right) \\
&= \int_X s\, \mathrm{d}\mu + \int_X t\, \mathrm{d}\mu
\end{aligned}$$

□

Theorem 2.4.4 ▸ Lebesgue's Monotone Convergence Theorem

Let $\{f_n\}$ be a sequence of measurable functions on X, and suppose that
(a) $0 \le f_1(x) \le f_2(x) \le \cdots \le \infty$ for every $x \in X$,
(b) $f_n(x) \to f(x)$ as $n \to \infty$, for every $x \in X$.
Then f is measurable, and

$$\int_X f_n\, d\mu \to \int_X f\, d\mu \quad \text{as } n \to \infty.$$

Proof. Since $\int_X f_n \,\mathrm{d}\mu \le \int_X f_{n+1} \,\mathrm{d}\mu$ for all n, there exists an $\alpha > 0$ such that

$$\int_X f_n \,\mathrm{d}\mu \to \alpha, \quad \text{as } n \to \infty$$

By Theorem 2.1.7, f is measurable, then by $\int_X f_n \,\mathrm{d}\mu \le \int_X f \,\mathrm{d}\mu \ \forall n$,

$$\alpha \le \int_X f \,\mathrm{d}\mu$$

Take any simple function $0 \le s \le f$, and $0 \le c < 1$. Define

$$E_n^s := \{x \in X : f_n(x) \ge cs(x)\}$$

Then since $\varphi(E) := \int_E s \,\mathrm{d}\mu$ is a measure ,

$$\alpha \ge \lim_{n\to\infty} \int_X f_n \,\mathrm{d}\mu \ge \lim_{n\to\infty} \int_{E_n^s} f_n \,\mathrm{d}\mu \ge c \lim_{n\to\infty} \int_{E_n^s} s \,\mathrm{d}\mu = c \lim_{n\to\infty} \varphi(E_n^s) = c\varphi(X)$$

That is

$$\alpha \ge c \int_X s \,\mathrm{d}\mu$$

Since s is arbitrary,

$$\alpha \ge c \int_X f \,\mathrm{d}\mu$$

and since c is arbitrary ,

$$\alpha \ge \int_X f \,\mathrm{d}\mu$$

which completes the proof. □

Corollary 2.4.5

Let $f, g : X \to [0, \infty]$ be mearuable functions on X . Then $f + g$ is mearuable and

$$\int_X (f + g) \,\mathrm{d}\mu = \int_X f \,\mathrm{d}\mu + \int_X g \,\mathrm{d}\mu$$

Proof. Let $\{s_n\}, \{s_n'\}$ be simple function sequences in Theorem 2.2.2 of f, g , respectively . Then by the theorem above and the Proposition 2.4.3

$$\begin{aligned}\int_X (f+g)\,\mathrm{d}\mu &= \lim_{n\to\infty}\int_X (s_n+s_n')\,\mathrm{d}\mu = \lim_{n\to\infty}\int_X s_n\,\mathrm{d}\mu + \lim_{n\to\infty}\int_X s_n'\,\mathrm{d}\mu \\ &= \int_X f\,\mathrm{d}\mu + \int_g \mathrm{d}\mu\end{aligned}$$

□

Corollary 2.4.6

If $f_n : X \to [0,\infty]$ is ***measurable***, for $n = 1, 2, 3, \dots$, and

$$f(x) = \sum_{n=1}^{\infty} f_n(x) \quad (x \in X),$$

then

$$\int_X f\,d\mu = \sum_{n=1}^{\infty}\int_X f_n\,d\mu.$$

Proof. Let $g_n(x) = \sum_{k=1}^{n} f_k(x)$. Then $g_1 \le g_2 \le \cdots \le \infty$ and $f = \lim_{n\to\infty} g_n(x)$. Applying Monotone Convergence Theorem to $\{g_n\}$ and f , we have

$$\int_X f\,\mathrm{d}\mu = \lim_{n\to\infty}\int_X \sum_{k=1}^{n} f_k(x)\,\mathrm{d}\mu = \lim_{n\to\infty}\sum_{k=1}^{n}\int_X f_k(x)\,\mathrm{d}\mu = \sum_{n=1}^{\infty}\int_X f_n\,\mathrm{d}\mu$$

□

Corollary 2.4.7

If $a_{ij} \ge 0$ for i and $j = 1, 2, 3, \dots$, then

$$\sum_{i=1}^{\infty}\sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty}\sum_{i=1}^{\infty} a_{ij}.$$

Proof. Let μ be counting measure on $\mathbb{Z}$. Let

$$f_i(j) = a_{ij}, \quad f(k) = \sum_{i=1}^{\infty} f_i(k) = \sum_{i=1}^{\infty} a_{ik}$$

By Monotone Convergence Theorem , for any measurable g ,

$$\int_{\mathbb{Z}} g \, \mathrm{d}\mu = \int_{\mathbb{Z}} \lim_{n\to\infty} \chi(\{1, \cdots, n\}) g \, \mathrm{d}\mu = \lim_{n\to\infty} \int_{\mathbb{Z}} \chi(\{1, \cdots, n\}) g \, \mathrm{d}\mu = \sum_{k=\infty}^{n} g(k)$$

Thus

$$\int_{\mathbb{Z}} f \, \mathrm{d}\mu = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}$$

And we have

$$\sum_{i=1}^{\infty} \int_X f_i \, \mathrm{d}\mu = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij}$$

The above corollary shows that

$$\int_{\mathbb{Z}} f \, \mathrm{d}\mu = \sum_{i=1}^{\infty} \int_X f_i \, \mathrm{d}\mu$$

That is

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}$$

□

Lemma 2.4.8 ▸ Fatou's Lemma

If $f_n : X \to [0, \infty]$ is ***measurable***, for each positive integer n, then

$$\int_X \left(\lim_{n\to\infty} \inf f_n \right) d\mu \leq \lim_{n\to\infty} \inf \int_X f_n d\mu. \tag{1}$$

Note.
有了 Monotone Convergence Theorem 之后, 关键的不等式就剩下

$$\int_X \inf_{n\geq k} f_n \,\mathrm{d}\mu \leq \inf_{n\geq k} \int_X f_n \,\mathrm{d}\mu$$

它的含义是局部最小的积分小于等于积分的最小.

Proof. Let

$$g_k(x) = \inf_{n\geq k} f_n(x)$$

Then

$$\lim_{n\to\infty} \inf f_n = \lim_{n\to\infty} g_n =: g$$

where $\{g_k\}, g$ are all measurable . It is obvious that $\{g_k\}$ is increasing . Thus

$$\lim_{n\to\infty} \int_X g_n \,\mathrm{d}\mu = \int_X g \,\mathrm{d}\mu = \int_X \left(\lim_{n\to\infty} \inf f_n\right) \mathrm{d}\mu$$

Only need to show that

$$\int_X \inf_{n\geq k} f_n \,\mathrm{d}\mu \leq \inf_{n\geq k} \int_X f_n \,\mathrm{d}\mu$$

which is true since

$$\int_X \inf_{n\geq k} f_n \,\mathrm{d}\mu \leq \int_X f_n \,\mathrm{d}\mu, \text{for all } n \geq k$$

□

Theorem 2.4.9

Suppose $f : X \to [0, \infty]$ is measurable, and

$$\varphi(E) = \int_E f \, d\mu \quad (E \in \mathfrak{M}).$$

Then φ is a measure on $\mathfrak{M}$, and

$$\int_X g\, d\varphi = \int_X gf\, d\mu$$

for every measurable g on X with range in $[0, \infty]$.

Note.
将 measurable function f 视为对 measure μ 的某种 countable 次算数加权操作. 那么得益于 μ 对 countable 操作的相容性, φ 也就成为一个测度. φ 的积分行为也验证了这种看法的合理性.

Proof.

- Let $E_1, \cdots, E_n, \cdots$ be disjoint measurable sets such that $E = \bigcup_{n=1}^{\infty} E_n$. Then $\left\{\sum_{k=1}^{n} \chi_{E_k} f\right\}$ be increasing measurable function sequence such that $\lim_{n\to\infty} \sum_{k=1}^{n} \chi_{E_k} f = f$. The Monotone Convergence Theorem shows that

$$\begin{aligned}\sum_{n=1}^{\infty} \varphi(E_k) = \lim_{n\to\infty} \sum_{k=1}^{n} \int_{E_k} f\, \mathrm{d}\mu &= \lim_{n\to\infty} \int_E \sum_{k=1}^{n} \chi_{E_k} f\, \mathrm{d}\mu \\ &= \int_E f\, \mathrm{d}\mu = \varphi(E)\end{aligned}$$

  Since $\varphi(\varnothing) = 0$. φ is a measure on X .

- To show the second results. See that it is true for $g = \chi_E$. Thus it is true for g is a simple function . And the general case follows from the Monotone Convergence Theorem.

□

2.5 Integration of Complex Functions

Definition 2.5.1 ▸ Lebegsgue Integrability

We define $L^1(\mu)$ to be the collection of all complex measurable functions f on X for which

$$\int_X |f|\, d\mu < \infty.$$

Note that the measurability of f implies that of $|f|$, hence the above integral is defined.
The members of $L^1(\mu)$ are called ***Lebesgue integrable functions*** (with respect to μ) or ***summable functions***.

Definition 2.5.2 ▶ Complex Integration

If $f = u + iv$, where u and v are real measurable functions on X, and if $f \in L^1(\mu)$, we define

$$\int_E f\,d\mu = \int_E u^+\,d\mu - \int_E u^-\,d\mu + i\int_E v^+\,d\mu - i\int_E v^-\,d\mu \tag{1}$$

for every measurable set E.

Remark.
We have $u^+ \leq |u| < |f|$, etc., so that each of these four integrals is finite. Thus (1) defines the integral on the left as a complex number.

Definition 2.5.3 ▶ Real Integration

Occasionally it is desirable to define the integral of a measurable function f with range in $[-\infty, \infty]$ to be

$$\int_E f\,d\mu = \int_E f^+\,d\mu - \int_E f^-\,d\mu, \tag{2}$$

provided that at least one of the integrals on the right of (2) is finite. The left side of (2) is then a number in $[-\infty, \infty]$.

Theorem 2.5.4

Suppose f and $g \in L^1(\mu)$ and α and β are complex numbers. Then $\alpha f + \beta g \in L^1(\mu)$, and

$$\int_X (\alpha f + \beta g)\,d\mu = \alpha\int_X f\,d\mu + \beta\int_X g\,d\mu$$

Proof sketch.
可积性来自于模长的三角不等式. 主要证明可加性. 将 $f + g = h$ 的正负部分到两边, 就都变成了非负函数. 积分即可.

Theorem 2.5.5

If $f \in L^1(\mu)$, then

$$\left|\int_X f d\mu\right| \leq \int_X |f| d\mu.$$

Note.
被缩放的部分实际上是 f 因改变了方向产生的损耗.

Proof. Let $z = \int_X f \,\mathrm{d}\mu$, z is a complex number . There exists a complex number α ($\alpha = \frac{\bar{z}}{|z|}$), such that $|\alpha| = 1$, and

$$\left|\int_X f \,\mathrm{d}\mu\right| = \alpha z = \alpha \int_X f \,\mathrm{d}\mu = \int_X \alpha f \,\mathrm{d}\mu$$

Suppose $\alpha f = u + iv$, since $\int_X \alpha f \,\mathrm{d}\mu$ is a real number , $\int_X v \,\mathrm{d}\mu = 0$. We have

$$\int_X \alpha f \,\mathrm{d}\mu = \int_X u \,\mathrm{d}\mu \leq \int_X |\alpha f| \,\mathrm{d}\mu = \int_X |f| \,\mathrm{d}\mu$$

□

Theorem 2.5.6 ▸ Lebesgue's Dominated Convergence Theorem

Suppose $\{f_n\}$ is a sequence of complex measurable functions on X such that

$$f(x) = \lim_{n\to\infty} f_n(x)$$

exists for every $x \in X$. If there is a function $g \in L^1(\mu)$ such that

$$|f_n(x)| \leq g(x) \quad (n = 1, 2, 3, \dots; x \in X),$$

then $f \in L^1(\mu)$,

$$\lim_{n\to\infty} \int_X |f_n - f| \,d\mu = 0,$$

and

$$\lim_{n\to\infty}\int_X f_n\, d\mu = \int_X f\, d\mu.$$

Proof sketch.
$\{f_n\}$ 不一定单调, 能将极限从积分外塞进积分里的工具, 目前我们只有 Fatou's Lemma , 但是它原始的样子跟我们需要的不等号是反向的. 自然的想法是填一个负号, 这又会导致函数失去非负性, 而控制函数 g 恰好能弥补这一点不足. 实现起来, 我们从 g 上扣去 f, f_n , 而非直接拿出 f, f_n 去做.

Proof. Since $|f| \leq g$ and f is measurable , $f \in L^1(\mu)$. Function $(2g - |f_n - f|)$ is a nonnegative measurable function . Applying Fatou's Lemma to it , we have

$$\int_X \lim_{n\to\infty}\inf(2g - |f_n - f|)\,\mathrm{d}\mu \leq \lim_{n\to\infty}\inf\int_X (2g - |f_n - f|)\,\mathrm{d}\mu$$

where

$$\lim_{n\to\infty}\inf(2g - |f_n - f|) = 2g - \lim_{n\to\infty}\sup|f_n - f| = 2g$$

$$\lim_{n\to\infty}\inf\int_X (2g - |f_n - f|)\,\mathrm{d}\mu = 2\int_X g\,\mathrm{d}\mu - \lim_{n\to\infty}\sup\int_X |f_n - f|\,\mathrm{d}\mu$$

The original inequality comes to

$$2\int_X g\,\mathrm{d}\mu \leq 2\int_X g\,\mathrm{d}\mu - \lim_{n\to\infty}\sup\int_X |f_n - f|\,\mathrm{d}\mu$$

Since $\int_X g\,\mathrm{d}\mu$ is finite, we can substract $2\int_X g\,\mathrm{d}\mu$ both sides and get

$$\lim_{n\to\infty}\sup\int_X |f_n - f|\,\mathrm{d}\mu \leq 0$$

Thus

$$\lim_{n\to\infty}\int_X |f_n - f|\,\mathrm{d}\mu = 0$$

Finnaly , we have

$$\begin{aligned}\left|\lim_{n\to\infty}\int_X f_n \, d\mu - \int_X f \, d\mu\right| &= \left|\lim_{n\to\infty}\int_X (f_n - f) \, d\mu\right| \\ &\leq \lim_{n\to\infty}\int_X |f_n - f| \, d\mu \\ &= 0\end{aligned}$$

which completes the proof. □

2.6 The Sets of Measure Zero

Definition 2.6.1

Let P be a property which a point x may or may not have. If μ is a measure on a σ-algebra $\mathfrak{M}$ and if $E \in \mathfrak{M}$, the statement "P holds ***almost everywhere on*** E" (abbreviated to "P holds a.e. on E") means that there exists an $N \in \mathfrak{M}$ such that $\mu(N) = 0$, $N \subset E$, and P holds at every point of $E - N$.

Remark.
This concept of a.e. depends of course very strongly on the given measure, and we shall write "a.e. $[\mu]$" whenever clarity requires that the measure be indicated.

Example 2.6.2

For example, if f and g are measurable functions and if

$$\mu(\{x : f(x) \neq g(x)\}) = 0,$$

we say that $f = g$ a.e. $[\mu]$ on X, and we may write $f \sim g$.

Remark.
This is easily seen to be an equivalence relation. The transitivity ($f \sim g$ and $g \sim h$ implies $f \sim h$) is a consequence of the fact that the union of two sets of measure 0 has measure 0.

Proposition 2.6.3 ▶ Completion of a Measure

If $f \sim g$, then, for every $E \in \mathfrak{M}$,

$$\int_E f d\mu = \int_E g d\mu.$$

Theorem 2.6.4

Let $(X, \mathfrak{M}, \mu)$ be a measure space, let $\mathfrak{M}^*$ be the collection of all $E \subset X$ for which there exist sets A and $B \in \mathfrak{M}$ such that $A \subset E \subset B$ and $\mu(B - A) = 0$, and define $\mu(E) = \mu(A)$ in this situation. Then $\mathfrak{M}^*$ is a σ-algebra, and μ is a measure on $\mathfrak{M}^*$.

Remark.
This extended measure μ is called ***complete***, since all subsets of sets of measure 0 are now measurable; the σ-algebra $\mathfrak{M}^*$ is called the ***μ-completion*** of $\mathfrak{M}$. The theorem says that every measure can be completed, so, whenever it is convenient, we may assume that any given measure is complete; this just gives us more measurable sets, hence more measurable functions.

Proof. We begin by checking that μ is well defined for every $E \in \mathfrak{M}^*$. If for $i = 1, 2$, $A_i \subseteq E \subseteq B_i$, where $A_i, B_i \in \mathfrak{M}, \mu(B_i - A_i) = 0$. We need to show that $\mu(A_1) = \mu(A_2)$. Since $A_1 \subseteq E \subseteq B_2$,

$$\mu(A_1) \leq \mu(B_2) = \mu(A_2) + \mu(B_2 - A_2) = \mu(A_2)$$

By symmetric ,

$$\mu(A_2) \leq \mu(A_2)$$

Thus $\mu(A_1) = \mu(A_2)$, $\mu(E)$ is well defined. Next , let us verify that $\mathfrak{M}^*$ has the three defining properties of a σ-algebra.

1. It is clearly that $X \in \mathfrak{M}^*$.
2. If $A \subseteq E \subseteq B$, then $B^c \subseteq E^c \subseteq A^c$. And since $\mu(A^c - B^c) = \mu(A^c \cap B) = \mu(B - A) = 0$, $E^c \in \mathfrak{M}^*$.
3. Suppose $A_i \subseteq E_i \subseteq B_i, i = 1, 2, \cdots$, where $A_i, B_i \in \mathfrak{M}$, $\mu(B_i - A_i) = 0$.

Let $E = \bigcup E_i, A = \bigcup A_i, B = \bigcup B_i$. Since

$$B - A = \bigcup_i (B_i - A) \subseteq \bigcup_i (B_i - A_i)$$

,

$$\mu(B - A) \leq \sum_i \mu(B_i - A_i) = 0$$

Then by $A \subseteq E \subseteq B$, it follows that $E \in \mathfrak{M}^*$.

Finally, we show the countable additive of μ . If the above sets E_i are disjoint, the smae is true of the sets A_i , then

$$\mu(E) = \mu(A) = \sum_i \mu(A_i) = \sum_i \mu(E_i)$$

□

Definition 2.6.5 ▶ More Mearuable Functions

We say a function f defined on a set $E \in \mathfrak{M}$ is ***measurable on*** X . If $\mu(E^c) = 0$ and if $f^{-1}(V) \cap E$ is a measurable for every open set V .

Remark.

- If we define $f(x) = 0$ for $x \in E^c$, we obtain a mearuable function on X , in the old sense.
- If our measure happens to be complete , we can define f on E^c in a perfrctly arbitrary manner, and we still get a measurable function . The integral of f over any set $A \in \mathfrak{M}$ is independent of the definition of f on E^c; therefore this definition need not even be specified at all.

Theorem 2.6.6

Suppose $\{f_n\}$ is a sequence of complex measurable functions defined a.e. on X such that

$$\sum_{n=1}^{\infty} \int_X |f_n| \, d\mu < \infty. \tag{1}$$

Then the series

$$f(x) = \sum_{n=1}^{\infty} f_n(x) \tag{2}$$

converges for almost all x, $f \in L^1(\mu)$, and

$$\int_X f \, d\mu = \sum_{n=1}^{\infty} \int_X f_n \, d\mu. \tag{3}$$

Remark.

- 现在，考察积分和求和是否可交换，只需要检查模长积分的级数是否收敛.
- 即便 f_n 在 X 上处处被定义，级数收敛的地方也只是 ***almost everywhere*** .

Note.
此时，控制函数就是绝对级数本身，它在一个满测集上将原级数控制住. 其特殊性间接给出了原级数的几乎处处收敛性质.

Proof. Let S_n be the set on which f_n is defined, so that $\mu\left(S_n^c\right) = 0$. Let $S = \bigcap S_n$. Put $\varphi\left(x\right) = \sum \left|f_n\left(x\right)\right|$, for $x \in S$ .By Monotone Convergence Theorem

$$\int_S \varphi \, \mathrm{d}\mu \leq \sum_{n=1}^{\infty} \int_S \left|f_n\right| \, \mathrm{d}\mu < \infty$$

Let $E = \{x \in S : \varphi\left(x\right) < \infty\}$. Then it follows that $\mu\left(E^c\right) = 0$. We have

$$\sum_{n=1}^{\infty} \left|f_n\left(x\right)\right| < \infty, \quad \forall x \in E$$

Thus

$$f\left(x\right) = \sum_{n=1}^{\infty} f_n\left(x\right)$$

converges on E , which is a set with its complement measure zero . That is to say f coverges for almost all x . And since $f \in L^1\left(\mu|_E\right)$, $\mu\left(E\right) = 0$, we have

$f \in L^1(\mu)$. Finally , by Monotone Convergence Theorem ,

$$\int_X f \, \mathrm{d}\mu = \int_E f \, \mathrm{d}\mu = \sum_{n=1}^{\infty} \int_E f_n \, \mathrm{d}\mu = \sum_{n=1}^{\infty} \int_X f_n \, \mathrm{d}\mu$$

□

Theorem 2.6.7

1. Suppose $f : X \to [0, \infty]$ is measurable , $E \in \mathfrak{M}$, and $\int_E f \, \mathrm{d}\mu = 0$. Then $f = 0$ a.e. on E.
2. Suppose $f \in L^1(\mu)$ and $\int_E f \, \mathrm{d}\mu = 0$ for every $E \in \mathfrak{M}$. Then $f = 0$ a.e. on X .
3. Suppose $f \in L^1(\mu)$ and

$$\left| \int_X f \, \mathrm{d}\mu \right| = \int_X |f| \, \mathrm{d}\mu$$

Then there is a constant α such that $\alpha f = |f|$ *a.e.* on X .

Note.
3. 这个积分等式的意义是不存在积分项因方向改变产生的损耗. 结论就是说如果这种损耗不存在, 那么 f 几乎不改变方向.

Proof. 1. Let $A_n = \left\{ x \in E : f(x) \geq \frac{1}{n} \right\}$. Then

$$0 = \int_E f \, \mathrm{d}\mu \geq \int_{A_n} f \, \mathrm{d}\mu \geq \mu(A_n) \frac{1}{n}.$$

It follows that

$$\mu(A_n) = 0, \quad \forall n = 1, 2, \cdots$$

Let $A = \bigcup_{n=1}^{\infty} A_n$. Then $\mu(A) = 0$, and $f \equiv 0$ on $E \setminus A$. Thus $f = 0$ a.e. on E .

2. Suppose $f = u + iv$. Let $E = \{x \in X : u(x) \geq 0\}$. Since $\mathrm{Re}\, f = u^+$ on E , then

$$\int_E u^+ \, \mathrm{d}\mu = 0$$

It follows by 1. that $\mu^+ = 0$ a.e. on E . Since $\mu^+(X \setminus E) \equiv 0$, $\mu^+ = 0$ a.e.

on X . We conclude similarly that

$$u^- = v^+ = v^- = 0, \quad a.e.$$

3. Let $z = \int_X f \, \mathrm{d}\mu$. Then z is a complex number . Take $\alpha = \bar{z}/\,|z|$ .Then

$$\alpha z = |z| = \left| \int_X f \, \mathrm{d}\mu \right|$$

Where

$$\alpha z = \alpha \int_X f \, \mathrm{d}\mu = \int_X \alpha f \, \mathrm{d}\mu$$

It follows that

$$\int_X \alpha f \, \mathrm{d}\mu = \left| \int_X f \, \mathrm{d}\mu \right| = \int_X |f| \, \mathrm{d}\mu \implies \int_X (\alpha f - |f|) \, \mathrm{d}\mu = 0$$

which shows that

$$\alpha f = |f|, \quad a.e. \text{ on } X$$

□

Theorem 2.6.8

Suppose $\mu(X) < \infty$, $f \in L^1(\mu)$, S is a closed set in the complex plane, and the averages

$$A_E(f) = \frac{1}{\mu(E)} \int_E f \, d\mu$$

lie in S for every $E \in \mathfrak{M}$ with $\mu(E) > 0$. Then $f(x) \in S$ for almost all $x \in X$.

Proof. Let Δ be circular disk , with centre α and radius r . And suppose that Δ lies in S^c . S^c can be written as the union of countably many such disks . Thus, we only need to show that $\mu(E) = 0$, where $E = f^{-1}(\Delta)$. If $\mu(E) > 0$, then

$$|A_E(f) - \alpha| = \frac{1}{\mu(E)} \left| \int_E (f - \alpha) \, \mathrm{d}\mu \right| \leq \frac{1}{\mu(E)} \int_E |f - \alpha| \, \mathrm{d}\mu \leq r$$

Which leads to a contradition, since $A_E(f) \in S$. Hence $\mu(E) = 0$. □

Theorem 2.6.9

Let $\{E_k\}$ be a sequence of measurable sets in X, such that

$$\sum_{k=1}^{\infty} \mu(E_k) < \infty.$$

Then almost all $x \in X$ lie in at most finitely many of the sets E_k.

Proof. Let

$$g(x) = \sum_{k=1}^{\infty} \chi_{E_k}(x)$$

Then g is measurable . By Monotone Convergence Theorem ,

$$\int_X g(x)\, \mathrm{d}\mu = \sum_{k=1}^{\infty} \int_X \chi_{E_k}(x)\, \mathrm{d}\mu = \sum_{k=1}^{\infty} \mu(E_k) < \infty$$

Thus $g \in L^1(\mu)$. There is a $E \in \mathfrak{M}$, $\mu(E^c) = 0$, such that $g^{-1}(\infty) \cap E = \varnothing$. If $x \in X$ lie in infinitely many of the sets E_k . Then $g(x) = \infty$, it follows that $x \in E^c$. Thus , points that lie in infinitely many of the sets E_k will always lie in E^c as well , whose measure is zero . □

2.7 Exersices

Note.
关于积分的问题有两种常用标准技巧. 一种是按取值划分集合. 一种用可控的函数列逼近.

Problem 2.7.1

Does there exist an infinite σ-algebra which has only countably many members?

Proof. □

Problem 2.7.2

Prove that if f is a real function on a measurable space X such that $\{x : f(x) \geq r\}$ is measurable for every rational r, then f is measurable.

Proof. Take any real number a . We have

$$f^{-1}((a,\infty]) = \bigcup_{r\in\mathbb{Q}, r>a} f^{-1}([r,\infty])$$

which is a countably many union of measurable sets . Thus $f^{-1}((a,\infty])$ is measurable for all a . It follows that f is measurable . $\square$

Problem 2.7.3

Let $\{a_n\}$ and $\{b_n\}$ be sequences in $[-\infty,\infty]$, and prove the following assertions:

(a)

$$\limsup_{n\to\infty}(-a_n) = -\liminf_{n\to\infty} a_n.$$

(b)

$$\limsup_{n\to\infty}(a_n+b_n) \leq \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n$$

provided none of the sums is of the form $\infty - \infty$.

(c) If $a_n \leq b_n$ for all n, then

$$\liminf_{n\to\infty} a_n \leq \liminf_{n\to\infty} b_n.$$

Show by an example that strict inequality can hold in (b).

Proof. 1.

$$\limsup_{n\to\infty}(-a_n) = \lim_{n\to\infty}\sup_{k\geq n}\{-a_k\} = -\lim_{n\to\infty}\inf_{k\geq n}\{a_k\} = -\liminf_{n\to\infty} a_n$$

2.
$$\begin{aligned}\limsup_{n\to\infty}(a_n+b_n) &= \lim_{n\to\infty}\sup_{k\geq n}\{a_k+b_k\}\\ &\leq \lim_{n\to\infty}\left(\sup_{k\geq n}\{a_k\}+\sup_{k\geq n}\{b_k\}\right)\\ &= \lim_{n\to\infty}\sup_{k\geq n}\{a_k\}+\lim_{n\to\infty}\sup_{k\geq n}\{b_k\}\\ &= \limsup_{n\to o} a_n+\limsup_{n\to\infty} b_n\end{aligned}$$

Let $\{a_n\}=\left\{(-1)^n\right\}$, $\{b_n\}=\left\{(-1)^{n+1}\right\}$. Then the RHS of the inequality is $1+1=2$. The LHS is 0.

3. If $a_n\leq b_n$ for all n, then
$$\inf_{k\geq n}\{a_k\}\leq\inf_{k\geq n}\{b_k\}$$
Thus
$$\liminf_{n\to\infty} a_n=\lim_{n\to\infty}\inf_{k\geq n}\{a_k\}\leq\lim_{n\to\infty}\inf_{k\geq n}\{b_k\}=\liminf_{n\to\infty} b_n$$

□

Problem 2.7.4

(a) Suppose $f : X\to[-\infty,\infty]$ and $g : X\to[-\infty,\infty]$ are measurable. Prove that the sets
$$\{x : f(x)<g(x)\},\{x : f(x)=g(x)\}$$
are measurable.

(b) Prove that the set of points at which a sequence of measurable real-valued functions converges (to a finite limit) is measurable.

Proof. 1. Let
$$h=\begin{cases} g-f, & x\in f^{-1}((-\infty,\infty))\\ 0, & f(x)=g(x)=-\infty\\ 0, & f(x)=g(x)=\infty\\ -1, & f(x)=\infty, g(x)<\infty\\ 1, & f(x)=-\infty, g(x)>-\infty\end{cases}$$

It is easy to see that h is measurable. Thus

$$\{x : f(x) < g(x)\} = h^{-1}((0, \infty])$$

is measruable . And

$$\{x : f(x) = g(x)\} = h^{-1}(0)$$

is measurable .

2. Let $\{f_n\}$ be a sequence of measurable real-valued functions . Take $x \in X$. $\{f_n(x)\}$ is converges. If and only if for any $k \in \mathbb{N}$, there exists $N \in \mathbb{N}$, such that for all $m, n \geq N$, $|f_m(x) - f_n(x)| \leq \frac{1}{k}$. That is to say

$$\{x : \{f_n(x)\} \text{ converges }\} = \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{m=N}^{\infty} \bigcap_{n=N}^{\infty} (|f_m - f_n|)^{-1} \left((-\infty, \frac{1}{k}]\right)$$

which is a measurable set .

□

Problem 2.7.5

Let X be an uncountable set, let $\mathfrak{M}$ be the collection of all sets $E \subset X$ such that either E or E^c is at most countable, and define $\mu(E) = 0$ in the first case, $\mu(E) = 1$ in the second. Prove that $\mathfrak{M}$ is a σ-algebra in X and that μ is a measure on $\mathfrak{M}$. Describe the corresponding measurable functions and their integrals.

Proof. We will first show $\mathfrak{M}$ is a σ-algebra step by step .

1. It is clear that if $E \in \mathfrak{M}$, E^c is as well .
2. Let $E_1, E_2, \cdots$ be members of $\mathfrak{M}$. If all of E_n is countable , then $\bigcup_{n=1}^{\infty} E_n$ is at most countable. Otherwise ,we may assume that E_1 is uncountable ,then E^c is at most countable.

$$X \setminus \bigcup_{n=1}^{\infty} E_n = E_1^c \cap \bigcap_{n=2}^{\infty} E_n^c$$

is at most countable . Hence $\bigcup_{n=1}^{\infty} E_n \in \mathfrak{M}$.

3. Since $\varnothing$ is at most coutable , $X \in \mathfrak{M}$.

The above shows that $\mathfrak{M}$ is a σ-algebra . To see μ is a measure , take $E_1, E_2, \cdots \in \mathfrak{M}$ with any two of them have empty intersection . If all of E_n is countable , then $\bigcup_{n=1}^{\infty} E_n$ is as well . We have

$$\sum_{n=1}^{\infty} \mu(E_n) = 0 = \mu\left(\bigcup_{n=1}^{\infty} E_n\right)$$

If one of the sets E_n is uncountable , say E_1 . If $E_i, i \geq 2$ is uncountable.

$$X = (\varnothing)^c = (E_1 \cap E_2)^c = E_1^c \cup E_2^c$$

is at most countable , which leads to a contradiction . Thus $E_i, i \geq 2$ is at most countable. Hence

$$\sum_{n=1}^{\infty} \mu(E_n) = 1 + 0 + \cdots + 0 = 1 = \mu\left(\bigcup_{n=1}^{\infty} E_n\right).$$

It follows that μ is a measure on $\mathfrak{M}$, since $\mu(\varnothing) = 0$.

To describe the measurable functions . If f is not *a.e.* constant , then there are two value a, b of f , such that $f^{-1}(a)$, $f^{-1}(b)$ is uncountable . Then

$$f^{-1}(b) \subseteq \left(f^{-1}(a)\right)^c$$

is at most countable , which leads to a contradiction. Thus f is *a.e.* constant. Take any measurable function f . Suppose that $E = f^{-1}(c)$, $\mu(E) \neq 0$. Then $\mu(E^c) = 0$. It follows that f is integrable and

$$\int_X f \, \mathrm{d}\mu = \int_E f \, \mathrm{d}\mu = c\mu(E) = c$$

$\square$

Problem 2.7.6

Suppose $f_n : X \to [0, \infty]$ is measurable for $n = 1, 2, 3, \ldots$, $f_1 \geq f_2 \geq f_3 \geq \cdots \geq 0$, $f_n(x) \to f(x)$ as $n \to \infty$, for every $x \in X$, and $f_1 \in L^1(\mu)$. Prove that then

$$\lim_{n\to\infty} \int_X f_n \, d\mu = \int_X f \, d\mu$$

and show that this conclusion does not follow if the condition ”$f_1 \in L^1(\mu)$” is omitted.

Proof. Let $g_n = f_1 - f_n$, then

$$0 = g_1 \leq g_2 \leq \cdots, \quad g_n(x) \to f_1(x) - f(x) \text{ as } n \to \infty$$

Applying the Monotone Convergence Theorem to $\{g_n\}$, we have

$$\int_X f_1 \, d\mu - \lim_{n\to\infty} \int_X f_n \, d\mu = \lim_{n\to\infty} \int_X g_n \, d\mu = \int_X (f_1 - f) \, d\mu = \int_X f_1 \, d\mu - \int_X f \, d\mu$$

Since $\int_X f_1 \, d\mu < \infty$, substracting $\int_X f_1 \, d\mu$ both sides , we have

$$\lim_{n\to\infty} \int_X f_n \, d\mu = \int_X f \, d\mu.$$

Let $X = \mathbb{R}$ with borel measure, and let $f_n = \chi_{\mathbb{R}\setminus[0,n]}\frac{1}{n}$. Then $f_n(x) \to f(x) = 0$, $\int_X f \, d\mu = 0$. But

$$\int_X f_n \, d\mu = \infty, \quad \forall n$$

Thus the condition “$f_1 \in L^1(\mu)$” can not be omitted. □

Problem 2.7.7

Put $f_n = \chi_E$ if n is odd, $f_n = 1 - \chi_E$ if n is even. What is the relevance of this example to Fatou’s lemma?

Proof.

$$\liminf_{n\to\infty} f_n(x) \equiv 0, \quad \int_X \liminf_{n\to\infty} f_n \, d\mu = 0$$

$$\int_X f_n \, d\mu = \begin{cases} \mu(E), & n \text{ is odd} \\ \mu(X \setminus E), & n \text{ is even} \end{cases}$$

$$\liminf_{n\to\infty} \int_X f_n \, d\mu = \min\{\mu(E), \mu(X \setminus E)\}$$

□

Problem 2.7.8

Suppose μ is a positive measure on X, $f : X \to [0, \infty]$ is measurable, $\int_X f\, d\mu = c$, where $0 < c < \infty$, and α is a constant. Prove that

$$\lim_{n\to\infty} \int_X n \log[1 + (f/n)^\alpha]\, d\mu = \begin{cases} \infty & \text{if } 0 < \alpha < 1, \\ c & \text{if } \alpha = 1, \\ 0 & \text{if } 1 < \alpha < \infty. \end{cases}$$

Hint: If $\alpha \geq 1$, the integrands are dominated by αf. If $\alpha < 1$, Fatou's lemma can be applied.

Proof. By differenting $\frac{1}{y} \log(1 + ky)$. We know $n \log\left(1 + \frac{f}{n}\right)$ is increasing as n is increasing. By Monotone Convergence Theorem ,

$$\lim_{n\to\infty} \int_X n \log\left[1 + \left(\frac{f}{n}\right)\right] \mathrm{d}\mu = \int_X \lim_{n\to\infty} n \log\left[1 + \frac{f}{n}\right] \mathrm{d}\mu = \int_X f\, \mathrm{d}\mu = c$$

. If $\alpha > 1$, then note that

$$0 \leq n \log\left[1 + \left(\frac{f}{n}\right)^\alpha\right] \leq \alpha f, \quad \int_X \alpha f\, \mathrm{d}\mu = \alpha c < \infty$$

The Dominated Convergence Theorem gives that

$$\lim_{n\to\infty} \int_X n \log\left[1 + \left(\frac{f}{n}\right)^\alpha\right] \mathrm{d}\mu = \int_X \lim_{n\to\infty} n \log\left[1 + \left(\frac{f}{n}\right)^\alpha\right] \mathrm{d}\mu = 0$$

If $\alpha < 1$, the Fatou;s Lemma gives that

$$\lim_{n\to\infty} \int_X n \log\left[1 + \left(\frac{f}{n}\right)^\alpha\right] \mathrm{d}\mu \geq \int_X \lim_{n\to\infty} n \log\left[1 + \left(\frac{f}{n}\right)^\alpha\right] \mathrm{d}\mu = \int_X \infty\, \mathrm{d}\mu = \infty$$

$\square$

Problem 2.7.9

Suppose $\mu(X) < \infty$, $\{f_n\}$ is a sequence of bounded complex measurable functions on X, and $f_n \to f$ uniformly on X. Prove that

$$\lim_{n\to\infty}\int_X f_n\,d\mu = \int_X f\,d\mu,$$

and show that the hypothesis "$\mu(X) < \infty$" cannot be omitted.

Proof. Take any $\varepsilon > 0$,

$$\lim_{n\to\infty}\int_X |f_n - f|\,\mathrm{d}\mu \le \mu\,(X)\,\varepsilon$$

Let $\varepsilon \to 0$,we have

$$\lim_{n\to\infty}\int_X |f_n - f|\,\mathrm{d}\mu = 0$$

Thus

$$\left|\lim_{n\to\infty}\int_X f_n\,\mathrm{d}\mu - \int_X f\,\mathrm{d}\mu\right| \le \lim_{n\to\infty}\int_X |f_n - f|\,\mathrm{d}\mu = 0$$

Let $f_n = \chi_{[0,n]}\frac{1}{n}$. Then $f = 0$,

$$\lim_{n\to\infty}\int_X f_n\,\mathrm{d}\mu = 1$$

$$\int_X f\,\mathrm{d}\mu = 0$$

Thus $\mu\,(X) < \infty$ cannot be omitted. □

Problem 2.7.10

Let $\{E_k\}$ be a sequence of measurable sets in X, such that

$$\sum_{k=1}^{\infty}\mu(E_k) < \infty.$$

Suppose A is the set of all x which lie in infinitely many E_k . Show that

$$A = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k$$

Then almost all $x \in X$ lie in at most finitely many of the sets E_k.

Proof. $x \in A^c$ iff there exists $k \in \mathbb{N}$, such that for all $n \geq k$, $x \in E_k^c$. That is

$$A^c = \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} E_k^c$$

Thus

$$A = \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} E_k$$

$$\mu(A) = \lim_{k\to\infty} \mu\left(\bigcup_{n=k}^{\infty} E_k\right) \leq \lim_{k\to\infty} \sum_{n=k}^{\infty} \mu(E_k) = \left(\sum_{k=1}^{\infty} \mu(E_k) - \lim_{k\to\infty} \sum_{m=1}^{k} \mu(E_k)\right) = 0$$

□

Problem 2.7.11

Suppose $f \in L^1(\mu)$. Prove that to each $\epsilon > 0$ there exists a $\delta > 0$ such that $\int_E |f|\, d\mu < \epsilon$ whenever $\mu(E) < \delta$.

Proof. Let

$$f_n := \min\{|f|, n\}$$

Then

$$0 \leq f_1 \leq f_2 \leq \cdots, \quad f_n \to |f| \text{ as } n \to \infty$$

$$\int_E |f| \,\mathrm{d}\mu \leq \lim_{n\to\infty} \int_X f_n \,\mathrm{d}\mu$$

There exists $N \in \mathbb{N}$,such taht

$$\int_E |f| \,\mathrm{d}\mu \leq \int_E f_N \,\mathrm{d}\mu + \frac{\varepsilon}{2} \leq \mu(E) N + \frac{\varepsilon}{2} \leq \varepsilon$$

whenever $\mu(E) < \delta := \frac{\varepsilon}{2N}$ □

Problem 2.7.12

If $f \geq 0$, then

$$\int_E \infty \cdot f \, d\mu = \infty \cdot \int_E f \, d\mu.$$

Proof. Let $E_1 = f^{-1}(0) \cap E$, $E_2 = E \setminus E_1$.

1. If $\mu(E_2) > 0$ Then

$$\int_E \infty \cdot f \, \mathrm{d}\mu \geq \int_{E_2} \infty \cdot f \, \mathrm{d}\mu = \infty$$

$$\int_E f \, \mathrm{d}\mu \geq \int_{E_2} f \, \mathrm{d}\mu > 0, \quad \infty \cdot \int_E f \, \mathrm{d}\mu = \infty$$

2. If $\mu(E_2) = 0$, then

$$\int_E \infty \cdot f \, \mathrm{d}\mu = \int_{E_1} \infty \cdot 0 \, \mathrm{d}\mu + \int_{E_2} \infty \cdot f \, \mathrm{d}\mu = 0$$

$$\int_E f \, \mathrm{d}\mu = \int_{E_1} 0 \, \mathrm{d}\mu + \int_{E_2} f \, \mathrm{d}\mu = 0, \quad \infty \cdot \int_E f \, \mathrm{d}\mu = 0$$

□

3 Measure Theory

3.1 Outer Measure

Definition 3.1.1 ▶ Outer Measure

A function φ defined for every subset A of an arbitrary set X is called an ***outer measure*** on X if the following conditions are satisfied:

(i) $\varphi(\varnothing) = 0$,

(ii) $0 \leq \varphi(A) \leq \infty$ whenever $A \subset X$,

(iii) $\varphi(A_1) \leq \varphi(A_2)$ whenever $A_1 \subset A_2$,

(iv) $\varphi(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \varphi(A_i)$ for every countable collection of sets $\{A_i\}$ in X.

Example 3.1.2 ▶ Outer Measure

(i) In an arbitrary set X, define $\varphi(A) = 1$ if A is nonempty and $\varphi(\varnothing) = 0$.

(ii) Let $\varphi(A)$ be the number (possibly infinite) of points in A.

(iii) Let $\varphi(A) = \begin{cases} 0 & \text{if card } A \leq \aleph_0, \\ 1 & \text{if card } A > \aleph_0. \end{cases}$

(iv) If X is a metric space, fix $\varepsilon > 0$. Let $\varphi(A)$ be the smallest number of balls of radius ε that cover A.

(v) Select a fixed x_0 in an arbitrary set X, and let

$$\varphi(A) = \begin{cases} 0 & \text{if } x_0 \notin A, \\ 1 & \text{if } x_0 \in A; \end{cases}$$

φ is called the ***Dirac measure*** concentrated at x_0.

Definition 3.1.3

Let φ be an outer measure on a set X. A set $E \subset X$ is called φ-***measurable*** if

$$\varphi(A) = \varphi(A \cap E) + \varphi(A - E)$$

for every set $A \subset X$. In view of property (iv) above, observe that φ-measurability requires only

$$\varphi(A) \geq \varphi(A \cap E) + \varphi(A - E).$$

Note.
也就是说, E 是可测的, 当且仅当任意集合 A 都可以被 E 按外测度被良好地拆解为 $A \cap E$ 和 $A \setminus E$.

Lemma 3.1.4

A set $E \subset X$ is φ-measurable if and only if

$$\varphi(P \cup Q) = \varphi(P) + \varphi(Q)$$

for all sets P and Q such that $P \subset E$ and $Q \subset E^c$.

Note.
也就是说 E 可测, 当且仅当落在 E 的内, 外的集合在外测度意义下是无关的.

Proof. For Necessity, if E is φ-measurable , then for all such P, Q,

$$\varphi(P \cup Q) = \varphi((P \cup Q) \cap E) + \varphi((P \cup Q) - E) = \varphi(P) + \varphi(Q)$$

For sufficiency , let $P = A \cap E$, $Q = A \cap E^c = A \setminus E$. Then

$$\varphi(A) = \varphi(P \cup Q) = \varphi(P) + \varphi(Q) = \varphi(A \cap E) + \varphi(A \setminus E)$$

□

Theorem 3.1.5

Suppose φ is an outer measure on an arbitrary set X. Then the following four statements hold:

(i) E is φ-measurable whenever $\varphi(E) = 0$.
(ii) $\varnothing$ and X are φ-measurable.
(iii) $E_1 - E_2$ is φ-measurable whenever E_1 and E_2 are φ-measurable.
(iv) If $\{E_i\}$ is a ***countable collection of disjoint φ-measurable sets***, then

$\bigcup_{i=1}^{\infty} E_i$ is φ-measurable and

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \varphi(E_i).$$

More generally, if $A \subset X$ is an arbitrary set, then

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c),$$

where

$$S = \bigcup_{i=1}^{\infty} E_i.$$

Proof. 1. Take any $A \subseteq X$. Since $A \cap E \subseteq E$, $\varphi(A \cap E) = 0$. Thus

$$\varphi(A \cap E) + \varphi(A \cap E^c) = \varphi(A \cap E^c) \leq \varphi(A)$$

2. Follows immediately frome lemma 3.1.4.
3. Take any P, Q such that $P \subseteq (E_1 \setminus E_2) = E_1 \cap E_2^c$, $Q \subseteq (E_1 \setminus E_2)^c = E_1^c \cup E_2$.

$$\varphi(Q) = \varphi(Q \cap E_2) + \varphi(Q - E_2)$$

Note that $Q - E_2 \subseteq E_1^c$, $Q \cap E_2 \subseteq E_2$. Since $P \subseteq E_1$, the measurability of E_1 implis that

$$\varphi(P) + \varphi(Q - E_2) = \varphi(P \cup (Q - E_2))$$

Note that $P \cup (Q - E_2) \subseteq E_2^c$. Since $Q \cap E_2 \subseteq E_2$, we have

$$\varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2) = \varphi(P \cup Q)$$

Finally , we have

$$\begin{aligned} \varphi(P) + \varphi(Q) &= \varphi(P) + \varphi(Q \cap E_2) + \varphi(Q - E_2) \\ &= \varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2) \\ &= \varphi(P \cup Q) \end{aligned}$$

4. Let $S_k = \bigcup_{i=1}^{k} E_i$. We proceed by finite induction and first note that the result is obviously true for $k = 1$. For $k > 1$ assume that S_k is φ-measurable and that

$$\varphi(A) \geq \sum_{i=1}^{k} \varphi(A \cap E_i) + \varphi\left(A \cap S_k^c\right), \quad \forall A \subseteq X$$

Then

$$\begin{aligned}
\varphi(A) &= \varphi(A \cap E_{k+1}) + \varphi\left(A \cap E_{k+1}^c\right) \\
&= \varphi(A \cap E_{k+1}) + \varphi\left(A \cap E_{k+1}^c \cap S_k^c\right) + \varphi\left(A \cap E_{k+1}^c \cap S_k\right) \\
&= \varphi(A \cap E_{k+1}) + \varphi\left(A \cap S_{k+1}^c\right) + \varphi(A \cap S_k)
\end{aligned}$$

Replace A by $A \cap S_k$, we have

$$\varphi(A \cap S_k) \geq \sum_{i=1}^{k} \varphi(A \cap S_k \cap E_i) + \varphi\left(A \cap S_k \cap S_k^c\right) = \sum_{i=1}^{k} \varphi(A \cap E_i)$$

Thus

$$\varphi(A) \geq \sum_{i=1}^{k+1} \varphi(A \cap E_{k+1}) + \varphi\left(A \cap S_{k+1}^c\right)$$

Then the subadditivity gives that

$$\varphi(A) \geq \varphi(A \cap S_{k+1}) + \varphi\left(A \cap S_{k+1}^c\right)$$

this , in turn , implies that S_{k+1} is φ-measurable. Since we now know for every set $A \subseteq X$ and for all positive integers k

$$\varphi(A) \geq \sum_{i=1}^{k} \varphi(A \cap E_i) + \varphi\left(A \cap S_k^c\right)$$

We have

$$\varphi(A) \geq \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c) \geq \varphi(A \cap S) + \varphi(A \cap S^c)$$

where $S = \bigcup_{i=1}^{\infty} E_k$. Thus S is measurable. The subadditivity gives that

$$\varphi(A) \leq \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c)$$

Finally, we have

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c)$$

Replace A by S , we can get

$$\varphi(S) = \sum_{i=1}^{\infty} \varphi(E_i)$$

$\square$

Lemma 3.1.6

Let $\{E_i\}$ be a sequence of arbitrary sets. Then there exists a sequence of disjoint sets $\{A_i\}$ such that each $A_i \subset E_i$ and

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i.$$

If each E_i is φ-measurable, and so is A_i.

Proof. Let $S_k = \bigcup_{i=1}^{k} E_i$. Let $A_k = S_k \setminus S_{k-1}, k \geq 1$, where $S_0 = \varnothing$. Then

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} (S_k \setminus S_{k-1}) = \bigcup_{i=1}^{\infty} A_i.$$

If each E_i is φ-measurable. $E_{i+1} \cup E_i = (E_{i+1} - E_i) \cup E_i$ is measurable follows from the (ii), (iii) of the above theorem . Inductively , every $S_k = (S_{k-1} \setminus E_k) \cup E_k$ is measurable. Then each A_k is measurable . $\square$

Theorem 3.1.7

If $\{E_i\}$ is a sequence of φ-***measurable*** sets in X, then $\cup_{i=1}^{\infty} E_i$ and $\cap_{i=1}^{\infty} E_i$ are φ-***measurable***.

Proof. By the lemma above, there exists a sequence of disjoint φ - measurable sets $\{A_i\}$ such that each $A_i \subseteq E_i$, and

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i$$

It follows from Theorem 3.1.5 that $\bigcup_{i=1}^{\infty} A_i$ is measurable , then $\bigcup_{i=1}^{\infty} E_i$ is as well . Since X is measurable , $E_i^c = X \setminus E_i$ is measurable , then $\bigcup_{i=1}^{\infty} E_i^c$ is measurable . Thus

$$\bigcap_{i=1}^{\infty} E_i = X \setminus \bigcup_{i=1}^{\infty} E_i^c$$

is measurable . □

Definition 3.1.8

A nonempty collection Σ of sets E satisfying the following two conditions is called a σ-***algebra***:

(i) if $E \in \Sigma$, then $E^c \in \Sigma$;

(ii) $\cup_{i=1}^{\infty} E_i \in \Sigma$ if each E_i is in Σ.

Definition 3.1.9

In a topological space, the elements of the smallest σ-algebra that contains all open sets are called ***Borel sets***. The term "smallest" is taken in the sense of inclusion.

Proof. We need to show that the "smallest" exists. □

Corollary 3.1.10

If φ is an ***outer measure*** on an arbitrary set X, then the class of φ-measurable sets forms a σ-***algebra***.

Proof. Follows immediately from Theorem 3.1.5 and Theorem 3.1.7. □

Corollary 3.1.11

Suppose φ is an outer measure on X and $\{E_i\}$ a countable collection of φ-measurable sets.

(i) If $E_1 \subset E_2$ with $\varphi(E_1) < \infty$, then

$$\varphi(E_2 \setminus E_1) = \varphi(E_2) - \varphi(E_1).$$

(ii) (Countable additivity) If $\{E_i\}$ is a disjoint sequence of sets, then

$$\varphi\Big(\bigcup_{i=1}^{\infty} E_i\Big) = \sum_{i=1}^{\infty} \varphi(E_i).$$

(iii) (Continuity from the left) If $\{E_i\}$ is an increasing sequence of sets, that is, if $E_i \subset E_{i+1}$ for each i, then

$$\varphi\Big(\bigcup_{i=1}^{\infty} E_i\Big) = \varphi\big(\lim_{i\to\infty} E_i\big) = \lim_{i\to\infty} \varphi(E_i).$$

(iv) (Continuity from the right) If $\{E_i\}$ is a decreasing sequence of sets, that is, if $E_i \supset E_{i+1}$ for each i, and if $\varphi(E_{i_0}) < \infty$ for some i_0, then

$$\varphi\Big(\bigcap_{i=1}^{\infty} E_i\Big) = \varphi\big(\lim_{i\to\infty} E_i\big) = \lim_{i\to\infty} \varphi(E_i).$$

(v) If $\{E_i\}$ is any sequence of φ-measurable sets, then

$$\varphi\big(\liminf_{i\to\infty} E_i\big) \le \liminf_{i\to\infty} \varphi(E_i).$$

(vi) If

$$\varphi\Big(\bigcup_{i=i_0}^{\infty} E_i\Big) < \infty$$

for some positive integer i_0, then

$$\varphi\big(\limsup_{i\to\infty} E_i\big) \ge \limsup_{i\to\infty} \varphi(E_i).$$

Proof. (v) Let $A_j = \bigcap_{k=j}^{\infty} E_k$, then

$$\liminf_{i\to\infty} E_i = \lim_{j\to\infty} A_j = \bigcup_{j=1}^{\infty} A_j$$

It follows from (iv) that

$$\varphi\left(\liminf_{i\to\infty} E_i\right) = \lim_{j\to\infty} \varphi\left(A_j\right)$$

Since $A_j \subseteq E_j$, we have $\varphi\left(A_j\right) \leq \varphi\left(E_j\right)$, then

$$\lim_{j\to\infty} \varphi\left(A_j\right) \leq \liminf_{i\to\infty} \varphi\left(E_j\right)$$

Thus

$$\varphi\left(\liminf_{i\to\infty} E_i\right) \leq \liminf_{i\to\infty} \varphi\left(E_i\right)$$

□

Definition 3.1.12

- An outer measure φ on a topological space X is called a ***Borel outer measure*** if all Borel sets are φ-measurable.
- A Borel outer measure is ***finite*** if $\varphi(X)$ is finite.
- An outer measure φ on a set X is called ***regular*** if for each $A \subseteq X$ there exists a φ-measurable set $B \supseteq A$ such that $\varphi(B) = \varphi(A)$.
- A ***Borel rugular outer measure*** is a Borel outer measure such that for each $A \subseteq X$, there exists a Borel set B such that $\varphi(B) = \varphi(A)$.
- A ***Radon outer measure*** is a Borel regular outer measure that is finite on compact sets.

Exercises for Outer Measure

Exercise 3.1.13

If X is a metric space, fix $\varepsilon > 0$. Let $\varphi(A)$ be the smallest number of balls of radius ε that cover A.

Let $\varphi_{\varepsilon}(A) := \varphi(A)$ denote the dependence on ε, and define

$$\psi(A) := \lim_{\varepsilon \to 0} \varphi_{\varepsilon}(A)$$

What is $\psi(A)$ and what are the correspoding ψ-measurable sets.

Proof. If there are only finite points in A. Let

$$l := \min\{d(x,y) : x, y \in A, x \neq y\}$$

Then for any $x, y \in A$, $x \neq y$. If B_1, B_2 are ball of radius $\frac{l}{2}$ such that $x \in B_1, y \in B_2$. Then $B_1 \cap B_2 = \varnothing$. But each point in A must covered by one of such ball, which shows that $\psi(A) \geq |A|$. Since it is obvious that $\psi(A) \leq |A|$, then $\psi(A) = |A|$.

Now suppose that A is a infinite set. If $\psi(A) < \infty$, say $\psi(A) = n$. Then for any sufficiently small ε, there are balls $B_1, \cdots, B_n$ with radius ε cover A. Take any $n+1$ different points $p_1, \cdots, p_{n+1}$ of A. Let $\varepsilon = \frac{1}{2}\min\left\{d\left(p_i, p_j\right) : i \neq j\right\}$, then $B_1, \cdots, B_n$ cannot cover these $n+1$ points, thus cannot cover A, which is a contradiction. Thus we have $\psi(A) = \infty$.

We find that ψ is just the counting measure on X.Take any $E \subseteq X$,and take any $A \subseteq X$, if A is finite, then $A \cap E, A - E$ are as well, and $|A| = |A \cap E| + |A - E|$, that is $\psi(A) = \psi(A \cap E) + \psi(A - E)$. If A is infinite, then one of the $A \cap E$ and $A - E$ is as well. Then

$$\psi(A) = \infty = \psi(A \cap E) + \psi(A - E)$$

The above shows that E is ψ-measurable. Thus all subset of X are ψ-measurable. □

Exercise 3.1.14

It was shown that $\varphi(E_2 \setminus E_1) = \varphi(E_2) - \varphi(E_1)$ if $E_1 \subseteq E_2$ are φ-measurable with $\varphi(E_1) < \infty$. Prove that this result remains true if E_2 is not assumed to be φ-measurable.

Proof. Since E_1 is measurable,

$$\varphi(E_2) = \varphi(E_2 \cap E_1) + \varphi(E_2 - E_1)$$

but

$$\varphi(E_2 \cap E_1) = \varphi(E_1)$$

which completes the proof. □

3.2 Carathéodory Outer Measure

Definition 3.2.1 ▸ Carathéodory outer measure

An outer measure φ defined on a metric space (X, ρ) is called a ***Carathéodory outer measure*** if

$$\varphi(A \cup B) = \varphi(A) + \varphi(B)$$

whenever A, B are arbitrary subsets of X with $d(A, B) > 0$. Where

$$d(A, B) := \inf\{\rho(a, b) : a \in A, b \in B\}$$

Lemma 3.2.2

Let φ is a ***Carathéodory outer measure*** on a metric space (X, ρ). $G \neq X$ is a open set, $E \subseteq G$, $\varphi(E) < \infty$. Let

$$E_k = \left\{x \in E : \rho(x, G^c) \geq \frac{1}{k}\right\}, \quad k = 1, 2, \cdots$$

Then

$$\lim_{k \to \infty} \varphi(E_k) = \varphi(E) = \varphi\left(\lim_{k \to \infty} E_k\right)$$

Proof. We have

$$E_1 \subseteq E_2 \subseteq \cdots \subseteq E$$

$\varphi(E_k) \leq \varphi(E)$ for each k. Define

$$T_i := \left\{x \in E : \frac{1}{i+1} < \rho(x, G^c) \leq \frac{1}{i}\right\}$$

Then for each $|i - j| \geq 2$, $\rho\left(T_i, T_j\right) > 0$. We can prove by induction that for each $n \in \mathbb{N}$,

$$\sum_{k=1}^{n} \varphi(T_{2k}) = \varphi\left(\bigcup_{k=1}^{n} T_{2k}\right) \leq \varphi(E)$$

Take $n \to \infty$, we have

$$\sum_{k=1}^{\infty} \varphi(T_{2k}) < \infty$$

Similarly,

$$\sum_{k=1}^{\infty} \varphi(T_{2k-1}) < \infty$$

Thus

$$\sum_{k=1}^{\infty} \varphi(T_k) < \infty \implies \lim_{j\to\infty} \sum_{k=j}^{\infty} \varphi(T_k) = 0$$

Then it follows that

$$\lim_{j\to\infty} \varphi\left(E \setminus E_j\right) \leq \lim_{j\to\infty} \sum_{k=j}^{\infty} \varphi(T_k) = 0$$

Since

$$\varphi(E) \leq \varphi(E \setminus E_k) + \varphi(E_k)$$

By taking $k \to \infty$, we have

$$\varphi(E) = \lim_{k\to\infty} \varphi(E_k)$$

□

Theorem 3.2.3

If φ is a Carathéodory outer measure on a metric space X, then all closed sets are φ-measurable.

Proof sketch.
用一列集合近似代替 C 是标准的想法. 证明的核心要素有: 1.C 是闭集, 使得按距离向外逼近的行为是良好的. 2. 距离这个属性与 Carathódory 条件是相适应的. 此外, 将命题转化为极限

$$\lim_{i\to\infty} \varphi(A \setminus C_i) = \varphi(A \setminus C)$$

是标准的流程. 要证明该极限成立, 则需要一定的技巧.
Carathéodory 条件允许我们拆开正距离集. 问题在于分解一个集合出来的东西

一定是“粘着的”，办法就是当我们对常数不敏感时，把集合拆开的部分每隔一块拿出来成一组. 这两个集合分别拿出来不同的一组，就拼成了原本的集合. 我们让每一组中的成员分别两两具有正距离就可以了.
这里就是把集合拆成“主要部分”和“尾巴项”，利用这种方法说明尾巴项对应的级数是收敛的，从而证明了尾巴项是可以甩掉的.

Proof. Let $C \subseteq X$ be any closed sets. Take $A \subseteq X$, only need to show that

$$\varphi(A) \geq \varphi(A \cap C) + \varphi(A \setminus C)$$

We assume that $\varphi(A) < \infty$. Define

$$C_i := \left\{x : d(x, C) \leq \frac{1}{i}\right\}$$

Since $d(A \setminus C_i, A \cap C) > 0$,

$$\varphi(A) \geq \varphi((A \setminus C_i) \cup (A \cap C)) = \varphi(A \setminus C_i) + \varphi(A \cap C)$$

Only need to show that

$$\lim_{i \to \infty} \varphi(A \setminus C_i) = \varphi(A \setminus C)$$

Define

$$T_i := \left\{x \in A : \frac{1}{i+1} < d(x, C) \leq \frac{1}{i}\right\}$$

, then

$$A \setminus C \subseteq (A \setminus C_i) \cup \bigcup_{j=i}^{\infty} T_i$$

Thus

$$\varphi(A \setminus C) \leq \varphi(A \setminus C_i) + \sum_{j=i}^{\infty} \varphi(T_i)$$

Once we have $\sum_{j=1}^{\infty} \varphi(T_i) < \infty$, then $\lim_{i \to \infty} \sum_{j=i}^{\infty} \varphi(T_i) = 0$, the conclusion follows from it. To show this, note that $\mathrm{d}\left(T_i, T_j\right) > 0$ whenever $|i - j| \geq 2$. Then it by induction follows from Carathódory condition that

$$\sum_{j=1}^{\infty} \varphi(T_{2i}) \leq \varphi(A) < \infty$$

$$\sum_{j=1}^{\infty} \varphi\left(T_{2j-1}\right) \leq \varphi(A) < \infty$$

Thus $\sum_{j=1}^{\infty} \varphi(T_i) < \infty$, which completes the proof. □

Theorem 3.2.4

Suppose $\mathcal{F}$ is a family of subsets of a topological space X that contains all open and all closed subsets of X. Suppose also that $\mathcal{F}$ is closed under countable unions and countable intersections. Then $\mathcal{F}$ contains all Borel sets; that is, $\mathcal{B} \subset \mathcal{F}$.

Note.
可以将包含 $\mathcal{B}$ 描述成对以下三种集合的封闭性: 1. 开集 2. 补集 3. 可数并.
如果我们事先没有对补集的封闭性, 可以通过加强 1.3. 的条件, 使得当我们强制使得补集封闭后, 得到的新集族依旧满足 1.3. 具体地, 我们希望原集族对于开集的补集, 也就是闭集也封闭; 对于可数并的补集, 也就是可数交也封闭.

Proof. Let

$$\mathcal{H} = \mathcal{F} \cap \{A : A^c \in \mathcal{F}\}$$

Obseve that $\mathcal{H}$ contains all closed sets. Moreover, it is easily seen that $\mathcal{H}$ is closed under complementation and countable unions. Thus $\mathcal{H}$ is a σ-algebra that contains the open sets and therefore contains all Borel sets. □

Theorem 3.2.5

If φ is a Carathéodory outer measure on a metric space X, then the Borel sets of X are φ-measurable.

Proof. Since the collection of φ-measurable sets is itself a σ-algebra , it is closed under complements and countable union and countable intersection. Theorem 3.2.3 gives that φ-measurable sets contains closed sets, thus contains open sets as well. Then the conclusion follows from Theorem 3.2.4 . □

Exercises for this section

Exercise 3.2.6 ▸ Smallest σ-algebra containing open sets

Prove that in every topological space X, there exists a smallest σ-algebra that contains all open sets in X. That is, prove that there is a σ-algebra Σ that contains all open sets and has the property that if Σ_1 is another σ-algebra containing all open sets, then $\Sigma \subset \Sigma_1$. In particular, for $X = \mathbb{R}^n$, note that there is a smallest σ-algebra that contains all the closed sets in $\mathbb{R}^n$.

Proof. Let

$$\Sigma := \bigcap_{\mathcal{M}_\alpha \text{ is a } \sigma \text{ algebra containing } X} \mathcal{M}_\alpha$$

Then it is obvious that Σ is closed under complement, countable union, and contains all open sets. Thus Σ is a σ-algebra containing all open sets. □

Exercise 3.2.7

In a topological space X the family of Borel sets, $\mathcal{B}$, is by definition the σ-algebra generated by the closed sets. The method below is another way of describing the Borel sets using transfinite induction. You are to fill in the necessary steps:

1. For an arbitrary family $\mathcal{F}$ of sets, let

$$\mathcal{F}^* = \left\{ \bigcup_{k=1}^{\infty} E_k \ : \ \text{where either } E_i \in \mathcal{F} \text{ or } \tilde{E}_i \in \mathcal{F} \text{ for all } i \in \mathbb{N} \right\}.$$

   Let Ω denote the smallest uncountable ordinal. We will use transfinite induction to define a family $\mathcal{E}_\alpha$ for each $\alpha < \Omega$.

2. Let $\mathcal{E}_0 :=$ all closed sets $:= \mathcal{K}$. Now choose $\alpha < \Omega$ and assume that $\mathcal{E}_\beta$ has been defined for each β such that $0 \le \beta < \alpha$. Define

$$\mathcal{E}_\alpha := \left(\bigcup_{0 \le \beta < \alpha} \mathcal{E}_\beta \right)^*$$

and define

$$\mathcal{A} := \bigcup_{0\le\alpha<\Omega} \mathcal{E}_\alpha.$$

3. Show that each $\mathcal{E}_\alpha \subseteq \mathcal{B}$.
4. Show that $\mathcal{A} \subset \mathcal{B}$.
5. Now show that $\mathcal{A}$ is a σ-algebra to conclude that $\mathcal{A} = \mathcal{B}$.
   (a) Show that $\varnothing, X \in \mathcal{A}$.
   (b) Let $A \in \mathcal{A} \implies A \in \mathcal{E}_\alpha$ for some $\alpha < \Omega$. Show that this implies $\tilde{A} \in \mathcal{E}_\alpha^* \subset \mathcal{E}_\beta$ for every $\beta > \alpha$.
   (c) Conclude that $\tilde{A} \in \mathcal{A}$ and thus conclude that $\mathcal{A}$ is closed under complementation.
   (d) Now let $\{A_k\}$ be a sequence in $\mathcal{A}$. Show that $\bigcup_{k=1}^\infty A_k \in \mathcal{A}$. (Hint: Each A_k is in $\mathcal{A}_{\alpha_k}$ for some $\alpha_k < \Omega$. We know that there is $\beta < \Omega$ such that $\beta > \alpha_k$ for each $k \in \mathbb{N}$.)

Proof. 3. Since $\mathcal{E}_0 \subseteq \mathcal{B}$, only need to show that if $\mathcal{E}_\beta \subseteq \mathcal{B}$ for all $0 \le \beta < \alpha$, then $\mathcal{E}_\alpha \subseteq \mathcal{B}$. Take any $E \in \mathcal{E}_\alpha$, then there are $E_1, \cdots, E_k, \cdots$, such that $E_i \in \bigcup_{0\le\beta<\alpha} \mathcal{E}_\beta$, or $E_i^c \in \bigcup_{0\le\beta<\alpha} \mathcal{E}_\beta$, and $E = \bigcup_{k=1}^\infty E_k$. We suppose that $E_i \in \mathcal{E}_{\beta_i}$ or $E_i^c \in \mathcal{E}_{\beta_i}$. Since $\mathcal{E}_{\beta_i} \subseteq \mathcal{B}$ for all β_i, and since $\mathcal{B}$ is closed under complement, then $E_i \in \mathcal{B}$. Since $\mathcal{B}$ is closed under countable union, $E = \bigcup_{k=1}^\infty E_k \in \mathcal{B}$. Thus $\mathcal{E}_\alpha \subseteq \mathcal{B}$.

4. We have shown that $\mathcal{E}_\alpha \subseteq \mathcal{B}$ for all $\alpha < \Omega$, then it is immediate that $A \subseteq \mathcal{B}$.

5. (a) Since $\varnothing \in \mathcal{E}_0$ and $X = (\varnothing)^c \in \mathcal{E}_1 = (\mathcal{E}_0)^*$, then $\varnothing, X \in \mathcal{A}$.

(b) Write

$$\mathcal{E}_\alpha^* = \left\{ \bigcup_{k=1}^\infty E_k \ : \ \text{where either } E_i \in \mathcal{E}_\alpha \text{ or } E_i^c \in \mathcal{E}_\alpha \text{ for all } i \in \mathbb{N} \right\}$$

We take $E_1 = A^c$, then $E_1^c = A \in \mathcal{E}_\alpha$, and take $E_2 = \cdots = \varnothing$.Then $A^c = \bigcup_{k=1}^\infty E_k \in \mathcal{E}_\alpha^*$. If $\beta > \alpha$, then

$$\mathcal{E}_\alpha \subseteq \bigcup_{0\le\gamma<\beta} \mathcal{E}_\gamma$$

Let $*$ act on both side, then it follows from the definition of $*$ that

$$(\mathcal{E}_\alpha)^* \subseteq \left(\bigcup_{0\le\gamma<\beta} \mathcal{E}_\gamma\right)^* = \mathcal{E}_\beta$$

Thus $A^c \in \mathcal{E}_\alpha^* \subseteq \mathcal{E}_\beta$ for every $\beta > \alpha$.

(c) It is immediate from (b) that $A^c \in \mathcal{E}_\beta \subseteq \mathcal{A}$.

(d) Suppose that $A_k \in \mathcal{E}_{\alpha_k}$ for some $\alpha_k < \Omega$. Let

$$\gamma := \bigcup_{k\in\mathbb{N}} \alpha_k, \quad \beta := \gamma + 1$$

Then $\gamma, \beta < \Omega$, and $\alpha_1, \cdots, \alpha_k < \beta$. Then

$$\bigcup_{k=1}^{\infty} A_k \in \left(\bigcup_{k=1}^{\infty} \mathcal{E}_{\alpha_k}\right)^* \subseteq \mathcal{E}_\beta \subseteq \mathcal{A}$$

□

Exercise 3.2.8 ▶ *

An outer measure φ on a space X is called σ-finite if there exists a countable number of sets A_i with $\varphi(A_i) < \infty$ such that $X \subseteq \bigcup_{i=1}^{\infty} A_i$. Assuming that φ is a σ-finite Borel regular outer measure on a metric space X, prove that $E \subseteq X$ is φ-measurable if and only if there exists an F_σ set $F \subseteq E$ such that $\varphi(E \setminus F) = 0$.

Proof. Let $E_i := E \cap A_i$. Then

$$E = \bigcup_{i=1}^{\infty} E_i, \quad \varphi(E_i) \le \varphi(A_i) < \infty$$

Once we show that there is a closed set $F_i \subseteq E_i$, such that $\varphi(E_i \setminus F_i) = 0$ for all i, then let $F = \bigcup_{i=1}^{\infty} F_i \in F_\sigma$, it follows that

$$\varphi(E \setminus F) \le \varphi\left(\bigcup (E \setminus F_i)\right) \le \varphi\left(\bigcup (E_i \setminus F_i)\right) \le \sum_i \varphi(E_i \setminus F_i) = 0$$

□

Exercise 3.2.9

Let φ be an outer measure on a space X. Suppose $A \subset X$ is an arbitrary set with $\varphi(A) < \infty$ such that there exists a φ-measurable set $E \supset A$ with $\varphi(E) = \varphi(A)$. Prove that $\varphi(A \cap B) = \varphi(E \cap B)$ for every φ-measurable set B.

Proof.

$$\varphi(A) = \varphi(E) = \varphi(E \cap B) + \varphi(E \setminus B)$$

$$\begin{aligned}\varphi(A \cap B) \geq \varphi(A) - \varphi(A \setminus B) &= \varphi(E) - \varphi(A \setminus B) \\ &= \varphi(E \cap B) + \varphi(E \setminus B) - \varphi(A \setminus B) \\ &\geq \varphi(E \cap B)\end{aligned}$$

Since $A \cap B \subseteq E \cap B$, $\varphi(A \cap B) \leq \varphi(E \cap B)$. Finally, we have $\varphi(A \cap B) = \varphi(E \cap B)$ □

Exercise 3.2.10

In $\mathbb{R}^2$, find two disjoint closed sets A and B such that $\mathbf{d}(A, B) = 0$. Show that this is not possible if one of the sets is compact.

Proof. 1. Let A be the graph of $y = \frac{1}{x}$. Let $B = \{(x, y) : y = 0\}$. Then $\left\{\left(n, \frac{1}{n}\right) : n \in \mathbb{N}\right\} \subseteq A$, $\{(n, 0) : n \in \mathbb{N}\} \subseteq B$, then

$$\mathbf{d}(A, B) \leq \inf\left\{d\left(\left(n, \frac{1}{n}\right), (n, 0)\right) : n \in \mathbb{N}\right\} = \lim_{n\to\infty} \frac{1}{n} = 0$$

2. If A is compact, B is closed. Since $\mathbf{d}(A, B) = 0$, there are convergent sequences $\{a_n\} \subseteq A$ and $\{b_n\} \subseteq B$ such that $\lim_{n\to\infty} d(a_n, b_n) = 0$. Since A is bounded and closed, there is a cluster point $a \in A$ of $\{a_n\}$. Suppose that $\lim_{k\to\infty} a_{n_k} = a$, then $\lim_{k\to\infty} b_{n_k} = a \in B$. $A \cap B \neq \emptyset$, which is a contradiction.

□

Exercise 3.2.11

Let φ be an outer Carathéodory measure on $\mathbb{R}$ and let $f(x) := \varphi(I_x)$, where I_x is an open interval of fixed length centered at x. Prove that f is lower semicontinuous. What can you say about f if I_x is taken as a closed interval? Prove the analogous result in $\mathbb{R}^n$; that is, let $f(x) := \varphi(B(x,a))$, where $B(x,a)$ is the open ball with fixed radius a centered at x.

Exercise 3.2.12

In a metric space X, prove that $\operatorname{dist}(A,B) = d(\bar{A},\bar{B})$ for arbitrary sets $A, B \in X$.

Exercise 3.2.13

Let A be a non-Borel subset of $\mathbb{R}^n$ and define for each subset E,

$$\varphi(E) = \begin{cases} 0 & \text{if } E \subset A, \\ \infty & \text{if } E \setminus A \neq \varnothing. \end{cases}$$

Prove that φ is an outer measure that is not Borel regular.

Exercise 3.2.14

Let $\mathcal{M}$ denote the class of φ-measurable sets of an outer measure φ defined on a set X. If $\varphi(X) < \infty$, prove that the family

$$\mathcal{F} := \{A \in \mathcal{M} : \varphi(A) > 0\}$$

is at most countable.

3.3 Lebesgue Measure

Theorem 3.3.1

Suppose each edge $I_k = [a_k, b_k]$ of an n-dimensional interval I is partitioned into α_k subintervals. The products of these intervals produce a partition of I into $\beta := \alpha_1 \cdot \alpha_2 \cdots \alpha_n$ subintervals I_i and

$$v(I) = \sum_{i=1}^{\beta} v(I_i).$$

Theorem 3.3.2

For each interval I and each $\epsilon > 0$, there exists an interval J whose interior contains I and

$$v(J) < v(I) + \epsilon.$$

Definition 3.3.3 ▶ Lebesgue outer measure

The ***Lebesgue outer measure*** of an arbitrary set $E \subseteq \mathbb{R}^n$, denoted by $\lambda^*(E)$, is defined by

$$\lambda^*(E) := \inf\left\{\sum_{k=1}^{\infty} v(I_k)\right\},$$

where the infimum is taken over all countable collections of closed intervals I_k such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k.$$

Remark.

- It may be necessary at times to emphasize the dimension of the Euclidean space in which Lebesgue outer measure if defined. When clarification is needed, we will write $\lambda_n^*(E)$ in plce place of $\lambda^*(E)$.
- We have not prove that ***Lebesgue outer measure*** is actually a outer measure till now.

Theorem 3.3.4

For a closed interval $I \subseteq \mathbb{R}^n$, $\lambda^*(I) = v(I)$.

Proof. The inequality $\lambda^*(I) \leq v(I)$ holds, since I itself givs a element of $\left\{\sum_{k=1}^{\infty} v(I_k)\right\}$ in the definition of $\lambda^*(I)$.

To prove the opposite inequality, choose $\varepsilon > 0$, and a sequence of closed intervals $I_1, I_2, \cdots$ such that

$$I \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) \le \lambda^*(E) + \varepsilon$$

We can choose closed intervals J_k such that

$$v(J_k) \le v(I_k) + \frac{\varepsilon}{2^k}, \quad I_k \subseteq J_k^\circ$$

Let $\mathcal{F} := \{J_k^\circ : k \in \mathbb{N}\}$. Then $\mathcal{F}$ is an open cover of I, since I is compact, it has a lebesgue number η. There is a finitely many partition $K_1, \cdots, K_m$ of I such that each with diameter less than η. Then K_i is contained in some element of $\mathcal{F}$, we say J_{k_i} althoung two K_i may be contained in a same J_{k_i}. Let N_m be the smallest number of J_{k_i} containing K_i, then

$$v(I) = \sum_{i=1}^{m} v(K_i) \le \sum_{i=1}^{N_m} v\left(J_{k_i}\right) \le \sum_{k=1}^{\infty} v(J_k) \le \lambda^*(E) + 2\varepsilon$$

Since ε is arbitrary, $v(I) \le \lambda^*(E)$. □

Lemma 3.3.5

Let $E \subseteq \mathbb{R}^n$, $\delta > 0$. Define

$$\lambda_\delta^*(E) = \inf\left\{\sum_{k=1}^{\infty} v(I_k) : \bigcup_{k=1}^{\infty} I_k \supseteq E, \text{with the side length of } I_k \text{ less than } \delta\right\}$$

Then

$$\lambda_\delta^*(E) = \lambda^*(E)$$

Proof. It follows from definition that $\lambda^*(E) \le \lambda_\delta^*(E)$. To show the converse, we assume that $\lambda^*(E) < \infty$. For each $\varepsilon > 0$, we take a countable collection of intervals $I_1, I_2, \cdots$ such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) < \lambda^*(E) + \varepsilon$$

Since $v(I_k) < \infty$ for each k, each side length of I_k is finite. Let l_k be the largest side length of the intervals I_k.

- If $l_k < \delta$, we do nothing. We keep I_k as it is.
- If $l_k \geq \delta$, we subdivide I_k. Choose an integer $M_k > \frac{l_k}{\delta}$. We can partition I_k into M_k^n smaller, congruent, and almost disjoint intervals $\{I_{k,m}\}_{m=1}^{M_k^n}$. Each small interval $I_{k,m}$ has side length l_k/M_k. Each length of each $I_{k,m}$ is $\frac{l_k}{M_k} < \delta$.

Thus , there are intervals $I_{k,1}, I_{k,2}, \cdots, I_{k,m_k}$ with side length less than δ such that

$$I_k = \bigcup_{m=1}^{m_k} I_{k,m}, \quad v(I_k) = \sum_{m=1}^{m_k} v\left(I_{k,m}\right)$$

Then $\left\{I_{k,j}\right\}$ is a contable collection of intervals covering E. We have

$$\lambda_\delta^*(E) \leq \sum_{k=1}^{\infty} \sum_{m=1}^{m_k} v\left(I_{k,m}\right) = \sum_{k=1}^{\infty} v(I_k) < \lambda^*(E) + \varepsilon$$

We have $\lambda_\delta^*(E) \leq \lambda^*(E)$ by taking $\varepsilon \to 0$. □

Theorem 3.3.6

Lebesgue outer measure, λ^*, defined on $\mathbb{R}^n$ is a Carathéorody outer measure.

Proof. We first verify that λ^* is an outer measure. We only show the subadditivity of λ^*, the other conditions are obvious. Let $\{A_i\}$ be a countable collection of arbitrary subsets of $\mathbb{R}^n$. Denote $A := \bigcup_{i=1}^{\infty} A_i$. There is a countable family of closed intervals $\left\{I_j^{(i)}\right\}_{j\in\mathbb{N}}$ such that

$$A_i \subseteq \bigcup_{j=1}^{\infty} I_j^{(i)}, \quad \sum_{j=1}^{\infty} v\left(I_j^{(i)}\right) < \lambda^*(A_i) + \frac{\varepsilon}{2^i}$$

Collection of closed interval $\left\{I_j^{(i)}\right\}_{i,j\in\mathbb{N}}$ can be aranged to a sequence of closed invervals, such that

$$A \subseteq \bigcup_{i,j\in\mathbb{N}} I_j^{(i)}$$

Thus

$$\lambda^*(A) \le \sum_{i,j} v\left(I_j^{(i)}\right) \le \sum_{i=1}^{\infty} \lambda^*(A_i) + \varepsilon$$

Since $\varepsilon > 0$ is arbitrary, the countable subadditivity of λ^* is established.

To show that λ^* is Carathéorody outer measure. We take any $\varepsilon > 0$, and take a countable conllection of closed intervals $I_1, I_2, \cdots$ such that

$$A \cup B \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) \subseteq \lambda^*(A \cup B).$$

We can subdividing each I_k into smaller invervals if necessary, we may assume that each interval I_k which diameter less that $\mathbf{d}(A, B)$. Then $\{I_k\}$ can be seperated into two subfamiles $\{I'_k\}$ and $\{I''_k\}$. The members in the former have empty intersection with B , and that in the later have empty intersection with A. $\{I_k\} = \{I'_k\} \sqcup \{I''_k\}$. Thus

$$A \subseteq \bigcup_{k=1}^{\infty} \{I'_k\}, \quad B \subseteq \bigcup_{k=1}^{\infty} \{I''_k\}$$

We have

$$\begin{aligned}
\lambda^*(A) + \lambda^*(B) &\le \sum_{k=1}^{\infty} v\left(I'_k\right) + \sum_{k=1}^{\infty} v\left(I''_k\right) \\
&= \sum_{k=1}^{\infty} v(I_k) \\
&< \lambda^*(A \cup B) + \varepsilon
\end{aligned}$$

Since $\varepsilon > 0$ is arbitrary, we have

$$\lambda^*(A) + \lambda^*(B) \le \lambda^*(A \cup B)$$

$\square$

Corollary 3.3.7

All Borel sets in $\mathbb{R}^n$ are Lebesgue measurable. In particular, each open set and each closed set are Lebesgue measurable.

Definition 3.3.8 ▶ Lebesgue Measure

Donote by λ the set function obtained by restricting λ^* to the family of Lebesgue measurable sets. λ is called ***Lebesgue measure***.

Corollary 3.3.9

Let J be an opoen interval in $\mathbb{R}^n$. Then

$$\lambda(J) = \operatorname{vol}(J)$$

Proof. There is a sequence of closed intervals I_k such taht $J = \bigcup_{k=1}^{\infty} I_k$, then

$$\lambda(J) = \lim_{k\to\infty} \lambda(I_k) = \lim_{k\to\infty} \operatorname{vol}(I_k) = \operatorname{vol}(J)$$

□

Theorem 3.3.10

The following five conditions are equivalent for Lebesgue outer measure, λ^*, on $\mathbb{R}^n$:

1. $E \subseteq \mathbb{R}^n$ is λ^*-measurable.
2. For each $\varepsilon > 0$, there is an open set $U \supseteq E$ such that $\lambda^*(U \setminus E) < \varepsilon$.
3. There is a G_δ set $U \supseteq E$ such that $\lambda^*(U \setminus E) = 0$.
4. For each $\varepsilon > 0$, there is a closed set $F \subseteq E$ such that $\lambda^*(E \setminus F) < \varepsilon$.
5. There is an F_σ set $F \subseteq E$ such that $\lambda^*(E \setminus F) = 0$.

Proof. 1. $\implies$ 2. : We first assmu that $\lambda^*(E) < \infty$. Take closed intervals $I_1, I_2, \cdots$ such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) < \lambda^*(E) + \frac{\varepsilon}{2}$$

Let I_k' be closed intervals whose interior contains I_k ,and

$$v\left(I_k'\right) < v\left(I_k\right) + \frac{\varepsilon}{2^{k+1}}$$

Let

$$U := \bigcup_{k=1}^{\infty} \left(I_k'\right)^{\circ}$$

is a open set.Since U, E is λ-measurable,

$$\begin{aligned}\lambda^*\left(U \setminus E\right) = \lambda\left(U \setminus E\right) &= \lambda\left(U\right) - \lambda\left(E\right)\\ &= \lambda\left(U\right) - \lambda^*\left(E\right)\\ &< \sum_{k=1}^{\infty} v\left(I_k'\right) - \left(\sum_{k=1}^{\infty} v\left(I_k\right) - \frac{\varepsilon}{2}\right)\\ &\leq \varepsilon\end{aligned}$$

Now, note that $\mathbb{R}^n$ is σ-finite, we suppose that $A_1, A_2, \cdots$ are measruable sets such that

$$\mathbb{R}^n = \bigcup_{j=1}^{\infty} A_j, \quad \lambda\left(A_j\right) < \infty, \forall j \in \mathbb{Z}_{>0}$$

Then $E_j := E \cap A_j$ is a λ^*-measurable set with $\lambda\left(E_j\right) < \infty$ for each j. For each $\varepsilon > 0$, and each $j \in \mathbb{Z}_{>0}$, there is an open set $U_j \supseteq E$ such that

$$\lambda^*\left(U_j \setminus E_j\right) < \frac{\varepsilon}{2^j}$$

Let $U := \bigcup_{j=1}^{\infty} U_j$,then

$$\lambda^*\left(U \setminus E\right) \leq \sum_{j=1}^{\infty} \lambda^*\left(U_j \setminus E\right) \leq \sum_{j=1}^{\infty} \lambda^*\left(U_j \setminus E_j\right) \leq \varepsilon$$

which completes the proof.

2. $\Longrightarrow$ 3. : For each $n \in \mathbb{Z}_{>0}$, there is a open set U_n, such that

$$\lambda^*\left(U_n \setminus E\right) < \frac{1}{n}$$

Let

$$U := \bigcap_{n=1}^{\infty} U_n$$

then U is a G_δ set, such that

$$\lambda^*(U \setminus E) \leq \lambda^*(U_n \setminus E) < \frac{1}{n}, \quad \forall n.$$

That is $\lambda^*(U \setminus E) = 0$.

3. $\Longrightarrow$ 1. : Since λ^* is a Carathéorody measure, then all Borel sets is measurable. Specially, the G_δ set U is measurable. And since λ^*-zero set is measurable, then

$$E = U \setminus (U \setminus E)$$

is measurable.

2. $\Longrightarrow$ 4. : There is an open set $U \supseteq E^c$ such that $\lambda^*(U \setminus E^c) < \varepsilon$. Let $F = U^c$, then

$$E \setminus F = E \cap F^c = E \cap U = U \setminus E^c$$

Thus

$$\lambda^*(E \setminus F) = \lambda^*(U \setminus E^c) < \varepsilon$$

4. $\Longrightarrow$ 5. : For each $n \in \mathbb{Z}_{>0}$, there is a closed set $F_n \subseteq E$ such taht

$$\lambda^*(E \setminus F_n) < \frac{1}{n}$$

Let

$$F := \bigcup_{n=1}^{\infty} F_n,$$

then F is a F_σ set, such that

$$\lambda^*(E \setminus F) \leq \lambda^*(E \setminus F_n) < \frac{1}{n}, \quad \forall n.$$

That is $\lambda^*(E \setminus F) = 0$.

5. $\Longrightarrow$ 1. : Since $E \setminus F$ and F is measurable, we have

$$E = (E \setminus F) \cup F$$

is measurable. □

Exercises for Lebesgue Measure

Exercise 3.3.11

With λ_t defined by

$$\lambda_t(E) = \lambda(|t|E),$$

prove that $\lambda_t(N) = 0$ whenever $\lambda(N) = 0$.

Proof. Since $\lambda(N) = 0$, for each ε, there is a sequence of closed intervals $I_1, \cdots$ such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) < \frac{\varepsilon}{|t|^n}$$

where n is the dimension of the Euclidean space we discuss. Then

$$|t|\, E \subseteq \bigcup_{k=1}^{\infty} |t|\, I_k, \quad \sum_{k=1}^{\infty} v(|t|\, I_k) = |t|^n \sum_{k=1}^{\infty} v(I_k) < \varepsilon$$

That is

$$\lambda^*(|t|\, E) < \varepsilon, \quad \forall \varepsilon > 0$$

Thus $|t|\, E$ is measurable, and

$$\lambda(|t|\, E) = \lambda^*(|t|\, E) = 0$$

□

Exercise 3.3.12

Let $I, I_1, I_2, \ldots, I_k$ be intervals in $\mathbb{R}$ such that $I \subset \bigcup_{i=1}^{k} I_i$. Prove that

$$v(I) \le \sum_{i=1}^{k} v(I_i),$$

where $v(I)$ denotes the length of the interval I.

Proof. We may assume that $v(I) < \infty$, that is I is bounded. Since $v(I) = v\left(\bar{I}\right)$, $v(I_i) = v\left(\bar{I}_i\right)$, and $\bar{I} \subseteq \bigcup_{i=1}^{k} \bar{I}_i$, we may assume that $I, I_1, \cdots, I_k$ are closed intervals. For any $\varepsilon > 0$, there are closed intervals I_i' such that

$$I_i \subseteq \left(I_i'\right)^\circ, \quad v\left(I_i'\right) \leq v\left(I_i\right) + \frac{\varepsilon}{2^i}$$

Then $\left\{\left(I_i'\right)^\circ\right\}$ is a open cover of I. Since I is compact, it has a Lebesgue number η. We can subdivide I into subintervals $K_1, \cdots, K_m$ with diameter less that η. Then each K_j is contained in some $(I_{i_j}')^\circ$.

We can formalize this by defining a map $f : \{1, \dots, m\} \to \{1, \dots, k\}$ such that for each j, $K_j \subset (I_{f(j)}')^\circ$. Since the subintervals K_j have disjoint interiors and their union is I, we can write the total length of I and group the sum according to the covering interval index:

$$v(I) = \sum_{j=1}^{m} v(K_j) = \sum_{i=1}^{k} \left(\sum_{j:f(j)=i} v(K_j) \right).$$

For any fixed i, the intervals $\{K_j\}_{j:f(j)=i}$ are a collection of disjoint intervals all contained within $(I_i')^\circ$. Therefore, the sum of their lengths cannot exceed the length of $(I_i')^\circ$:

$$\sum_{j:f(j)=i} v(K_j) = v\left(\bigcup_{j:f(j)=i} K_j \right) \leq v((I_i')^\circ) = v(I_i').$$

Applying this inequality to our grouped sum, we obtain a single chain of inequalities:

$$v(I) = \sum_{i=1}^{k} \left(\sum_{j:f(j)=i} v(K_j) \right) \leq \sum_{i=1}^{k} v(I_i') \leq \sum_{i=1}^{k} \left(v(I_i) + \frac{\epsilon}{2^i} \right) = \sum_{i=1}^{k} v(I_i) + \epsilon \sum_{i=1}^{k} \frac{1}{2^i}.$$

Since $\sum_{i=1}^{k} \frac{1}{2^i} < 1$, we have $v(I) < \sum_{i=1}^{k} v(I_i) + \epsilon$. As $\epsilon > 0$ was arbitrary, we conclude that

$$v(I) \leq \sum_{i=1}^{k} v(I_i).$$

□

Exercise 3.3.13

Let $E \subset \mathbb{R}$, and for each real number t, let $E + t = \{x + t : x \in E\}$. Prove that $\lambda^*(E) = \lambda^*(E + t)$. From this show that if E is Lebesgue measurable, then so is $E + t$.

Proof. Once we show that

$$\lambda^*(E + t) \leq \lambda^*(E), \quad \forall t$$

. By replacing t by $-t$, E by $E + t$, we have

$$\lambda^*(E) \leq \lambda^*(E + t), \forall t$$

Then we get $\lambda^*(E) = \lambda^*(E + t)$. To show this, for each $\varepsilon > 0$, we take closed intervals $I_1, \cdots$ such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) < \lambda(E) + \varepsilon$$

Then

$$E + t \subseteq \bigcup_{k=1}^{\infty} (I_k + t), \quad \sum_{k=1}^{\infty} v(I_k + t) = \sum_{k=1}^{\infty} v(I_k) < \lambda(E) + \varepsilon.$$

Thus

$$\lambda(E + t) \leq \lambda(E) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, $\lambda(E + t) \leq \lambda(E)$.

To show that $E + t$ is Lebesgue measurable. There is a G_δ set G, such that

$$G \supseteq E, \quad \lambda^*(G \setminus E) = 0.$$

Since open sets is closed under translation and since intersection is kept, then $G_\delta + t$ is a G_δ set as well. We have

$$(G + t) \supseteq (E + t), \quad (G + t) \setminus (E + t) = (G \setminus E) + t$$

Thus

$$\lambda^*((G + t) \setminus (E + t)) = \lambda^*((G \setminus E) + t) = \lambda^*(G \setminus E) = 0$$

Then it follows from Theorem 3.3.10 that $E + t$ is measurable. $\square$

Exercise 3.3.14 ▶ *

Prove that Lebesgue measure on $\mathbb{R}^n$ is independent of the choice of coordinate system. That is, prove that Lebesgue outer measure is invariant under rigid motions in $\mathbb{R}^n$.

Remark.
或许需要积分的替换公式, 之后再来看.

Proof. Since we have shown that Lebesgue otuer measure is invariant under translation, we only need to show that is it also invariant under Orthogonal Transformation. □

Exercise 3.3.15

Let P denote an arbitrary $(n-1)$-dimensional hyperplane in $\mathbb{R}^n$. Prove that $\lambda(P) = 0$.

Proof. Since Lebesgue measure is invariant under rigid motions. We may assume that $P = \{x_n = 0\}$. For each $\varepsilon > 0$, let

$$I_k := \left[-\frac{k}{2}, \frac{k}{2}\right]^{n-1} \times \left[\frac{-\varepsilon}{2k^{n-1}} \cdot \frac{1}{2^k}, \frac{\varepsilon}{2k^{n-1}} \cdot \frac{1}{2^k}\right]$$

Then

$$P \subseteq \bigcup_{k=1}^{\infty} I_k, \quad v(I_k) = \frac{\varepsilon}{2^k}, \quad \forall k$$

Thus

$$\lambda(P) \leq \sum_{k=1}^{\infty} v(I_k) \leq \varepsilon$$

Since ε is arbitrary, $\lambda(P) = 0$. □

Exercise 3.3.16

In this problem, we want to show that every Lebesgue measurable subset of $\mathbb{R}$ must be "densely populated" in some interval. Thus, let $E \subset \mathbb{R}$ be a Lebesgue measurable set, $\lambda(E) > 0$. For each $\varepsilon > 0$, show that there exists an interval I such that

$$\frac{\lambda(E \cap I)}{\lambda(I)} > 1 - \varepsilon.$$

Note.
我们发现构造这样一个 I 是及其困难的. 转而使用反证法, 去论证“如果一个建筑的每一块砖都不合格, 那么整个建筑的质量必然有问题”.

Proof. Suppose conversely, there exists an $\varepsilon > 0$, such that for each interval I, we have

$$\frac{\lambda(E \cap I)}{\lambda(I)} \leq 1 - \varepsilon.$$

Let U be a open set such that

$$U \supseteq E, \quad \lambda(U \setminus E) < \delta.$$

Suppose

$$U = \bigcup_{k=1}^{\infty} I_k$$

where I_k are disjoint open intervals. Then

$$\lambda(E \cap U) = \sum_{k=1}^{\infty} \lambda(E \cap I_k) \leq (1 - \varepsilon) \sum_{k=1}^{\infty} \lambda(I_k) = (1 - \varepsilon) \lambda(U)$$

Since

$$\lambda(U) = \lambda(E \cap U) + \lambda(U \setminus E)$$

, we have

$$\lambda(E \cap U) + \lambda(U \setminus E) \geq \frac{1}{1 - \varepsilon} \lambda(E \cap U)$$

Thus

$$\delta \geq \left(1 - \frac{1}{1 - \varepsilon}\right) \lambda(E \cap U) = \frac{\varepsilon}{1 - \varepsilon} \lambda(E)$$

since $E \cap U = E$. Note that δ can be choosed arbitrarily and indepent with ε and $\lambda(E)$, then it leads to a contradiction, which completes the proof. $\square$

Exercise 3.3.17

Prove that every set $A \subset \mathbb{R}^n$ is contained within a G_δ-set G with the property $\lambda(G) = \lambda^*(A)$.

Proof. For any $n > 0$, there are closed intervals $I_1^{(n)}, I_2^{(n)}, \cdots$ such that

$$A \subseteq \bigcup_{k=1}^{\infty} I_k^{(n)}, \quad \sum_{k=1}^{\infty} v\left(I_k^{(n)}\right) \leq \lambda^*(A) + \frac{1}{2n}$$

For each k, there is a closed interval $\left(I_k^{(n)}\right)'$ such that

$$I_k^{(n)} \subseteq \left(\left(I_k^{(n)}\right)'\right)^\circ, \quad v\left(\left(I_k^{(n)}\right)'\right) < v\left(I_k^{(n)}\right) + \frac{1}{2^{k+1}n}$$

Let

$$G := \bigcap_{n=1}^{\infty} \bigcup_{k=1}^{\infty} \left(\left(I_k^{(n)}\right)'\right)^\circ$$

is a G_δ set. Then

$$\lambda(G) \leq \lambda\left(\bigcup_{k=1}^{\infty} \left(I_k^{(n)}\right)'\right) \leq \sum_{k=1}^{\infty} v\left(\left(I_k^{(n)}\right)'\right) < \lambda^*(A) + \frac{1}{n}, \quad \forall n$$

Thus

$$\lambda(G) \leq \lambda^*(A)$$

The other hand, since $G \supseteq A$, it is immediate that

$$\lambda(G) = \lambda^*(G) \geq \lambda^*(A).$$

Finally,

$$\lambda(G) = \lambda^*(A)$$

□

Exercise 3.3.18

Let $\{E_k\}$ be a sequence of Lebesgue measurable sets contained in a compact set $K \subset \mathbb{R}^n$. Assume for some $\varepsilon > 0$ that $\lambda(E_k) > \varepsilon$ for all k. Prove that there is some point that belongs to infinitely many E_k's.

Note.
对于由一列成员组成的族类，一个点属于无穷多个成员，当且仅当它永远无法仅被有限多个成员“杀死”，当且仅当它是“反复出现”的，当且仅当它属于集合列的极限上集.

Proof. A point belongs to infinitely many E_k, if and only if it belongs to

$$\limsup_{k\to\infty} E_k := \bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} E_k$$

Since $\lambda\left(\bigcup_{k=1}^{\infty} E_k\right) \leq \lambda(K) < \infty$. It follows from Corollary 3.1.11 that

$$\lambda\left(\lim_{k\to\infty}\sup E_k\right) \geq \limsup_{k\to\infty}\lambda(E_k) \geq \varepsilon > 0$$

Thus

$$\lim_{k\to\infty}\sup E_k \neq \varnothing$$

which completes the proof. □

Exercise 3.3.19

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a Lipschitz map. Prove that if $\lambda(E) = 0$, then $\lambda(T(E)) = 0$.

Proof. Suppose that

$$d(T(x_1), T(x_2)) \leq L \cdot d(x_1, x_2)$$

Since $\lambda(E) = 0$, for any $\varepsilon > 0$, there are closed intervals $I_1, \cdots$, such that

$$E \subseteq \bigcup_{k=1}^{\infty} I_k, \quad \sum_{k=1}^{\infty} v(I_k) < \varepsilon$$

There are finitely many Cubes $Q_{k1}, \cdots, Q_{km_k}$, such that

$$I_k \subseteq \bigcup_{i=1}^{m_k} Q_{ki}, \quad \sum_{i=1}^{m_k} v(Q_{ki}) < 2v(I_k)$$

Thus we may assume that I_k are all cubes. Denote the diameter of I_k by $\sqrt{n}a_k$, and denote the centre of I_k by c_k, then

$$d\left(T\left(x\right),T\left(c_k\right)\right)\leq L\cdot d\left(x,c_k\right)\leq\sqrt{n}L\cdot a_k$$

Thus $T\left(I_k\right)$ is contained by the cube centering $T\left(c_k\right)$ with diameter $nL\cdot a_k$, which we denote by I'_k, then

$$v\left(I'_k\right)\leq n^nL^nv\left(I_k\right)$$

. We have

$$T\left(E\right)\subseteq\bigcup_{k=1}^{\infty}I'_k,\quad\sum_{k=1}^{\infty}v\left(I'_k\right)<n^nL^n\varepsilon$$

, which concludes that $\lambda\left(T\left(E\right)\right)=0$. □

3.4 The Contor Set

Definition 3.4.1 ▶ Contor set

The Cantor set is a subset of the interval $[0,1]$. We will desccribe it by constructing its complement in $[0,1]$.

1. At the fist step, let $I_{1,1}$ denote the open interval $\left(\frac{1}{3},\frac{2}{3}\right)$. Thus $I_{1,1}$ is the open middle third of the interval $I=[0,1]$.
2. The second step involves performing the first tep on each of the two remaining intervals of $I-I_{1,1}$. That is, We produce two open intervals, $I_{2,1}$ and $I_{2,2}$, each being the open middle third of one of the two intervals that constitute $I-I_{1,1}$.
3. At the ith step we produce 2^{i-1} open intervals $I_{i,1},I_{i,2},\cdots,I_{i,2^{i-1}}$, each of length $\left(\frac{1}{3}\right)^i$. The $(i+1)$ th step consists in producing mid-ddle thirds of each of the intervals of

$$I-\bigcup_{j=1}^{i}\bigcup_{k=1}^{2^{j-1}}I_{j,k}.$$

4. With C denoting the Cantor set, we define its complement by

$$I \setminus C = \bigcup_{j=1}^{\infty} \bigcup_{k=1}^{2^{j-1}} I_{j,k}.$$

Proposition 3.4.2

Let C be the Contor set. Then the following holds:
1. C is closed, that is $C = \bar{C}$.
2. Let λ be the Lebesgue measure, then

$$\lambda(\bar{C}) = \lambda(C) = 0$$

3. C is nowhere dense.
4. The cardinality of C is

$$|C| = 2^{\aleph_0} = |R| = c$$

Proof. 1. $I \setminus C$ is open by definition. Thus C is closed.

2.
$$\lambda(I \setminus C) = \sum_{j=1}^{\infty} \sum_{k=1}^{2^{j-1}} \lambda(I_{j,k}) = \sum_{j=1}^{\infty} 2^{j-1} \left(\frac{1}{3}\right)^j = \frac{1}{2} \sum_{j=1}^{\infty} \left(\frac{2}{3}\right)^j = 1$$

Thus
$$\lambda(C) = 1 - \lambda(I \setminus C) = 1 - 1 = 0$$

3.
> *Note.*
> $[0,1]$ 上的每一个值通过修饰存在唯一的二进制表示. 同样地也存在唯一的三进制表示. Contor 集就是把每个三进制表示中, 字符含 1 的数都去掉.

Every number $x \in [0,1]$ has a ternary expansion of the form

$$x = \sum_{i=1}^{\infty} \frac{x_i}{3^i}$$

where each x_i is $0, 1$, or 2 and we write $x = 0, x_1 x_2 \cdots$ This expansion unique except when

$$x = \frac{a}{3^n}$$

(Since $\frac{1}{3^n}$ has another expresion $\sum_{i=n+1}^{\infty} \frac{2}{3^i}$)where a and n are positive integers with $0 < a < 3^n$ and where 3 does not divide a. In this case, x has the form

$$x = \sum_{i=1}^{n} \frac{x_i}{3^i}$$

where x_i is either 1 or 2. If $x_n = 2$, we will use this expression to represent x. However, if $x_n = 1$, we will use the following representation for x:

$$x = \frac{x_1}{3} + \frac{x_2}{3^2} + \cdots + \frac{x_{n-1}}{3^{n-1}} + \frac{0}{3^n} + \sum_{i=n+1}^{\infty} \frac{2}{3^i}.$$

Thus , with this convention, each number $x \in [0, 1]$ has a uniue ternary expansion.

Let $x \in I$ and conssider its ternary expansion $x = 0.x_1x_2 \cdots$, bearing in mind the convention we have adopted above. Obseve that $x \notin I_{1,1}$ if and only if $x_1 \neq 1$. Also , if $x_1 \neq 1$, then $x \notin I_{2,1} \cup I_{2,2}$ if and only if $x_2 \neq 1$. Continuing in this way, we see that $x \in C$ if and only if $x_i \neq 1$ for each positive integer i. Thus, there is a one-to-one correspondence between elements of C and all sequences $\{x_i\}$ in which each x_i is either 0 or 2. The cardinality of the latter is $2^{\aleph_0}$, which is $|\mathbb{R}| = c$.

□

Exercises for Cantor set

Exercise 3.4.3

Prove that the Cantor-type set C_α (The method of construction is the same as C,except that at the ith step, each of the intervals removed has length $\alpha 3^{-i}$) is nowhere dense, has cardinality c, and has Lebesgue measure $1 - \alpha$.

Proof.

- $\overline{C}_\alpha = C_\alpha$. Only need to show that

$$\operatorname{int}(C_\alpha) = \emptyset$$

With E_k denoting the result of the kth step constructing the C_α, we define its complements by

$$I \setminus E_k := \bigcup_{j=1}^{k} \bigcup_{m=1}^{2^{j-1}} I_{m,j}$$

Then $E_1 \supseteq E_2 \supseteq \cdots$, and $C_\alpha = \bigcap_{k=1}^{\infty} E_k$.If there is a nonempty open inverval $(a,b) \subseteq C_\alpha$. Then

$$(a,b) \subseteq E_k, \quad \forall k$$

Note that E_k is a disjoint union of closed intervals with length l_k, which satisfies

$$l_k = \frac{1}{2}\left(l_{k-1} - \alpha_3^{-k}\right) < \frac{1}{2} l_{k-1} \implies l_k < \frac{1}{2^k}$$

. There is a k_0 such that $l_{k_0} < b - a$. Since

$$(a,b) \subseteq E_{k_0}$$

is a connected set lying in E_{k_0}, E_{k_0} has a component with length more than $b-a$, which is a contradiction. Thus there is no open interval contained in C_α, that is $C_\alpha = \bar{C}_\alpha$ is no where dense.

- We denote the two closed intervals of $I \setminus I_{1,1}$ by J_1, J_2. Denote the two closed intervals of $J_1 - I_{2,1}$ by $J_{1,1}, J_{1,2}$, and denote the two that of $J_2 - I_{2,2}$ by $J_{2,1}, J_{2,2}$. Inductively, denote the two intervals of $J_{i_1,i_2,\cdots,i_{j-1}} - I_{j,2^{j-1}i_1+\cdots+2^0 i_{j-1}}$ by $J_{i_1,\cdots,i_{j-1},1}, J_{i_1,\cdots,i_{j-1},2}$. Then

$$E_k = \bigcup_{i_1,\cdots,i_k \in \{1,2\}^k} J_{i_1,\cdots,i_k}$$

Let

$$\Sigma := \{1,2\}^{\mathbb{N}}$$

Given a $(s_1, s_2, \cdots) \in \Sigma$, we can get a sequence of nested closed intervals

$$J_{s_1} \supseteq J_{s_1,s_2} \supseteq \cdots$$

From Nested Interval Theorem, there is a unique point $s \in \bigcap_{k=1}^{\infty} J_{s_1,\cdots,s_k}$. We define a f by mapping $(s_1, s_2, \cdots) \in \Sigma$ to $s \in C_\alpha$. To show that f is injective, we take

$$s=(s_1,s_2,\cdots),t=(t_1,t_2,\cdots)\in\Sigma,\quad s\neq t$$

Then there is a smallest $k\in\mathbb{Z}_{>0}$, such that $s_k\neq t_k$. We may assume that $s_k=1$, $t_k=2$. Then

$$f(s)\in J_{s_1,s_2,\cdots,s_{k-1},1},\quad f(t)\in J_{t_1,t_2,\cdots,t_{k-1},2}=J_{s_1,s_2,\cdots,s_{k-1},2}$$

Since $J_{s_1,s_2,\cdots,s_{k-1},1}\cap J_{s_1,s_2,\cdots,s_{k-1},2}=\varnothing$, $f(s)\neq f(t)$. Thus f is injective. To show that f is surjective. Take any $x\in C_\alpha$. Since

$$C_\alpha=\bigcap_{k=1}^{\infty}E_k$$

, we have

$$x\in E_k,\quad\forall k$$

And since

$$E_1=J_1\cup J_2$$

is a disjoint union. There exists a unique $i_1\in\{1,2\}$, such that

$$x\in J_{i_1}$$

Then

$$x\in E_2\cap J_{i_1}=J_{i_1,1}\cup J_{i_1,2}$$

There exists a unique $i_2\in\{1,2\}$ such that

$$x\in J_{i_1,i_2}$$

Inductively, once we show that there exists a unique $(i_1,\cdots,i_k)\in\{1,2\}^k$ such that

$$x\in J_{i_1,i_2,\cdots,i_k}$$

From

$$x\in E_{k+1}\cap J_{i_1,\dots,i_k}=J_{i_1,\cdots,i_k,1}\cup J_{i_1,\cdots,i_k,2}$$

we know that there exists a unique $(i_1,\cdots,i_{k+1})\in\{1,2\}^{k+1}$, such that

$$x\in J_{i_1,\cdots,i_{k+1}}$$

Thus we have determined a unique $(i_1, i_2, \cdots) \in \Sigma$, such that

$$x \in \bigcap_{k=1}^{\infty} J_{i_1,i_2,\cdots,i_k}$$

That is

$$f(i_1, i_2, \cdots) = x$$

Thus f is surjective. Finally, we have

$$c = 2^{\aleph_0} = |\Sigma| = |C_\alpha|$$

- $$\lambda(I \setminus C_\alpha) = \lambda\left(\bigcup_{j=1}^{\infty} \bigcup_{k=1}^{2^{j-1}} I_{k,j}\right) = \sum_{j=1}^{\infty} 2^{j-1}\alpha\left(\frac{1}{3}\right)^j = \alpha$$

  Thus

  $$\lambda(C_\alpha) = \lambda(I) - \lambda(I \setminus C_\alpha) = 1 - \alpha$$

□

Exercise 3.4.4

Construct an open set $U \subset [0, 1]$ such that U is dense in $[0, 1]$, $\lambda(U) < 1$, and $\lambda(U \cap (a, b)) > 0$ for every interval $(a, b) \subset [0, 1]$.

Proof. Define

$$U := I \setminus C_\alpha, \quad 0 < \alpha < 1$$

Where C_α is the Cantor type set with Lebesgue measure $1 - \alpha$. Then

$$\lambda(U) = \lambda(I) - \lambda(C_\alpha) = \alpha > 0$$

If U is not dense. Then there is a nonempty open subset $Y \subseteq [0, 1]$ such that

$$Y \subseteq U^c = C_\alpha$$

which is a contradicion to C_α is no where dense. Since U is dense,

$$U \cap (a, b) \neq \varnothing$$

That is $U \cap (a, b)$ is a nonempty open set, thus its Lebesgue measure is nonzero.

□

Exercise 3.4.5

Prove that the family of Borel subsets of $\mathbb{R}$ has cardinality c. From this deduce the existence of a Lebesgue measurable set that is not a Borel set.

Exercise 3.4.6

Let E be the set of numbers in $[0,1]$ whose ternary expansions have only finitely many 1's. Prove that $\lambda(E)=0$.

Proof. Let E_k be the set of numbers in $I=[0,1)$ whose ternary expansions have no $1's$ after the kth digit . Then

$$E=\bigcup_{k=1}^{\infty}E_k,\quad E_1\subseteq E_2\subseteq\cdots$$

Let I_1,I_2,I_3 be the fomer, middle, latter third of the $I=[0,1)$ respectively, specially, we say $[0,\frac{1}{3}),[\frac{1}{3},\frac{2}{3}),[\frac{2}{3},1)$. Once we have defined $I_{i_1,i_2,\cdots,i_k}$, then we define $I_{i_1,i_2,\cdots,i_k,1}$ by the first third of the interval $I_{i_1,i_2,\cdots,i_k}$ as before , the remainings are similar. We have

$$E_k=I\setminus\bigcup_{j=k}^{\infty}\bigcup_{i_1,i_2,\cdots,i_k\in\{1,2,3\}^k,i_{k+1},\cdots,i_j\in\{1,3\}^{j-k}}I_{i_1,i_2,\cdots,i_j,2}$$

Let

$$A_{k,m}:=\bigcup_{j=k}^{m}\bigcup_{i_1,i_2,\cdots,i_k\in\{1,2,3\}^k,i_{k+1},\cdots,i_j\in\{1,3\}^{j-k}}I_{i_1,i_2,\cdots,i_j,2}$$

Then

$$A_{k,k}\subseteq A_{k,k+1}\subseteq\cdots$$

Denote $\bigcup_{m=k}^{\infty}A_{k,m}$ by A_k, then $E_k=I\setminus A_k$. We have

$$\lambda\left(A_k\right)=\lim_{m\to\infty}\lambda\left(A_{k,m}\right)=\sum_{j=k}^{\infty}3^k2^{j-k}\left(\frac{1}{3}\right)^{j+1}=\frac{1}{3}\left(\frac{3}{2}\right)^k\sum_{j=k}^{\infty}\left(\frac{2}{3}\right)^j=1$$

Thus

$$\lambda(E_k) = \lambda(I) - \lambda(A_k) = 1 - 1 = 0$$

It follows from the continuity of λ from below that

$$\lambda(E) = \lim_{k\to\infty} \lambda(E_k) = 0$$

□

Proof. Another proof, use Cantor set.

> *Note.*
> - 标准 Cantor 集合描述了 [0,1] 内三进制表示不含 1 的那些数.
> - 通过对 Cantor 集 $\frac{1}{3^k}$ 倍的缩放, 可以实现所描述的三进制范围表示相对于小数点的滞后 (k 个字符).
> - 缩放后 Cantor 集的起点, 决定了新 Cantor 集开始作用的之前的三进制字符都是哪些.
> - 一切的一切, 源于 Cantor 集的自相似性, 也就是说
>
> $$C = \left(\frac{1}{3}C\right) + \left(\frac{1}{3}C + \frac{2}{3}\right)$$

E_k is defined as above. Let $P_k := \{0,1,2\}^k$. For $p = (d_1, \cdots, d_k) \in P_k$, define

$$C_p := \frac{1}{3^k}C + \sum_{j=1}^{k} \frac{d_j}{3^j}$$

Then the similar process we describe C gives that

$$C_p = \left\{x \in [0,1] : x = (0.m_1m_2\cdots)_{ternary}, m_i = d_i \text{ for } 1 \le i \le k, m_j \ne 1 \text{ for } j > k\right\}$$

We have

$$E_k = \bigcup_{p\in P_k} C_p$$

Since

$$\lambda\left(C_p\right) = \frac{1}{3^k}\lambda(C) = 0$$

We have

$$E_k \le \sum_{p\in P_k} \lambda\left(C_p\right) = 0$$

Then

$$\lambda(E) = \lim_{k\to\infty} \lambda(E_k) = 0$$

□

3.5 Existence of Nonmeasurable Sets

Lemma 3.5.1

If $S \subset \mathbb{R}$ is a Lebesgue measurable set of positive and finite measure, then the set of differences $D_S := \{x - y : x, y \in S\}$ contains an interval.

Proof sketch.
Since

$$\begin{aligned} x \in D_S &\iff S \cap (S + x) \neq \varnothing \\ &\iff \text{There exists } T \subseteq S \text{ such that } T \cap (T + x) \neq \varnothing \end{aligned}$$

Only need to find a such T, and a interval I, such that $T \cap (T + x) \neq \varnothing$ for $\forall x \in I$.
When $T \cap (T + x) = \varnothing$, we have $\lambda(T \cup (T + x)) = 2\lambda(T)$. Once we find a not too large (less than $2\lambda(T)$) set containing $T \cup (T + x)$ for all $x \in I$.
From another perspective, we need to find an T occupise at least half of a set (relative to Lebesgue measure) and dose not escape it under a tiny transformation.

Proof. For each $\varepsilon > 0$, there exists an open set $U \supseteq S$ with $\lambda(U) < (1 + \varepsilon)\lambda(S)$. We write U as a countable disjoint union of open intervals:

$$U = \bigcup_{k=1}^{\infty} I_k$$

Then

$$\lambda(S) = \lambda(S \cap U) = \sum_{k=1}^{\infty} \lambda(S \cap I_k)$$

We have

$$\sum_{k=1}^{\infty} \lambda(I_k) < (1 + \varepsilon) \sum_{k=1}^{\infty} \lambda(S \cap I_k)$$

Then there is some k_0 such that

$$\lambda(I_k) < (1+\varepsilon)\lambda\left(S \cap I_{k_0}\right)$$

We take $\varepsilon = \frac{1}{3}$, then

$$\lambda\left(S \cap I_{k_0}\right) > \frac{3}{4}\lambda\left(I_{k_0}\right)$$

For any $|t| < \frac{1}{2}\lambda\left(I_{k_0}\right)$, $S \cap I_{k_0} \cup \left(S \cap I_{k_0} + t\right)$ is contained in some open interval with Lebesgue measure $\frac{3}{2}\lambda\left(I_{k_0}\right)$. Thus

$$\lambda\left(S \cap I_{k_0} \cup \left(S \cap I_{k_0} + t\right)\right) \leq \frac{3}{2}\lambda\left(I_{k_0}\right)$$

Then it follows that it is impossible that

$$\left(S \cap I_{k_0}\right) \cap \left(S \cap I_{k_0} + t\right) = \varnothing, \quad \text{for some } |t| < \frac{1}{2}\lambda\left(I_{k_0}\right)$$

since otherwise the Lebsgue measure of the union of the two is greater than $2\lambda\left(S \cap I_{k_0}\right) > \frac{3}{2}\lambda\left(I_{k_0}\right)$, which is a contradiction. Now we have

$$\left(-\frac{1}{2}\lambda\left(I_{k_0}\right), \frac{1}{2}\lambda\left(I_{k_0}\right)\right) \subseteq D_{S \cap I_{k_0}} \subseteq D_S$$

□

Exercise 3.5.2

There exists a set $E \subseteq \mathbb{R}$ that is not Lebesgue measurable.

Proof. We define a ralation on the real line by

$$x \sim y :\iff x - y \in \mathbb{Q}$$

Then it is easy to verify that $\sim$ is a equivalent relation. Then $\mathbb{R}$ can be seprated by the class of the relation. Note that every equivalent class is countable. But $\mathbb{R}$ is uncountable, there are uncountably many equivalent class. Now we use the axiom of choice, to get a set S by selecting precisely one point from each equivalent class. Then

$$\mathbb{R} = S + \mathbb{Q} := \bigcup_{r\in\mathbb{Q}} (S + r)$$

If S is Lebesgue measurable, then $\lambda(S) > 0$ since

$$\infty = \lambda(\mathbb{R}) = \sum_{r\in\mathbb{Q}} \lambda(S + r) = \sum_{r\in\mathbb{Q}} \lambda(S)$$

Now, considering

$$D_S = \{x - y \,:\, x, y \in S\}$$

If $x \neq y$, since x, y belong to different class, $x - y \notin \mathbb{Q}$. Then we have

$$D_S \subseteq 0 \cup (\mathbb{R} \setminus \mathbb{Q})$$

which cannot contain any inverval. Then it follows from the lemma above that $\lambda(S) = \infty$. But then there exists a closed inverval I such that

$$0 < \lambda(S \cap I) < \infty$$

and $S \cap I$ is measurable, it follows fome the lemma again that there is a interval

$$J \subseteq D_{S\cap I} \subseteq D_S$$

, which is a contradiction. □

Exersices for Existence of Nonmeasurable Sets

Exercise 3.5.3

Prove that there every subset of $\mathbb{R}$ with positive outer Lebesgue measure contains a nonmeasurable subset.

Proof. Let $T \subseteq \mathbb{R}$ be any subset of $\mathbb{R}$ with positive outer Lebesgue measure. The $\sim$ defined as above is also a equivalent relation on T. Let S be the same set we just defined in the above theorem.

$$T = T \cap \mathbb{R} = T \cap \left(\bigcup_{r\in\mathbb{Q}} (S + r) \right) = \bigcup_{r\in\mathbb{Q}} (T \cap (S + r))$$

which is a disjoint union. We have

$$0 < \lambda^* (T) \le \sum_{r \in \mathbb{Q}} \lambda^* (T \cap (S + r))$$

There exists an $r_0 \in \mathbb{Q}$ such that

$$\lambda^* (T \cap (S + r_0)) > 0$$

We denote $T \cap (S + r_0)$ by S'. If S' is Lebesgue measurable. Note that $D_{S'} \subseteq (\mathbb{R} \setminus \mathbb{Q}) \cup \{0\}$. If $\lambda (S') < \infty$, then it is a contradiction to Lemma 3.5.1. If $\lambda (S') = \infty$. Then there exists a closed interval I such that

$$0 < \lambda (S' \cap I) < \infty$$

then there is an interval J such that

$$J \subseteq D_{S' \cap I} \subseteq D_{S'}$$

which is a contradicion as well. □

Exercise 3.5.4

Let $\mathcal{N}$ denote the nonmeasurable set constructed in this section. Show that if E is a measurable subset of $\mathcal{N}$, then $\lambda(E) = 0$.

Proof. We still use the notation S to represent that nonmeasurable set. If

$$E \subseteq S, \quad \lambda (E) > 0$$

If $\lambda (E) < \infty$, then from Lemma 3.5.1 there is an interval J such that

$$J \subseteq D_E \subseteq D_S$$

which is a contradiction. Again, if $\lambda (E) = \infty$, then there exists a closed interval I such that

$$0 < \lambda (E \cap I) < \infty$$

Then again from Lemma 3.5.1 there exists an interval J such that

$$J \subseteq D_{E \cap I} \subseteq D_E \subseteq D_S,$$

which is a contradicion. □

3.6 Lebesgue-Stieljes Measure

Definition 3.6.1

If f is a nondecreasing function defined on $\mathbb{R}$, then the "length" of a half-open interval $(a,b]$, denoted by $\alpha_f((a,b])$, can be defined by

$$\alpha_f((a,b]) = f(b) - f(a)$$

Definition 3.6.2 ▶ Lebesgue-Stieltjes outer measure

The Lebesgue-Stieltjes outer measure of an arbitrary set $E \subseteq \mathbb{R}$ is defined by

$$\lambda_f^*(E) := \inf\left\{\sum_{h_k \in \mathcal{F}} \alpha_f(h_k)\right\}$$

where the infimum is take over all countable collections $\mathcal{F}$ of half-open intervals h_k of the form $(a_k, b_k]$ such that

$$E \subseteq \bigcup_{h_k \in \mathcal{F}} h_k$$

Remark.
If f is right-continuous, then the (Lebesgue) length of each interval $(a_k, b_k]$ that appears above can be assumed to be arbitrarily small, beacause

$$\alpha_f((a,b]) = f(b) - f(a) = \sum_{k=1}^{N} [f(a_k) - f(a_{k-1})] = \sum_{k=1}^{N} \alpha_f((a_{k-1}, a_k])$$

whenever $a = a_0 < a_1 < \cdots < a_N = b$.

Theorem 3.6.3

If $f : \mathbb{R} \to \mathbb{R}$ is a nondecreasing function, then λ_f^* is a Carathéodory outer measure. We denote the restriction of λ_f^* on measruable sets by λ_f.

Proof. To show that λ_f^* is a outer measure, we only need to show the it is monotone and countably subadditive.

For the proof of monotonicity, let $A_1 \subseteq A_2$ be any subset of $\mathbb{R}$. Take any $\varepsilon > 0$. There is a countable collections $h_1, h_2, \cdots$ of $\mathcal{F}$, such that

$$A_2 \subseteq \bigcup_{k=1}^{\infty} h_k, \quad \sum_{k=1}^{\infty} \alpha_f(h_k) < \lambda_f^*(A_2) + \varepsilon$$

Then

$$A_1 \subseteq \bigcup_{k=1}^{\infty} h_k, \quad \sum_{k=1}^{\infty} \alpha_f(h_k) < \lambda_f^*(A_2) + \varepsilon$$

That is

$$\lambda_f^*(A_1) \leq \lambda_f^*(A_2) + \varepsilon$$

Since $\varepsilon > 0$ is arbitrary, $\lambda_f^*(A_1) \leq \lambda_f^*(A_2)$.

The proof of countable subadditivity is virtually identical to the proof the corresponding result of Lebesgue measure given in Theorem 3.3.6, since we didn't use any explicit expresion of the length of an interval there.

The same idea of proving Lebesgue outer measure to be Carathéodory is effective for that of the Lebesgue-Stieljes oute measure, since the only thing we should check to repeate that proof here is whether we can subdividing each h_k into smaller members of $\mathcal{F}$ whose diameter less than $\mathbf{d}(A, B)$. But that was shown in the remark of the definition. □

Theorem 3.6.4

If $f : \mathbb{R} \to \mathbb{R}$ is nondecreasing and right-continuous then

$$\lambda_f((a, b]) = f(b) - f(a)$$

Note.
回忆 Lebesgue 测度的情况, 我们将闭集构成的覆盖, 通过微小的几乎不影响测度的 (向两边) 扩张, 其内部形成一组开覆盖. 我们是利用这组开覆盖来覆盖身为紧集的闭区间, 以处理问题的.
这里, 我们由于选择了半开区间, 只需要对右侧进行微小的扩张, 作为代价, 我们研究的集合本身不再是紧集. 这里假设 f 是右连续, 就同时解决了两个问题:

1. 向右微小扩张, 得益于右连续, 对测度的影响也变成是无限小的.
2. 得益于右连续, 用闭集 $[a', b]$, 向左逼近 $(a, b]$, $a' \to a^+$ 的行为也是良好的.

Proof. It suffices to show that

$$\lambda_f((a,b]) \geq f(b) - f(a)$$

For each $\varepsilon > 0$, we take a sequence of half-invervals $h_1 = (a_1, b_1], h_2 = (a_2, b_2], \cdots$ such that

$$(a,b] \subseteq \bigcup_{k=1}^{\infty} h_k, \quad \sum_{k=1}^{\infty} \alpha_f(h_k) < \lambda_f((a,b]) + \varepsilon.$$

Since f is right-continuous , we have

$$\lim_{t \to 0^+} \alpha_f((a_k, b_k + t]) = \alpha_f((a_k, b_k])$$

Consequently, we may replace each $(a_k, b_k]$ with $(a_k, b_k']$ such that

$$b_k < b_k', \quad \alpha_f((a_k, b_k']) < \alpha_f((a_k, b_k]) + \frac{\varepsilon}{2^k}$$

Let $a' \in (a,b)$. The compact set $[a', b]$ is covered by a finite family of open intervals, which we denote $\mathcal{C}_{fin} = \{C_j\}_{j=1}^N$, where $C_j = (a_j, b_j')$. $[a', b] \subseteq \bigcup_{j=1}^N C_j$. By the Lebesgue number lemma, there exists an $\eta > 0$ such that any subset of $[a', b]$ with diameter less than η is contained in at least one C_j. We partition $[a', b]$ into subintervals $I_k = [t_{k-1}, t_k]$ for $k = 1, \dots, m$, each with length less than η. Thus, for each I_k, we can choose an interval from $\mathcal{C}_{fin}$ that contains it.

Let's formalize this choice. We define an assignment function $\phi : \{1, \dots, m\} \to \{1, \dots, N\}$ such that for each k, $I_k \subseteq C_{\phi(k)}$.

We can now write the total increase of f over $[a', b]$ and group the terms according to our assignment:

$$f(b) - f(a') = \sum_{k=1}^{m} (f(t_k) - f(t_{k-1})) = \sum_{j=1}^{N} \left(\sum_{k \text{ s.t. } \phi(k)=j} (f(t_k) - f(t_{k-1})) \right)$$

The outer sum is over the indices of the open cover $\mathcal{C}_{fin}$, ensuring no open interval is counted more than once.

Now, let's analyze the inner sum for a fixed j. Let $\mathcal{P}_j = \{I_k \mid \phi(k) = j\}$ be the set of subintervals assigned to C_j. These are a finite number of disjoint subintervals, all contained within $C_j = (a_j, b'_j)$. Let $t_{\min,j}$ be the minimum of all left endpoints of intervals in $\mathcal{P}_j$, and $t_{\max,j}$ be the maximum of all right endpoints. Then $\bigcup_{I_k \in \mathcal{P}_j} I_k \subseteq [t_{\min,j}, t_{\max,j}]$. Since f is nondecreasing, the sum of increases over disjoint intervals is less than or equal to the increase over their convex hull:

$$\sum_{I_k \in \mathcal{P}_j} (f(t_k) - f(t_{k-1})) \leq f(t_{\max,j}) - f(t_{\min,j})$$

Furthermore, since all these I_k are in $C_j = (a_j, b'_j)$, we have $a_j < t_{\min,j}$ and $t_{\max,j} < b'_j$. The nondecreasing property of f implies:

$$f(t_{\max,j}) - f(t_{\min,j}) \leq f(b'_j) - f(a_j)$$

Combining these, for each j, we have established:

$$\sum_{k \text{ s.t. } \phi(k)=j} (f(t_k) - f(t_{k-1})) \leq f(b'_j) - f(a_j)$$

Substituting this back into our grouped sum:

$$f(b) - f(a') \leq \sum_{j=1}^{N} (f(b'_j) - f(a_j))$$

This sum is over a finite subset of the original infinite cover used to define the 2ε bound. Therefore:

$$\sum_{j=1}^{N} (f(b'_j) - f(a_j)) \leq \sum_{i=1}^{\infty} (f(b'_i) - f(a_i)) < \lambda_f((a, b]) + 2\varepsilon$$

So we have $f(b) - f(a') < \lambda_f((a, b]) + 2\varepsilon$.

Since ε is arbitrary, we have $f(b) - f(a') \leq \lambda_f((a, b])$. Furthermore, the right continuity of f implies $\lim_{a' \to a^+} f(a') = f(a)$, and hence

$$f(b) - f(a) \leq \lambda_f((a, b])$$

as desired. $\square$

Theorem 3.6.5

Suppose μ is a finite Borel outer measure on $\mathbb{R}$ and let

$$f(x) = \mu((-\infty, x]).$$

Then the Lebesgue-Stieltjes measure, λ_f, agrees with μ on all Borel sets.

Proof. * f is nondecreasing from definition. Note that

$$(-\infty, x] = \bigcap_{n\in\mathbb{N}} (-\infty, x + \frac{1}{n}]$$

Since μ is finite, the continuity from below gives that

$$\lim_{n\to\infty} \mu\left((-\infty, x + \frac{1}{n}]\right) = \mu\left((-\infty, x]\right)$$

That is

$$\lim_{n\to\infty} f\left(x + \frac{1}{n}\right) = f\left(x\right)$$

Since f is nondecreasing ,

$$\lim_{t\to 0^+} f\left(x + t\right) = f\left(x\right)$$

Since

$$(-\infty, a] \cup (a, b] = (-\infty, b]$$

then

$$f\left(b\right) - f\left(a\right) = \mu((-\infty, b]) - \mu((-\infty, a]) = \mu((a, b])$$

thus proving that μ and λ_f agree on all half-open intervals. Since very open set in $\mathbb{R}$ is a countable union of disjoint half-open intervals, it follows that μ and λ_f agree on all open sets. It will be shown later that μ, λ_f agree on all Borel sets . □

Exercises for Lebesgue-Stieltjes Measure

Exercise 3.6.6

Let $f : \mathbb{R} \to \mathbb{R}$ be a nondecreasing function and let λ_f be the Lebesgue-Stieltjes measure generated by f. Prove that $\lambda_f(\{x_0\}) = 0$ if and only if f is left-continuous at x_0.

Proof.

$$\{x_0\} = \bigcap_{n\in\mathbb{N}} (x_0, x_0 + \frac{1}{n}]$$

It follows from the continuity from above that

$$\lim_{n\to\infty} \left(f\left(x_0 + \frac{1}{n}\right) - f(x_0) \right) = \lambda_f(\{x_0\})$$

Since f is nondecreasing,

$$\lim_{t\to 0^+} f(x_0 + t) = \lim_{n\to\infty} f\left(x_0 + \frac{1}{n}\right)$$

We have

$$\lim_{t\to 0^+} f(x_0 + t) - f(x_0) = \lambda_f(\{x_0\})$$

The LHS$= 0$ iff f is left-continuous, which completes the proof. □

Exercise 3.6.7

Let f be a nondecreasing function defined on $\mathbb{R}$. Define a Lebesgue-Stieltjes-type measure as follows: For $A \subset \mathbb{R}$ an arbitrary set,

$$\Lambda_f^*(A) = \inf\left\{ \sum_{h_k\in\mathcal{F}} [f(b_k) - f(a_k)] \right\},$$

where the infimum is taken over all countable collections $\mathcal{F}$ of closed intervals of the form $h_k := [a_k, b_k]$ such that

$$E \subset \bigcup_{h_k\in\mathcal{F}} h_k.$$

In other words, the definition of $\Lambda_f^*(A)$ is the same as $\lambda_f^*(A)$ except that closed intervals $[a_k, b_k]$ are used instead of half-open intervals $(a_k, b_k]$. As in the case of Lebesgue-Stieltjes measure it can be easily seen that Λ_f^* is a Carathéodory measure. (You need not prove this.)

(a) Prove that $\Lambda_f^*(A) \le \lambda_f^*(A)$ for all sets $A \subset \mathbb{R}^n$.

(b) Prove that $\Lambda_f^*(B) = \lambda_f^*(B)$ for all Borel sets B if f is left-continuous.

Proof. 1. Let $(a_1, b_1], (a_2, b_2], \cdots$ be a sequence of half-open intervals , such that

$$A \subseteq \bigcup_{k=1}^{\infty}(a_k, b_k], \quad \sum_{k=1}^{\infty}(f(b_k) - f(a_k)) < \lambda_f^*(A) + \varepsilon$$

Then we have

$$A \subseteq \bigcup_{k=1}^{\infty}[a_k, b_k], \quad \sum_{k=1}^{\infty}(f(b_k) - f(a_k)) < \lambda_f^*(A) + \varepsilon$$

Thus

$$\Lambda_f^*(A) \le \lambda_f^*(A) + \varepsilon$$

Since $\varepsilon > 0$ is arbitrary, we have

$$\Lambda_f^*(A) \le \lambda_f^*(A)$$

2. Let $[a_1, b_1], [a_2, b_2], \cdots$ be a sequence of closed intervals , such that

$$B \subseteq \bigcup_{k=1}^{\infty}[a_k, b_k], \quad \sum_{k=1}^{\infty}(f(b_k) - f(a_k)) < \Lambda_f^*(B) + \varepsilon$$

If f is left-continuous, for each $k \in \mathbb{N}$, there is a'_k, such that

$$f(a_k) - f\left(a'_k\right) < \frac{\varepsilon}{2^k}$$

Then

$$B \subseteq \bigcup_{k=1}^{\infty}(a'_k, b_k], \quad \sum_{k=1}^{\infty}\left(f(b_k) - f\left(a'_k\right)\right) < \Lambda_f^*(B) + 2\varepsilon$$

That is

$$\lambda_f^*(B) \le \Lambda_f^*(B) + 2\varepsilon$$

Since $\varepsilon > 0$ is arbitrary, we have

$$\lambda_f^*(B) \le \Lambda_f^*(B)$$

□

3.7 Hausdorff Measure

Definition 3.7.1 ▶ Hausdorff Outer Measure

Hausdorff measure is defined in terms of an auxiliary set function that we introduce first. Let $0 \le s < \infty$, $0 < \varepsilon \le \infty$, and let $A \subset \mathbb{R}^n$. Define

$$H_\varepsilon^s(A) = \inf\left\{\sum_{i=1}^{\infty} \alpha(s) 2^{-s} (\operatorname{diam} E_i)^s \, : \, A \subset \bigcup_{i=1}^{\infty} E_i, \operatorname{diam} E_i < \varepsilon\right\},$$

where $\alpha(s)$ is a normalization constant defined by

$$\alpha(s) = \frac{\pi^{\frac{s}{2}}}{\Gamma(\frac{s}{2}+1)},$$

with

$$\Gamma(t) = \int_0^\infty e^{-x} x^{t-1} \, dx, \quad 0 < t < \infty.$$

It follows from the definition that if $\varepsilon_1 < \varepsilon_2$, then $H_{\varepsilon_1}^s(E) \ge H_{\varepsilon_2}^s(E)$. This allows the following, which is the definition of ***s-dimensional Hausdorff measure***:

$$H^s(A) = \lim_{\varepsilon \to 0} H_\varepsilon^s(A) = \sup_{\varepsilon > 0} H_\varepsilon^s(A).$$

Remark.
- Hausdorff Measure can be defined in any metric space.
- The sets E_i in the definition of $H_\varepsilon^s(A)$ are arbitrary subsets of $\mathbb{R}^n$. However, they could be taken to be closed sets, since $\operatorname{diam} E_i =$

diam $\overline{E_i}$.

Theorem 3.7.2

For each nonnegative number s, H^s is a Carathéodory outer measure.

Proof. First we show that H^s is an outer measure. We only show that H^s is subadditive, the other properties are obvious. Let $A_1, A_2, \cdots$ be a sequence of sets with H^s- outer measure finite. Let $A = \bigcup_{k=1}^{\infty} A_k$. For any $\varepsilon > 0$, and for each k, there are sets $E_1^{(k)}, \cdots$ such that

$$A_k \subseteq \bigcup_{i=1}^{\infty} E_i^{(k)}, \quad \sum_{i=1}^{\infty} \alpha(s) 2^{-s} \left(\operatorname{diam} E_i^{(k)}\right)^s < H_\varepsilon^s(A_k) + \frac{\varepsilon}{2^k}$$

We can arange $\left\{E_i^{(k)}\right\}_{i,k \in \mathbb{Z}_{>0}}$ to be a sequence of sets such that

$$A \subseteq \bigcup_{i,k \in \mathbb{Z}_{>0}} E_i^{(k)}, \quad \sum_{i,k \in \mathbb{Z}_{>0}} \alpha(s) 2^{-s} \left(\operatorname{diam} E_i^{(k)}\right)^s < \sum_{k=1}^{\infty} H_\varepsilon^s(A_k) + \varepsilon$$

Thus

$$H_\varepsilon^s(A) \leq \sum_{k=1}^{\infty} H_\varepsilon^s(A_k) + \varepsilon$$

Since $H_\varepsilon^s(A_k) \leq H^s(A_k)$ for each k, we have

$$H_\varepsilon^s(A) \leq \sum_{k=1}^{\infty} H^s(A_k) + \varepsilon$$

Let $\varepsilon > 0$, we have

$$H^s(A) \leq \sum_{k=1}^{\infty} H^s(A_k)$$

Now we show that H^s is a Carathéodory outer measure. Suppose that A, B be two sets with $\mathbf{d}(A, B) > 0$, and suppose $0 < \varepsilon < \mathbf{d}(A, B)$. Let $\{E_k\}$ be a covering of $A \cup B$ with $\operatorname{diam}(E_i) < \varepsilon$ .Then each E_k cannot intersect with both A and B. Let $\mathcal{A}$ be the collection of E_k intersecting with A , and Let $\mathcal{B}$

be that of intersecting with B. Then

$$\sum_{k=1}^{\infty} \alpha(s) 2^{-s} (\operatorname{diam} E_i)^s = \sum_{E_k \in \mathcal{A}} \alpha(s) 2^{-s} (\operatorname{diam} E_k)^s + \sum_{E_k \in \mathcal{B}} \alpha(s) 2^{-s} (\operatorname{diam} E_k)^s$$
$$\geq H_\varepsilon^s(A) + H_\varepsilon^s(B)$$

We can take $\{E_i\}$ such that

$$\sum_{k=1}^{\infty} \alpha(s) 2^{-s} (\operatorname{diam} E_i)^s < H_\varepsilon^s(A \cup B) + \varepsilon$$

Thus we have

$$H_\varepsilon^s(A \cup B) \geq H_\varepsilon^s(A) + H_\varepsilon^s(B) + \varepsilon$$

Since $0 < \varepsilon < \mathbf{d}(A, B)$ is arbitrary, take $\varepsilon \to 0^+$, it follows that

$$H^s(A \cup B) \geq H^s(A) + H^s(B)$$

, which shows that H^s is a Carathéodory outer measure. □

Theorem 3.7.3

For each $A \subset \mathbb{R}^n$, there exists a Borel set $B \supset A$ such that

$$H^s(B) = H^s(A).$$

Proof. Let $\{\varepsilon_j\}$ be a decreasing suquence of positive numbers such that $\varepsilon_j \to 0$ as $j \to \infty$. For each j, there is a suquence of closed set (Remark of Definition 3.7.1) $E_{i,j}$, such that

$$B \subseteq \bigcup_{i=1}^{\infty} E_{i,j}, \quad \sum_{i=1}^{\infty} \alpha(s) 2^{-s} (\operatorname{diam}(E_{i,j}))^s < H_{\varepsilon_j}^s(B) + \varepsilon_j$$

Let $A_j := \bigcup_{i=1}^{\infty} E_{i,j}$, and let $A := \bigcap_{j=1}^{\infty} A_j$. Then A is a Borel set such that $B \subseteq A$. Note that

$$H_{\varepsilon_j}^s(B) + \varepsilon_j > \sum_{i=1}^{\infty} \alpha(s) 2^{-s} (\operatorname{diam}(E_{i,j}))^s \geq H_{\varepsilon_j}^s(A_j) \geq H_{\varepsilon_j}^s(A)$$

Let $j \to \infty$, we have

$$H^s(B) \geq H^s(A) \implies H^s(B) = H^s(A)$$

$\square$

Theorem 3.7.4

Suppose $A \subset \mathbb{R}^n$ and $0 \leq s < t < \infty$. Then
(i) If $H^s(A) < \infty$ then $H^t(A) = 0$.
(ii) If $H^t(A) > 0$ then $H^s(A) = \infty$.

Proof. (ii) is just a restatement of (i). We now show that (i) holds. Take any $\varepsilon > 0$, there are $\{E_i\}$ with $\operatorname{diam}(E_i) \leq \varepsilon$ such that

$$A \subseteq \bigcup_{i=1}^{\infty} E_i, \quad \sum_{i=1}^{\infty} \alpha(s) 2^{-s} (\operatorname{diam}(E_i))^s < H_\varepsilon^s(A) + \varepsilon \leq H^s(A) + \varepsilon$$

Then

$$\begin{aligned} H_\varepsilon^t(A) &\leq \sum_{i=1}^{\infty} \alpha(t) 2^{-t} (\operatorname{diam}(E_i))^t \\ &= \frac{\alpha(t)}{\alpha(s)} 2^{s-t} \sum_{i=1}^{\infty} \alpha(s) 2^{-s} (\operatorname{diam}(E_i))^s (\operatorname{diam}(E_i))^{t-s} \\ &\leq \frac{\alpha(s)}{\alpha(t)} 2^{s-t} \varepsilon^{t-s} (H^s(A) + \varepsilon) \end{aligned}$$

Let $\varepsilon \to 0$, we have

$$H^t(A) \leq 0 \implies H^t(A) = 0$$

$\square$

Exercises for Hausdorff Measure

Exercise 3.7.5

If $A \subset \mathbb{R}$ is an arbitrary set, show that $H^1(A) = \lambda^*(A)$.

Remark.
In this problem, you will see the importance of the constant that appears in the definition of Hausdorff measure. The constant $\alpha(s)$ that appears in the definition of H^s-measure is equal to 2 when $s = 1$. That is, $\alpha(1) = 2$, and therefore, the definition of $H^1(A)$ can be written as

$$H^1(A) = \lim_{\varepsilon\to 0} H^1_\varepsilon(A),$$

where

$$H^1_\varepsilon(A) = \inf\left\{\sum_{i=1}^{\infty} \operatorname{diam} E_i \,:\, A \subset \bigcup_{i=1}^{\infty} E_i \subset \mathbb{R}, \operatorname{diam} E_i < \varepsilon\right\}.$$

This result is also true in $\mathbb{R}^n$ but is more difficult to prove. The isodiametric inequality $\lambda^*(A) \le \alpha(n)\left(\frac{\operatorname{diam} A}{2}\right)^n$ (whose proof is omitted in this book) can be used to prove that $H^n(A) = \lambda^*(A)$ for all $A \subset \mathbb{R}^n$.

Proof. For any $\varepsilon > 0$, there are closed intervals $I_1, \cdots$ with $v(I_i) < \varepsilon$ such that

$$A \subseteq \bigcup_{i=1}^{\infty} I_i, \quad \sum_{i=1}^{\infty} v(I_i) < \lambda^*(A) + \varepsilon$$

Then

$$\sum_{i=1}^{\infty} \alpha(1)\, 2^{-1} (\operatorname{diam} I_i)^1 = \sum_{i=1}^{\infty} v(I_i) < \lambda^*(A) + \varepsilon$$

Thus

$$H^1_\varepsilon(A) < \lambda^*(A) + \varepsilon$$

Take $\varepsilon \to 0$, we have

$$H^1(A) \le \lambda^*(A)$$

The other hand, take $\varepsilon > 0$, there are closed set E_i with $\operatorname{diam} E_i < \varepsilon$, such that

$$A \subseteq \bigcup_{i=1}^{\infty} E_i, \quad \sum_{i=1}^{\infty} \operatorname{diam} E_i < H^1_\varepsilon(A) + \varepsilon$$

Note that

$$\operatorname{diam} E_i = \sup E_i - \inf E_i$$

Let

$$I_i := \left[\inf E_i - \frac{\varepsilon}{2^{i+1}}, \sup E_i + \frac{\varepsilon}{2^{i+1}}\right]$$

Then

$$E_i \subseteq I_i, \quad v(I_i) < \operatorname{diam} E_i + \frac{\varepsilon}{2^i}$$

Thus

$$A \subseteq \bigcup_{i=1}^{\infty} I_i, \quad \sum_{i=1}^{\infty} v(I_i) < H_\varepsilon^1(A) + 2\varepsilon < H^1(A) + 2\varepsilon$$

Since $\varepsilon > 0$ is arbitrary, we have

$$\lambda^*(A) \leq H^1(A)$$

□

Exercise 3.7.6

For $A \subset \mathbb{R}^n$, use the isodiametric inequality introduced in Exercise 4.1 to show that

$$\lambda^*(A) \leq H^n(A) \leq \alpha(n)\left(\frac{\sqrt{n}}{2}\right)^n \lambda^*(A).$$

Proof. For each $\varepsilon > 0$, there are sets $E_1, E_2, \cdots$ with $\operatorname{diam} E_i < \varepsilon$ such that

$$A \subseteq \bigcup_{i=1}^{\infty} E_i, \quad \sum_{i=1}^{\infty} \alpha(n) 2^{-n} (\operatorname{diam}(E_i))^n < H_\varepsilon^n(A) + \varepsilon$$

Then

$$\lambda^*(A) \leq \sum_{i=1}^{\infty} \lambda^*(E_i) \leq \sum_{i=1}^{\infty} \alpha(n) 2^{-n} (\operatorname{diam}(E_i))^n < H_\varepsilon^n(A) + \varepsilon$$

Take $\varepsilon \to 0$, then

$$\lambda^*(A) \leq H^n(A)$$

For each $\varepsilon > 0$, there are closed intervals $I_1, I_2, \cdots$ such that

$$A \subseteq \bigcup_{i=1}^{\infty} I_i, \quad \sum_{i=1}^{\infty} v(I_i) < \lambda^*(A) + \varepsilon$$

Since for each I_i, there are finitely may squares $J_{i,j}$, such that

$$I_i \subseteq \bigcup_{j=1}^{\infty} J_{i,j}, \quad \sum_{j=1}^{\infty} v\left(J_{i,j}\right) < v\left(I_i\right) + \frac{\varepsilon}{2^i}$$

We can arange $\left\{J_{i,j}\right\}_{i,j\in\mathbb{Z}_{>0}}$ into a sequence of squares $\{K_m\}$ such that

$$A \subseteq \bigcup_{m=1}^{\infty} K_m, \quad \sum_{i=1}^{\infty} v\left(K_m\right) < \lambda^*\left(A\right) + 2\varepsilon$$

Note that

$$\left(\frac{1}{\sqrt{n}}\operatorname{diam}\left(K_m\right)\right)^n = v\left(K_m\right)$$

We have

$$\begin{aligned} H_\varepsilon^n\left(A\right) \leq \sum_{i=1}^{\infty} \alpha\left(n\right) 2^{-n}\left(\operatorname{diam}\left(K_m\right)\right)^n &= \alpha\left(n\right)\left(\frac{\sqrt{n}}{2}\right)^n \sum_{i=1}^{\infty} v\left(K_m\right) \\ &\leq \alpha\left(n\right)\left(\frac{\sqrt{n}}{2}\right)^n \left(\lambda^*\left(A\right) + 2\varepsilon\right) \end{aligned}$$

Take $\varepsilon \to 0$, we have

$$H^n\left(A\right) \leq \alpha\left(n\right)\left(\frac{\sqrt{n}}{2}\right)^n \lambda^*\left(A\right)$$

□

Exercise 3.7.7

Show that $H^s \equiv 0$ on $\mathbb{R}^n$ for all $s > n$.

Proof. Let

$$I_k := [-k, k]^n$$

For any $A \subseteq \mathbb{R}^n$, we have

$$A \subseteq \bigcup_{k=1}^{\infty} I_k$$

Isodiametric inequality gives that

$$H^n(I_k) \leq \alpha(n)\left(\frac{\sqrt{n}}{2}\right)^n \lambda^*(I_k) < \infty$$

Then Theorem 3.7.4 gives that

$$H^s(I_k) = 0$$

Thus we have

$$H^s(A) \leq \sum_{k=1}^{\infty} H^s(I_k) = 0$$

□

Exercise 3.7.8

Let $C \subset \mathbb{R}^2$ denote the circle of radius 1 and regard C as a topological space with the topology induced from $\mathbb{R}^2$. Define an outer measure $\mathcal{H}$ on C by

$$\mathcal{H}(A) := \frac{1}{2\pi} H^1(A) \quad \text{for any set } A \subset C.$$

Later in the book, we will prove that $H^1(C) = 2\pi$. Thus, you may assume that. Show that

(i) $\mathcal{H}(C) = 1$.
(ii) Prove that $\mathcal{H}$ is a Borel regular outer measure.
(iii) Prove that $\mathcal{H}$ is rotationally invariant; that is, prove that for every set $A \subset C$, one has $\mathcal{H}(A) = \mathcal{H}(A')$, where A' is obtained by rotating A through an arbitrary angle.
(iv) Prove that $\mathcal{H}$ is the only outer measure defined on C that satisfies the previous three conditions.

3.8 Hausdorff Dimension of Cantor Sets

Definition 3.8.1 ▸ General Cantor set

Let $0 < \gamma < 1/2$ and set $I_{0,1} = [0,1]$. Let $I_{1,1}$ and $I_{1,2}$ denote the intervals $[0,\gamma]$ and $[1-\gamma,1]$ respectively. They result by deleting the open middle interval of length $1-2\gamma$. At the next stage, delete the open middle interval of length $\gamma(1-2\gamma)$ of each of the intervals $I_{1,1}$ and $I_{1,2}$. There remain 2^2 closed intervals each of length γ^2. Continuing this process, at the kth stage there are 2^k closed intervals each of length γ^k. Denote these intervals by $I_{k,1}, \ldots, I_{k,2^k}$. We define the generalized Cantor set as

$$C(\gamma) = \bigcap_{k=0}^{\infty} \bigcup_{j=1}^{2^k} I_{k,j}.$$

Theorem 3.8.2 ▸ Hausdorff Dimension of Cantor Sets

The Hausdorff Diemension of $C(\gamma)$, denoted by s, is

$$s = \log 2/\log\left(\frac{1}{\gamma}\right)$$

that is the unique s such that

$$2\gamma^s = 1$$

Note.
通过选取特定的 s, 使得 $C(\gamma)$ 构造过程中的每一个“丢弃”中间区间的过程都是恰好完全不损失“s-长度的”. 这样, $C(\gamma)$ 中 $I_{k,j}$ 的 s-长度可以完全它内部更小的组件的 s-长度和来代替.
由于 $C(\gamma)$ 的自相似性, 每个区间 I, 在盖住了最大的一个小组件 I_{k,k_1} 时, 也会同时连同被跟它一样大的另外三个组件 $I_{k,k_2}, I_{k,k_3}, I_{k,k_4}$ 所含住. 这样就提供了上下两边包夹的不等式关系.

Proof. Since $C(\gamma) \subseteq \bigcup_{j=1}^{2^k} I_{k,j}$ for each k. We have

$$H^s_{\gamma^k}(C(\gamma)) \leq \sum_{j=1}^{2^k} \ell\left(I_{k,j}\right)^s = 2^k\left(\gamma^k\right)^s = \left(2\gamma^s\right)^k$$

We choose s such that $2\gamma^s = 1$, then

$$H^s_{\gamma^k}(C(\gamma)) \leq 1, \quad \forall k$$

Since $\lim_{k\to\infty}\gamma^k = 0$, we have

$$H^s(C(\gamma)) = \lim_{k\to\infty} H^s_{\gamma^k}(C(\gamma)) \leq 1$$

It is important to observe that we choose our s such that

$$\sum_{j=1}^{2^k} \ell\left(I_{k,j}\right)^s = 1$$

Next, we show that $H^s(C(\gamma)) \geq \frac{1}{4}$, which along with above implies that the Hausdorff dimension of $C(\gamma)$ equals $\log 2/\log\left(\frac{1}{\gamma}\right)$. We will establish this by showing that if

$$C(\gamma) \subseteq \bigcup_{j=1}^{\infty} J_i$$

is an open covering $C(\gamma)$ by intervals J_i, then

$$\sum_{i=1}^{\infty} \ell(J_i)^s \geq \frac{1}{4}$$

Firstly, we will show that for any open interval I and fixed l that

$$\sum_{I_{i,l}\subseteq I} \ell\left(I_{l,i}\right)^s \leq 4\ell(I)^s$$

Let k be the smallest number such that I contains some $I_{k,j}$. Then there are at most for of $I_{k,j}$'s intersect with I, otherwise there are some $I_{k-1,j}$ contained in I. We call them I_{k,k_m}, $m = 1,2,3,4$. Then

$$4\ell(I)^s \geq \sum_{m=1}^{4} \ell\left(I_{k,k_m}\right)^s = \sum_{m=1}^{4} \sum_{I_{l,i}\subseteq I_{k,k_m}} \ell\left(I_{l,i}\right)^s$$

Since

$$\ell\left(I_{k,k_m}\right)^s = \frac{1}{2^k}, \qquad \sum_{I_{l,i}\subseteq I_{k,k_m}} \ell\left(I_{l,i}\right)^s = 2^{l-k}\frac{1}{2^l} = \frac{1}{2^k}$$

Thus

$$4\ell(I)^s \geq \sum_{I_{l,i}\subseteq I} \ell\left(I_{l,i}\right)^s$$

From this, we can get

$$\sum_{i=1}^{\infty} \ell(J_i)^s \geq \frac{1}{4}\sum_{i=1}^{\infty}\sum_{I_{l,k}\subseteq J_i} \ell\left(I_{l,k}\right)^s \geq \frac{1}{4}\sum_{k=1}^{2^l} \ell\left(I_{l,k}\right)^s = \frac{1}{4}$$

since the covering has an Lebesgue number , thus for sufficiently large l, each $I_{l,k}$ is contained in some J_i. □

3.9 Measures on Abstract Spaces

Definition 3.9.1

Let X be a set and $\mathcal{M}$ a σ-algebra of subsets of X. A ***measure*** on $\mathcal{M}$ is a function $\mu : \mathcal{M} \to [0,\infty]$ satisfying the properties

(i) $\mu(\varnothing) = 0$;

(ii) if $\{E_i\}$ is a sequence of disjoint sets in $\mathcal{M}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i).$$

Thus, a measure is a countably additive set function defined on $\mathcal{M}$.

- Sometimes the notion of ***finite additivity*** is useful. It states that

(ii)′ If $E_1, \cdots, E_k$ is any finite family of disjoint sets in $\mathcal{M}$, then

$$\mu\left(\bigcup_{i=1}^{k} E_i\right) = \sum_{i=1}^{k} \mu\left(E_i\right).$$

- If μ satisfies (i) and (ii′) but not necessarily (ii), then μ is called a ***finitely additive measure***.
- Then triple $(M, \mathcal{M}, \mu)$ is called a ***measure space***, and sets should be referred to as $\mathcal{M}$-measurable, to indicate their dependence on $\mathcal{M}$.
- If M consitutes the family of Borel sets in a meatric space X, then μ is called a ***Borel measure***.
- A measure μ is said to be ***finite*** if $\mu(X) < \infty$, and σ-***finite*** if X can

be wriiten as $X = \bigcup_{i=1}^{\infty} E_i$, where $\mu(E_i) < \infty$ for each i.

- A measure μ with the property that all subsets of sets of μ-measure zero are measurable is said to be ***complete***, and $(X, \mathcal{M}, \mu)$ is called a ***complete measure space***.
- A Borel measure on a topological space X that is finite on compact sets is called a ***Radon measure***.

Example 3.9.2 ▶ EXAMPLES

Here are some examples of measures.

1. $(\mathbb{R}^n, \mathcal{M}, \lambda)$, where λ is Lebesgue measure and $\mathcal{M}$ is the family of Lebesgue measurable sets.
2. $(X, \mathcal{M}, \varphi)$, where φ is an outer measure on an abstract set X and $\mathcal{M}$ is the family of φ-measurable sets.
3. $(X, \mathcal{M}, \delta_{x_0})$, where X is an arbitrary set and δ_{x_0} is an outer measure defined by
$$\delta_{x_0}(E) = \begin{cases} 1 & \text{if } x_0 \in E, \\ 0 & \text{if } x_0 \notin E. \end{cases}$$
The point $x_0 \in X$ is selected arbitrarily. It can easily be shown that all subsets of X are δ_{x_0}-measurable, and therefore $\mathcal{M}$ is taken as the family of all subsets of X.
4. $(\mathbb{R}, \mathcal{M}, \mu)$, where $\mathcal{M}$ is the family of all Lebesgue measurable sets, and $x_0 \in \mathbb{R}$ and μ is defined by
$$\mu(E) = \lambda(E \setminus \{x_0\}) + \delta_{x_0}(E)$$
whenever $E \in \mathcal{M}$.
5. $(X, \mathcal{M}, \mu)$, where $\mathcal{M}$ is the family of all subsets of an arbitrary space X and where $\mu(E)$ is defined as the number (possibly infinite) of points in E.

Theorem 3.9.3

Let $(X, \mathcal{M}, \mu)$ be a measure space and suppose $\{E_i\}$ is a sequence of sets in $\mathcal{M}$.

1. (Monotonicity) If $E_1 \subset E_2$, then $\mu(E_1) \leq \mu(E_2)$.
2. (Subtractivity) If $E_1 \subset E_2$ and $\mu(E_1) < \infty$, then $\mu(E_2 - E_1) = \mu(E_2) - \mu(E_1)$.

3. (Countable subadditivity)

$$\mu(\bigcup_{i=1}^{\infty} E_i) \leq \sum_{i=1}^{\infty} \mu(E_i).$$

4. (Continuity from the left) If $\{E_i\}$ is an increasing sequence of sets, that is, if $E_i \subset E_{i+1}$ for each i, then

$$\mu(\bigcup_{i=1}^{\infty} E_i) = \mu(\lim_{i\to\infty} E_i) = \lim_{i\to\infty} \mu(E_i).$$

5. (Continuity from the right) If $\{E_i\}$ is a decreasing sequence of sets, that is, if $E_i \supset E_{i+1}$ for each i, and if $\mu(E_{i_0}) < \infty$ for some i_0, then

$$\mu(\bigcap_{i=1}^{\infty} E_i) = \mu(\lim_{i\to\infty} E_i) = \lim_{i\to\infty} \mu(E_i).$$

6. $$\mu(\liminf_{i\to\infty} E_i) \leq \liminf_{i\to\infty} \mu(E_i).$$

7. If

$$\mu(\bigcup_{i=i_0}^{\infty} E_i) < \infty$$

for some positive integer i_0, then

$$\mu(\limsup_{i\to\infty} E_i) \geq \limsup_{i\to\infty} \mu(E_i).$$

Proof. Only 1. and 3. have not been established in Corollary 3.1.11. For 1. , note that $E_2 \setminus E_1 \in \mathcal{M}$ and

$$\mu(E_2) = \mu(E_2 \setminus E_1) + \mu(E_1) \geq \mu(E_1)$$

To show 3., we start by defining

$$A_i := E_i \setminus \left(\bigcup_{k=1}^{i-1} E_k\right), \quad i = 2, 3, \cdots, \quad A_1 := E_1$$

Then $A_1, A_2, \cdots$ are disjoint sets in $\mathcal{M}$. And

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i, \quad A_i \subseteq E_i$$

Thus

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i) \leq \sum_{i=1}^{\infty} \mu(E_i)$$

□

Theorem 3.9.4

Suppose $(X, \mathcal{M}, \mu)$ is a measure space. Define $\overline{\mathcal{M}} = \{A \cup N : A \in \mathcal{M}, N \subset B$ for some $B \in \mathcal{M}$ such that $\mu(B) = 0\}$ and define $\bar{\mu}$ on $\overline{\mathcal{M}}$ by $\bar{\mu}(A \cup N) = \mu(A)$. Then $\overline{\mathcal{M}}$ is a σ-algebra, $\bar{\mu}$ is a complete measure on $\overline{\mathcal{M}}$, and $(X, \overline{\mathcal{M}}, \bar{\mu})$ is a complete measure space. Moreover, $\bar{\mu}$ is the only complete measure on $\overline{\mathcal{M}}$ that is an extension of μ.

Proof. It is easy to proof that $\overline{\mathcal{M}}$ is closed under uncountable unions since this is true for measure zero sets. To show that $\overline{\mathcal{M}}$ is closed under complementation. We take $A \in \mathcal{M}, N \subseteq B$ such that $\mu(B) = 0$. We may assume that $A \cap N = \varnothing$ since $A \cup N = A \cup (N \setminus A)$, where $N \setminus A \subseteq B$. Thus

$$A \cup N = (A \cup B) \cap ((B^c \cup N) \cup (A \cup B))$$

Then

$$(A \cup N)^c = (A \cup B)^c \cup \left((B \cap N^c) \cap (A \cup B)^c\right)$$

Where $(A \cup B)^c \in \mathcal{M}$, and

$$(B \cap N^c) \cap (A \cup B)^c \subseteq B$$

To show that $\bar{\mu}$ is not unambiguous, we assume that $A_1 \cup N_1 = A_2 \cup N_2$, where $A_1, A_2 \in \mathcal{M}$, $N_1 \subseteq B_1, N_2 \subseteq B_2, \mu(B_1) = \mu(B_2) = 0$. Since $A_1 \subseteq A_2 \cup N_2$, we have

$$\mu(A_1 \cup N_1) = \mu(A_1) \leq \mu(A_2 \cup N_2)$$

Similarly, $\mu(A_2 \cup N_2) \subseteq \mu(A_1 \cup N_1)$. It is easily verified that $\bar{\mu}$ is complete since

$$\bar{\mu}(N) = \bar{\mu}(\varnothing \cup N) = \mu(\varnothing) = 0$$

Suppose ν is another complete measure on $\overline{\mathcal{M}}$ that is an extension of μ . Let $A \cup N \in \overline{\mathcal{M}}$, $N \subseteq B$, $\mu(B) = 0$. Then

$$\nu(A \cup N) \geq \nu(A) = \mu(A) = \bar{\mu}(A \cup N)$$

The other hand,

$$\nu(A \cup N) = \nu(A \cup (N \setminus A)) = \nu(A) + \nu(N \setminus A)$$

since $N \setminus A \subseteq B$ is ν - measurable for completeness. Then

$$\begin{aligned} \nu(A \cup N) = \nu(A) + \nu(N \setminus A) = \mu(A) + \nu(N \setminus A) &\leq \mu(A) + \nu(B) \\ &\leq \mu(A) + \mu(B) \\ &= \mu(A) \\ &= \bar{\mu}(A \cup N) \end{aligned}$$

We have

$$\nu(A \cup N) = \mu(A \cup N)$$

for all $A \cup N \in \overline{\mathcal{M}}$ □

Exercises for Abstract Measure Space

Exercise 3.9.5

Let $\{\mu_k\}$ be a sequence of measures on a measure space such that $\mu_{k+1}(E) \geq \mu_k(E)$ for each measurable set E. With μ defined as $\mu(E) = \lim_{k\to\infty} \mu_k(E)$, prove that μ is a measure.

Proof. It is obvious that $\mu(\varnothing) = 0$ and μ is nonnegative . If $\{E_i\}$ is a sequence of measurable sets. Then

$$
\begin{aligned}
\mu\left(\bigcup_{i=1}^{\infty} E_i\right) &= \lim_{k\to\infty} \mu_k\left(\bigcup_{i=1}^{\infty} E_i\right) \\
&= \lim_{k\to\infty} \sum_{i=1}^{\infty} \mu_k(E_i) \\
&= \sum_{i=1}^{\infty} \mu(E_i)
\end{aligned}
$$

□

3.10 Regular Outer Measure

Theorem 3.10.1

If φ is a ***regular outer measure*** on X, then

(i) If $A_1 \subset A_2 \subset \dots$ is an increasing sequence of arbitrary sets, then

$$\varphi\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{i\to\infty} \varphi(A_i).$$

(ii) If $A \cup B$ is φ-measurable, $\varphi(A) < \infty$, $\varphi(B) < \infty$, and $\varphi(A \cup B) = \varphi(A) + \varphi(B)$, then both A and B are φ-measurable.

Note.

1. 自下连续对于正则的外测度也成立.
2. 大可测集可以在外测度意义下被完全分为两个小集合,则这两个小集合也可测.

Proof. (i) For each $k \in \mathbb{Z}_{>0}$, thers exists a φ-measurable set C_k, such that $C_k \supseteq A_k$ and $\varphi(C_k) = \varphi(A_k)$. Then we define

$$B_i := \bigcap_{j=i}^{\infty} C_j.$$

$\{B_k\}$ forms a increasing sequence of φ-measurable sets such that

$$A_k \subseteq B_k \subseteq C_k$$

It follows that

$$\varphi\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \varphi\left(\bigcup_{i=1}^{\infty} B_i\right) = \lim_{i\to\infty} \varphi(B_i) = \lim_{i\to\infty} \varphi(A_i)$$

Since $\varphi(A_k) \leq \varphi(B_k) \leq \varphi(C_k) = \varphi(A_k)$ for each k.
(ii) There exists φ-measurable set $C' \supseteq A$ such that $\varphi(C') = \varphi(A)$. Let

$$C := C' \cap (A \cup B)$$

then C is φ-measurable. Note that

$$A \subseteq C \subseteq C'$$

We have $\varphi(C) = \varphi(A)$, then

$$\begin{aligned}
\varphi(C) + \varphi(B) &= \varphi(A) + \varphi(B) \\
&= \varphi(A \cup B) \\
&= \varphi((A \cup B) \cap C) + \varphi((A \cup B) \setminus C) \\
&= \varphi(C) + \varphi(B \setminus C) \\
&= \varphi(C) + \varphi(B) - \varphi(B \cap C)
\end{aligned}$$

where we use the measurability of C on the third and fifth equality. The above implies that $\varphi(B \cap C) = 0$, $B \cap C$ is φ-measurable. Since $C \setminus A \subseteq (A \cup B) \setminus A \subseteq B$, we have $C \setminus A \subseteq B \cap C$. Then it follows that $C \setminus A$ is φ-measruable, thus $A = C \setminus (C \setminus A)$ is as well. The symmetry gives that B is φ-measruable . □

Theorem 3.10.2

Suppose φ is an ***Borel outer measure*** on a metric space X. Then for each Borel set $B \subseteq X$ with $\varphi(B) < \infty$ and each $\varepsilon > 0$, there exists a closed set $F \subseteq B$ such that

$$\varphi(B \setminus F) < \varepsilon$$

Furthermore, suppose

$$B \subseteq \bigcup_{i=1}^{\infty} V_i$$

where each V_i is an open set with $\varphi(V_i) < \infty$. Then for each $\varepsilon > 0$, there is an open set $W \supseteq B$ such that

$$\varphi(W \setminus B) < \varepsilon$$

Note.
Method of part I : 题目对于 B 的条件给予了“额外”(脱离 σ-代数体系) 的增强条件 $\varphi(B) < \infty$. 通过定义新的测度 $\mu(A) := \varphi(A \cap B)$. 将增强条件“腾挪”到变成测度的增强条件 $\mu(X) < \infty$. 以增加命题的整体性, 使得证明不拘泥于某个特定集合的 $\varphi(B) < \infty$.
Trick of part II: 我们通过挖去闭集的方法“雕刻”向外逼近的开集. “雕刻”的原材料就是题目给出的 φ-有限开集 V_i. 由于要求得到 $W \supseteq B$, 我们必须在没有 B 的集合上“雕刻”. 自然想到在每个 V_i 中没有 B 的地方, 也就是 $V_i \setminus B$ 上, 利用 Part I 挖去一个很大的闭集 C_i.

Proof. Select a Borel set B with $\varphi(B) < \infty$, and define

$$\mu(A) := \varphi(A \cap B)$$

It is easy to verify that μ is an outer measure on X, and also all Borel set is μ-measurable. Furthermore, $\mu(X) < \infty$. Once we show that the first part is true for μ , then for each $\varepsilon > 0$

$$\varphi(B \setminus F) = \mu(B \setminus F) < \varepsilon$$

is true for some closed set F and the specific Borel set B we have just selected. Since B is arbitray, we may assume that $\varphi(X) < \infty$ in the following proccess. Let

$$\mathcal{D} := \{A : \forall \varepsilon > 0, \exists \text{ closed set } F \subseteq A \text{ s.t. } \varphi(A \setminus F) < \varepsilon\}$$

Only need to show that $\mathcal{D}$ is a σ-algebra containing all closed sets. That is to say from Theorem 3.2.4 that $\mathcal{D}$ contains all closed and open sets and is closed under countable union and countable intersection. Take any open set $U \subseteq X$. Let

$$F_i := \left\{x : d(x, U^c) \geq \frac{1}{i}\right\}$$

be a closed set for each i. Then

$$F_1 \subseteq F_2 \subseteq \cdots, \quad U = \bigcup_{i=1}^{\infty} F_i$$

Thus

$$\lim_{k\to\infty} \varphi\left(U \setminus \bigcup_{i=1}^{k} F_i\right) = \varphi\left(U \setminus \bigcup_{i=1}^{\infty} F_i\right) = \varphi(\emptyset) = 0$$

For each $\varepsilon > 0$, there exists k such that

$$\varphi\left(U \setminus \bigcup_{i=1}^{k} F_i\right) < \varepsilon$$

, where $\bigcup_{i=1}^{k} F_i$ is a closed set.

Now, we take a sequence of membser $A_1, A_2, \cdots$ in $\mathcal{D}$. For each i, there is a closed set $C_i \subseteq A_i$, such that

$$\varphi(A_i \setminus C_i) < \frac{\varepsilon}{2^{i+1}}$$

Note that

$$\bigcup_{i=1}^{\infty} A_i \setminus \bigcup_{i=1}^{\infty} C_i \subseteq \bigcup_{i=1}^{\infty} (A_i \setminus C_i), \quad \bigcap_{i=1}^{\infty} A_i \setminus \bigcap_{i=1}^{\infty} C_i \subseteq \bigcup_{i=1}^{\infty} (A_i \setminus C_i)$$

We have

$$\varphi\left(\bigcap_{i=1}^{\infty} A_i \setminus \bigcap_{i=1}^{\infty} C_i\right) \leq \sum_{i=1}^{\infty} \varphi(A_i \setminus C_i) < \varepsilon$$

$$\lim_{k\to\infty} \varphi\left(\bigcup_{i=1}^{\infty} A_i \setminus \bigcup_{i=1}^{k} C_i\right) = \varphi\left(\bigcup_{i=1}^{\infty} A_i \setminus \bigcup_{i=1}^{\infty} C_i\right) \leq \sum_{i=1}^{\infty} \varphi(A_i \setminus C_i) < \frac{\varepsilon}{2}$$

Then there exists k such that

$$\varphi\left(\bigcup_{i=1}^{\infty} A_i \setminus \bigcup_{i=1}^{k} C_i\right) < \varepsilon$$

Thus $\bigcap_{i=1}^{\infty} A_i, \bigcup_{i=1}^{\infty} A_i \in \mathcal{D}$.

It follows from the first part that for each i, there exists a closed set $C_i \subseteq V_i \backslash B$, such that

$$\varphi((V_i \setminus C_i) \setminus B) = \varphi((V_i \setminus B) \setminus C_i) < \frac{\varepsilon}{2^i}$$

Let

$$W := \bigcup_{i=1}^{\infty} (V_i \setminus C_i)$$

, which is an open set. We have

$$\varphi(W \setminus B) = \varphi\left(\bigcup_{i=1}^{\infty} (V_i \setminus C_i) \setminus B\right) \leq \sum_{i=1}^{\infty} \varphi((V_i \setminus C_i) \setminus B) < \varepsilon$$

□

Corollary 3.10.3

In the previous theorem, if φ is assumed to be a ***Borel regular outer measure***, then the conclusions remain valid if the phrase "for each Borel set B" is replaced by "for each φ-measurable set B."

Theorem 3.10.4

Let φ be a ***Carathéodory outer measure***. For each set $A \subset X$, define

$$\psi(A) = \inf\{\varphi(B) : B \supset A, B \text{ a Borel set}\}.$$

Then ψ is a ***Borel regular outer measure*** on X that agrees with φ on all ***Borel sets***.

Proof. We first show that ψ is an outer measure on X. The monotonicity and $\psi(\varnothing) = 0$ is obvious. We only show that ψ is countably subadditive. Let $A_1, A_2, \cdots$ be a sequence of subsets of X. Take any $\varepsilon > 0$. By the definition of ψ, for each A_i, there is a Borel set $B_i \supseteq A_i$ such that

$$\psi(A_i) > \varphi(B_i) - \frac{\varepsilon}{2^i}$$

Then

$$\sum_{i=1}^{\infty} \psi(A_i) \geq \sum_{i=1}^{\infty} \varphi(B_i) - \varepsilon \geq \varphi\left(\bigcup_{i=1}^{\infty} B_i\right) - \varepsilon$$

Since $\bigcup_{i=1}^{\infty} B_i$ is a Borel set contaning $\bigcup_{i=1}^{\infty} A_i$, then

$$\varphi\left(\bigcup_{i=1}^{\infty} B_i\right) \geq \psi\left(\bigcup_{i=1}^{\infty} A_i\right)$$

from the definition of $\psi\left(\bigcup_{i=1}^{\infty} A_i\right)$. Thus

$$\sum_{i=1}^{\infty} \psi(A_i) \geq \psi\left(\bigcup_{i=1}^{\infty} A_i\right) - \varepsilon$$

Since ε is arbitrary, the countably subadditive is true for ψ.

To show that all Borel sets are ψ-measurable, we suppose $D \subseteq X$ is a Borel set. Only need to show that

$$\psi(A) \geq \psi(A \cap D) + \psi(A \setminus D)$$

is true for all sets $A \subseteq X$ with $\psi(A) < \infty$. For each $\varepsilon > 0$, there exsist a Borel set $B \supseteq A$ such that

$$\psi(A) > \varphi(B) - \varepsilon$$

Then

$$\begin{aligned} \psi(A) &> \varphi(B) - \varepsilon \\ &\geq \varphi(B \cap D) + \varphi(B \setminus D) - \varepsilon \\ &\geq \psi(A \cap D) + \psi(A \setminus D) - \varepsilon \end{aligned}$$

D is ψ-measurable since $\varepsilon > 0$ is arbitrary. To show that ψ agrees with φ on all Borel sets. Take any Borel set $B \subseteq X$. It is obvious that

$$\psi(B) \leq \varphi(B)$$

As for the opposite inequality, choose a suquence of Borel sets $D_i \supseteq B$ such that $\lim_{i\to\infty} \varphi(D_i) = \psi(B)$. Then, with $D := \liminf_{i\to\infty} D_i$, we have by Corollary 3.1.11 (v) ,

$$\varphi(B) \leq \varphi(D) \leq \liminf_{i\to\infty} \varphi(D_i) = \psi(B)$$

Then we have

$$\psi(A) = \inf\{\psi(B) : B \supseteq A, B \text{ a Borel set}\}$$

For each positive i, there exists Borel set B_i such that

$$B_i \supseteq A, \quad \psi(A) > \psi(B_i) - \frac{1}{i}$$

Let $B := \bigcap_{i=1}^{\infty} B_i$, which is a Borel set. We have

$$\psi(A) > \psi(B_i) - \frac{1}{i} \geq \psi(B) - \frac{1}{i}, \quad \forall i$$

That is $\psi(A) \geq \psi(B)$, thus $\psi(A) = \psi(B)$ from $B \supseteq A$. □

3.11 Outer Measure Generated by Measure

Definition 3.11.1 ▶ Measure on Algebra

An ***algebra*** in a space X is defined as a nonempty collection of subsets of X that is closed under the operations of finite unions and complements. Thus, the only difference between an algebra and a σ-algebra is that the latter is closed under countable unions. By a ***measure on an algebra***, $\mathcal{A}$, we mean a function $\mu : \mathcal{A} \to [0, \infty]$ satisfying the properties

(i) $\mu(\varnothing) = 0$,

(ii) if $\{A_i\}$ is a disjoint sequence of sets in $\mathcal{A}$ whose union is also in $\mathcal{A}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

Remark.
A measure on the algebra $\mathcal{A}$ is a real measure, if and only if $\mathcal{A}$ is a σ-algebra.

Definition 3.11.2

A measure μ on an algebra $\mathcal{A}$ is called σ-***finite*** if X can be written

$$X = \bigcup_{i=1}^{\infty} A_i$$

with $A_i \in \mathcal{A}$ and $\mu(A_i) < \infty$.

Theorem 3.11.3

Let μ be a measure on an algebra $\mathcal{A}$ and let μ^* be the corresponding set function generated by μ. Then

(i) μ^* is an outer measure.

(ii) μ^* is an extension of μ; that is, $\mu^*(A) = \mu(A)$ whenever $A \in \mathcal{A}$.

(iii) Each $A \in \mathcal{A}$ is μ^*-measurable.

Proof. The proof of (i) is similar to the proof of λ^* is an outer measure.

Now we prove (ii). Take $A \in \mathcal{A}$. It follows from definition that

$$\mu^*(A) \leq \mu(A)$$

For the opposite of the inequality, we take any sequence $A_1, A_2, \cdots \in \mathcal{A}$, such that

$$A \subseteq \bigcup_{i=1}^{\infty} A_i$$

We define

$$B_i := (A \cap A_i) \setminus \left(\bigcup_{j \neq i} A_j \right)$$

Then $B_1, B_2, \cdots$ are disjoint members of $\mathcal{A}$ and

$$A = \bigcup_{i=1}^{\infty} B_i$$

Since μ is additive on $\mathcal{A}$, we have

$$\mu(A) = \sum_{i=1}^{\infty} \mu(B_i) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

Since the RHS tends to $\mu^*(A)$, we have $\mu(A) \leq \mu^*(A)$.

For (iii), we need to show

$$\mu^*(E) \geq \mu^*(A \setminus E) + \mu^*(A \cap E)$$

for each $E \subseteq X$. We may assume that $\mu^*(E) < \infty$. We take $A_1, A_2, \cdots \in \mathcal{A}$ such that

$$E \subseteq \bigcup_{i=1}^{\infty} A_i, \quad \sum_{i=1}^{\infty} \mu(A_i) < \mu^*(E) + \varepsilon$$

Then

$$A \cap E \subseteq \bigcup_{i=1}^{\infty} (A_i \cap A) \implies \mu^*(A \cap E) \leq \sum_{i=1}^{\infty} \mu(A_i \cap A)$$

Since μ is additive on $\mathcal{A}$

$$\mu(A_i) = \mu(A_i \cap A) + \mu(A_i \setminus A)$$

And since μ agrees with μ^* on $\mathcal{A}$

$$\begin{aligned}
\mu^*(E) + \varepsilon &> \sum_{i=1}^{\infty} \mu(A_i) \\
&= \sum_{i=1}^{\infty} \mu(A_i \cap A) + \sum_{i=1}^{\infty} \mu(A_i \setminus A) \\
&\geq \mu^*(A \cap E) + \mu^*\left(\bigcup_{i=1}^{\infty} A_i \setminus A\right) \\
&\geq \mu^*(A \cap E) + \mu^*(E \setminus A)
\end{aligned}$$

Let $\varepsilon \to 0$, we get

$$\mu^*(E) \geq \mu^*(A \cap E) + \mu^*(E \setminus A)$$

Thus A is μ^*-measurable. $\square$

Example 3.11.4

Let us see how the previous result can be used to produce Lebesgue–Stieltjes measure. Let $\mathcal{A}$ be the algebra formed by including $\varnothing, \mathbb{R}$, all intervals of the form $(-\infty, a], (b, +\infty)$, along with all possible finite disjoint unions of these and intervals of the form $(a, b]$. Suppose that f is a nondecreasing right-continuous function and define μ on intervals $(a, b]$ in $\mathcal{A}$ by

$$\mu((a, b]) = f(b) - f(a),$$

and then extend μ to all elements of $\mathcal{A}$ by additivity. Then we see that the outer measure μ^* generated by μ agrees with the definition of Lebesgue–Stieltjes measure . Our previous result states that $\mu^*(A) = \mu(A)$ for all $A \in \mathcal{A}$, which agrees with Theorem above.

Remark.
In the previous example, the right continuity of f is needed to ensure that μ is in fact a measure on $\mathcal{A}$. For example, if

$$f(x) := \begin{cases} 0 & x \leq 0, \\ 1 & x > 0, \end{cases}$$

then $\mu((0,1]) = 1$. But $(0,1] = \bigcup_{k=1}^{\infty}(\frac{1}{k+1}, \frac{1}{k}]$ and

$$\begin{aligned} \mu\left(\bigcup_{k=1}^{\infty}(\frac{1}{k+1}, \frac{1}{k}]\right) &= \sum_{k=1}^{\infty} \mu((\frac{1}{k+1}, \frac{1}{k}]) \\ &= 0, \end{aligned}$$

which shows that μ is not a measure.

Theorem 3.11.5 ▸ Carathéodory–Hahn extension theorem

Let μ be a measure on an algebra $\mathcal{A}$, let μ^* be the outer measure generated by μ, and let $\mathcal{A}^*$ be the σ-algebra of μ^*-measurable sets.

(i) Then $\mathcal{A}^* \supset \mathcal{A}$ and $\mu^* = \mu$ on $\mathcal{A}$.

(ii) Let $\mathcal{M}$ be a σ-algebra with $\mathcal{A} \subset \mathcal{M} \subset \mathcal{A}^*$ and suppose ν is a measure on $\mathcal{M}$ that agrees with μ on $\mathcal{A}$. Then $\nu = \mu^*$ on $\mathcal{M}$ if μ is σ-finite.

Proof. (i) is just a restatement of Theorem 3.11.2.

(ii) Take any $E \in \mathcal{M}$, if $A_1, A_2, \cdots \in \mathcal{A}$ is a countable collection covering E, then

$$\nu(E) \leq \sum_{i=1}^{\infty} \nu(A_i) = \sum_{i=1}^{\infty} \mu^*(A_i)$$

Since $\left\{\sum_{i=1}^{\infty} \mu^*(A_i)\right\}$ tends to $\mu^*(E)$. We have $\nu \leq \mu^*$ on $\mathcal{M}$. To show the equality. Since μ is σ-finite, there are disjoint $X_1, X_2, \cdots \in \mathcal{A}$ such that

$$X \subseteq \bigcup_{i=1}^{\infty} X_i, \quad \mu(X_i) < \infty, \quad \forall i$$

Take any $A \in \mathcal{A}$, such that $\mu(A) < \infty$, since $A \setminus E \in \mathcal{M}$, we have

$$\mu^*(A \setminus E) \geq \nu(A \setminus E) = \nu(A) - \nu(E \cap A) = \mu(A) - \nu(E \cap A)$$

That is

$$\nu(E \cap A) \geq \mu(A) - \mu^*(A \setminus E) \geq \mu^*(E \cap A)$$

From μ is σ-finite, we know that there are disjoint $X_1, X_2, \cdots \in \mathcal{A}$ such that

$$X \subseteq \bigcup_{i=1}^{\infty} X_i, \quad \mu(X_i) < \infty, \forall i$$

Thus

$$\begin{aligned}
\nu(E) &= \nu\left(E \cap \left(\bigcup_{i=1}^{\infty} X_i\right)\right) \\
&= \sum_{i=1}^{\infty} \nu(E \cap X_i) \\
&\geq \sum_{i=1}^{\infty} \mu^*(E \cap X_i) \\
&\geq \mu^*(E)
\end{aligned}$$

□

Theorem 3.11.6

Suppose $(X, \mathcal{M}, \mu)$ is a measure space, where X is a metric space and μ is a finite Borel measure (that is, $\mathcal{M}$ denotes the Borel sets of X and

$\mu(X) < \infty$). Then for each $\varepsilon > 0$ and each Borel set B, there exist an open set U and a closed set F such that $F \subset B \subset U$, $\mu(B \setminus F) < \varepsilon$, and $\mu(U \setminus B) < \varepsilon$.

Proof. It is a immediate corollary of Theorem 3.10.2 and the agreements between μ and μ^* on $\mathcal{M}$. ☐

Definition 3.11.7

If μ is a measure defined on a σ-algebra $\mathcal{M}$ rather than on an algebra $\mathcal{A}$, there is another method for generating an outer measure. In this situation, we define μ^{**} on an arbitrary set $E \subset X$ by

$$\mu^{**}(E) = \inf\{\mu(B) : B \supset E, B \in \mathcal{M}\}.$$

Remark.
We know $\mu^* \leq \mu^{**}$ from definition. For $A_1, A_2, \cdots \in \mathcal{M}$ covering E , let $B_i = A_i \setminus A_{i-1}$, $B = \bigcup_{i=1}^{\infty} B_i$.

$$E \subseteq B = \bigcup_{i=1}^{\infty} A_i, \quad \mu(B) = \sum_{i=1}^{\infty} \mu(B_i) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

Which shows that $\mu^*(E) \leq \mu^{**}(E)$

Theorem 3.11.8

Consider a measure space $(X, \mathcal{M}, \mu)$. The set function μ^{**} defined above is an outer measure on X. Moreover, μ^{**} is a regular outer measure and $\mu(B) = \mu^{**}(B)$ for each $B \in \mathcal{M}$.

Proof. The proof proceeds exactly as in Theorem 3.10.4. One need only replace each reference to a Borel set in that proof with M-measurable set. Since there we don't use any relation between Borel sets and open/closed sets at all. All we need there is Borel sets is a σ-algebra . ☐

Theorem 3.11.9

Suppose $(X, \mathcal{M}, \mu)$ is a measure space and let μ^* and μ^{**} be the outer measures generated by μ as described above in diffenrent ways, respectively. Then for each $E \subset X$ with $\mu(E) < \infty$ there exists $B \in \mathcal{M}$ such that $B \supset E$,

$$\mu^*(B) = \mu^*(E) = \mu^{**}(E).$$

Proof. We shall show the part that $\mu(B) = \mu^*(B) = \mu^*(E)$. Then the ramaining had been shown above.

For each $k \in \mathbb{Z}_{>0}$ and every set E, there exists a sequence $\{A_i\} \in \mathcal{M}$ such that

$$E \subseteq \bigcup_{i=1}^{\infty} A_i, \quad \sum_{i=1}^{\infty} \mu(A_i) < \mu^*(E) + \frac{1}{k}$$

Setting $B_k = \bigcup_{i=1}^{\infty} A_i$ for that. Then

$$E \subseteq B_k, \quad \mu(B_k) < \mu^*(E) + \frac{1}{k}$$

And setting

$$B := \bigcap_{k=1}^{\infty} B_k$$

, we get

$$E \subseteq B, \quad \mu(B) \leq \mu^*(E) \implies \mu(B) = \mu^*(E)$$

□

4 Measurable Functions

4.1 Elementray Properties of Measruable Functions

Definition 4.1.1

We endow $\overline{\mathbb{R}}$ with a topology called the order topology in the following manner. For each $a \in \mathbb{R}$ let

$$L_a = \overline{\mathbb{R}} \cap \{x : x < a\} = [-\infty, a) \quad \text{and} \quad R_a = \overline{\mathbb{R}} \cap \{x : x > a\} = (a, \infty].$$

The collection $\mathcal{S} = \{L_a : a \in \mathbb{R}\} \cup \{R_a : a \in \mathbb{R}\}$ is taken as a subbasis for this topology. A basis for the topology is given by

$$\mathcal{S} \cup \{R_a \cap L_b : a, b \in \mathbb{R}, a < b\}.$$

Observe that the topology on $\mathbb{R}$ induced by the order topology on $\overline{\mathbb{R}}$ is precisely the usual topology on $\mathbb{R}$.

Definition 4.1.2 ▶ Measurable Functions

- Suppose $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ are measure spaces. A mapping $f : X \to Y$ is called ***measurable with respect to*** $\mathcal{M}$ ***and*** $\mathcal{N}$ if $f^{-1}(E) \in \mathcal{M}$ whenever $E \in \mathcal{N}$. If there is no danger of confusion, reference to $\mathcal{M}$ and $\mathcal{N}$ will be omitted, and we will simply use the term "measurable mapping."
- If Y is a topological space, a restriction is placed on $\mathcal{N}$. In this case it is always assumed that $\mathcal{N}$ is the σ-algebra of Borel sets $\mathcal{B}$. Thus, in this situation, a mapping $(X, \mathcal{M}) \xrightarrow{f} (Y, \mathcal{B})$ is measurable if

$$f^{-1}(E) \in \mathcal{M} \text{ whenever } E \in \mathcal{B}.$$

- When Y is taken as $\overline{\mathbb{R}}$ (endowed with the order topology) and $(X, \mathcal{M})$ is a topological space with $\mathcal{M}$ the collection of Borel sets. Then f is called a ***Borel measurable function***.

- When $X = \mathbb{R}^n$, $\mathcal{M}$ is the class of Lebesgue measurable sets and $Y = \overline{\mathbb{R}}$. Here, the measurability of f required that $f^{-1}(E)$ be Lebesgue measurable whenever $E \subset \overline{\mathbb{R}}$ is Borel, in which case f is called a ***Lebesgue measurable function***. The definitions imply that E is a measurable set if and only if χ_E is a measurable function.

Theorem 4.1.3

If the mapping $(X, \mathcal{M}) \xrightarrow{f} (Y, \mathcal{B})$ is measurable, where $\mathcal{B}$ is the σ-algebra of Borel sets. Define

$$\Sigma = \{E : E \subset Y \text{ and } f^{-1}(E) \in \mathcal{M}\}.$$

Then Σ is a σ-algebra.

Proof. Easy to verify. □

Corollary 4.1.4

A continuous mapping is a Borel measurable function.

Proof. If f is continuous, Σ contains all open open sets. Since Σ is a σ-algebra, Σ contains all Borel sets. □

Definition 4.1.5

If $f : X \to \mathbb{R}$, we call the sets

$$\{f > a\} := \{x \in X : f(x) > a\}$$

the ***superlevel sets of f***.

Theorem 4.1.6

Let $f : X \to \overline{\mathbb{R}}$, where $(X, \mathcal{M})$ is a measure space. The following conditions are equivalent:

(i) f is measurable.
(ii) $\{f > a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.
(iii) $\{f \geq a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.
(iv) $\{f < a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.

(v) $\{f \le a\} \in \mathcal{M}$ for each $a \in \mathbb{R}$.

Theorem 4.1.7

A function $f : X \to \overline{\mathbb{R}}$ is measurable if and only if
(i) $f^{-1}\{-\infty\} \in \mathcal{M}$ and $f^{-1}\{\infty\} \in \mathcal{M}$ and
(ii) $f^{-1}(a,b) \in \mathcal{M}$ for all open intervals $(a,b) \subset \mathbb{R}$.

Lemma 4.1.8

If f and g are measurable functions, then the following sets are measurable:
(i) $X \cap \{x : f(x) > g(x)\}$,
(ii) $X \cap \{x : f(x) \ge g(x)\}$,
(iii) $X \cap \{x : f(x) = g(x)\}$.

Proof. If $f(x) > g(x)$, then there is a rational number r such that $f(x) > r > g(x)$. Therefore, it follows that

$$\{f > g\} = \bigcup_{r \in \mathbb{Q}} (\{f > r\} \cap \{g < r\}),$$

which is a measurable set , and easily (i) holds. (ii) is just the complements of (i) with f and g interchanged. (iii) is just the intersection two of two measruable sets of type (ii), and so it too is measurable. □

Definition 4.1.9

If f and g are real-valued measurable functions, then $f+g$ is undefined at points where it would be of the form $\infty - \infty$. This difficulty is overcome if we define

$$(f+g)(x) := \begin{cases} f(x) + g(x), & x \in X - B, \\ \alpha, & x \in B, \end{cases}$$

where $\alpha \in \overline{\mathbb{R}}$ is chosen arbitrarily and where

$$B := (f^{-1}\{\infty\} \cap g^{-1}\{-\infty\}) \cup (f^{-1}\{-\infty\} \cap g^{-1}\{\infty\}).$$

Theorem 4.1.10

If $f, g : X \to \overline{\mathbb{R}}$ are measurable functions, then $f + g$ and fg are measurable.

Proof. We will treat the case that f and g have values in $\mathbb{R}$. The proof is similar in the general case and is left as an exercise (see Exercise 2, Section 5.1).

To prove that the sum is measurable, define $F : X \to \mathbb{R} \times \mathbb{R}$ by

$$F(x) = (f(x), g(x))$$

and $G : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ by

$$G(x, y) = x + y.$$

Then $G \circ F(x) = f(x) + g(x)$, so it suffices to show that $G \circ F$ is measurable. Referring to Theorem 4.1.7, we need only show that $(G \circ F)^{-1}(J) \in \mathcal{M}$ whenever $J \subset \mathbb{R}$ is an open interval. Now $U := G^{-1}(J)$ is an open set in $\mathbb{R}^2$, since G is continuous. Furthermore, U is the union of a countable family, $\mathcal{F}$, of 2-dimensional intervals I of the form $I = I_1 \times I_2$, where I_1 and I_2 are open intervals in $\mathbb{R}$. Since

$$F^{-1}(I) = f^{-1}(I_1) \cap g^{-1}(I_2),$$

we have

$$F^{-1}(U) = F^{-1}\left(\bigcup_{I \in \mathcal{F}} I\right) = \bigcup_{I \in \mathcal{F}} F^{-1}(I),$$

which is a measurable set. Thus $G \circ F$ is measurable, since

$$(G \circ F)^{-1}(J) = F^{-1}(U).$$

The product is measurable by essentially the same proof. □

Definition 4.1.11 ▸ Cantor Function

Recall that the Cantor set C can be expressed as

$$C = \bigcap_{j=1}^{\infty} C_j$$

Where C_j is the union of the 2^j closed intervals that remain after the jth step of the construction. The set

$$D_j = [0,1] - C_j$$

consists of the 2^{j-1} open intervals that are deleted at the jth step. Let there intervals be denoted by $I_{j,k}$, $k = 1, 2, \cdots, 2^{j-1}$, and order them in the obvious way from left to right. Now define a continuous function f_j on $[0,1]$ by

$$\begin{aligned} f_j(0) &= 0, \\ f_j(1) &= 1, \\ f_j(x) &= \frac{k}{2^j}, \quad x \in I_{j,k} \end{aligned}$$

and define f_j linearly on each interavl of C_j. The function f_j is continuous and nondecreasing, and it satisfies

$$|f_j(x) - f_{j+1}(x)| < \frac{1}{2^j}, \quad x \in [0,1]$$

Since

$$|f_j - f_{j+m}| < \sum_{i=j}^{j+m-1} \frac{1}{2^i} < \frac{1}{2^{j-1}}$$

it follows that the sequence $\{f_j\}$ converges uniformly to a continuous function f, called ***Cantor-Lebesgue function***.

Note.
f_j 在 $I_{j,k}$ 上的设定是端点的平均值.

Corollary 4.1.12

The Cantor-Lebesgue function f and its associate $h(x) := f(x) + x$ described above have the following properties:

(i) $f(C) = [0,1]$; that is, f maps a set of measure 0 onto a set of positive measure.

(ii) h maps a Lebesgue measurable set onto a nonmeasurable set.

(iii) The composition of Lebesgue measurable functions need not be Lebesgue measurable.

(iv) Lebesgue measurable function dose not preserve Borel sets.

Proof. 1. $f : [0,1] \to [0,1]$ is onto. It is easy to see that $f(C) = [0,1]$, because $f(C)$ is compact and $f([0,1] - C)$ is countable.

2.

Proof sketch.
把 C 看成是集合 $[0,1]$ 测度为零的“骨”. $[0,1] \setminus C$ 是 $[0,1]$ 的测度为 1 的“肉”. f 的构造是对“骨”的展开: $f(C) = [0,1]$; 同时对“肉”的压缩: 令 f 在 $I_{j,k}$ 上取常值. 通过令 $h : [0,1] \to [0,2]$. $h(x) = f(x) + x$, 我们变换映射的尺度, 不再将“肉”压缩, 而是忠实地搬运, 将每个 $I_{j,k}$ 映到大小相同的一个区间. 从而间接的看到这两个操作对“骨”的影响区别不大 (测度都是 1).

Note.
$h(C)$ 和 $f(C)$ 测度相同在直觉上也是存在依据的.
Cantor 函数 f 是一列分段线性函数 f_n 的一致极限. 在构成 Cantor 集第 n 步近似 C_n 的那些区间上, f_n 的斜率为 $(3/2)^n$, 这是一个趋向于无穷大的值.
函数 $h(x) = f(x) + x$ 可以看作是 $h_n(x) = f_n(x) + x$ 的极限. 在同样的区间上, h_n 的斜率变为 $(3/2)^n + 1$.
为一个趋于无穷的量加上 1, 并不会显著改变它的数量级. 两个斜率的比值为 $\frac{(3/2)^n+1}{(3/2)^n} = 1 + (2/3)^n \to 1$.
这便解释了我们的直觉: ”+x” 这个操作虽然修复了 f 对 C 补集测度的” 压缩”, 但对于 f 自身在 C 上的” 测度生成” 效应来说, 其 * 相对 * 影响是微不足道的. 因此, f 和 h 都将测度为零的集合 C ” 拉伸” 成了一个测度为 1 的集合.

h is trictly increasing. Thus , h is a homeomorphism from $[0,1]$ onto $[0,2]$. Then h carries the open set $[0,1] \setminus C$ to an open set. And we have

$$\begin{aligned} m(h([0,1] \setminus C)) &= m\left(h\left(\bigcup_{j,k} I_{j,k}\right)\right) \\ &= \sum_{j,k} m(h(I_{j,k})) \\ &= \sum_{j,k} m(I_{j,k}) \\ &= 1 \end{aligned}$$

Therefore, h maps the Cantor set to a set P with measure 1. From Exercise **??** , there exists a non-measurable set $N \subseteq P$. Set $A = h^{-1}(N)$. Since $h^{-1}(N) \subseteq h^{-1}(P) = C$, A is a subset of measure-zero set C. Thus A is Lebesgue-measurable. But $h(A) = N$ is not measurable.

3. Note that h^{-1} is measurable, since it is continuous. Let $F := h^{-1}$. Observe that χ_A is measurable, but

$$\chi_A \circ F = \chi_A \circ h^{-1} = \chi_N$$

is not Lebesgue-measurable.

4. Observe that A is not a Borel set, for if it were, then $F^{-1}(A)$ would be a Borel set. But $F^{-1}(A) = h(A) = N$ and N is not a Borel set. Now $g := \chi_A$ is a Lebesgue measurable function such that $g^{-1}(\{1\}) = A$, where $\{1\}$ is a Borel sets but A is not.

□

Theorem 4.1.13

Suppose $f : X \to \overline{\mathbb{R}}$ is measurable and $g : \overline{\mathbb{R}} \to \overline{\mathbb{R}}$ is Borel measurable. Then $g \circ f$ is measurable. In particular, if $X = \mathbb{R}^n$ and f is Lebesgue measurable, then $g \circ f$ is Lebesgue measurable.

Corollary 4.1.14

Let $f : X \to \overline{\mathbb{R}}$ be a measurable function.

(i) Let $\varphi(x) = |f(x)|^p$, $0 < p < \infty$, and let φ assume arbitrary extended values on the sets $f^{-1}(\infty)$ and $f^{-1}(-\infty)$. Then φ is measurable.

(ii) Let $\varphi(x) = \dfrac{1}{f(x)}$, and let φ assume arbitrary extended values on the sets $f^{-1}(0)$, $f^{-1}(\infty)$ and $f^{-1}(-\infty)$. Then φ is measurable.

In particular, if $X = \mathbb{R}^n$ and f is Lebesgue measurable, then φ is Lebesgue measurable in (i) and (ii).

Proof. 1. Define $g(t) = |t|^p$ for $t \in \mathbb{R}$, and assign arbitrary caluse to $g(\infty)$ and $g(-\infty)$. Now apply the prvious theorem.

2. Proceed in a similar way by defining $g(t) = \frac{1}{t}$ when $t \neq 0, \infty, -\infty$ and assigning arebitrary values to $g(0)$, $g(\infty)$, and $g(-\infty)$.

□

Theorem 4.1.15

Let $(X, \mathcal{M}, \mu)$ be a complete measure space and let f, g be extended real-valued functions defined on X. If f is measurable and $f = g$ almost everywhere, then g is measurable.

Remark.
In a complete measure space, we can consider a function f that just define on almost everywhere on X, by extending f into $\overline{f}$ with arbitrary value taken on where f was not defined. .

Proof. Let $N = \{x : f(x) = g(x)\}$. Then $\mu(\not N) = 0$, and thus $\not N$ as well as all subsets of $\not N$ are measurable. For $a \in \mathbb{R}$, we have

$$\begin{aligned}\{g > a\} &= (\{g > a\} \cap N) \cup (\{g > a\} \cap \not N) \\ &= (\{f > a\} \cap N) \cup (\{g > a\} \cap \not N) \in \mathcal{M}.\end{aligned}$$

□

Theorem 4.1.16

Suppose φ is an outer measure on a space X. Then an extended real-valued function f on X is φ-measurable if and only if

$$\varphi(A) \geq \varphi(A \cap \{f \leq a\}) + \varphi(A \cap \{f \geq b\})$$

whenever $A \subset X$ and $a < b$ are real numbers.

Proof. The "only if" is immediate from

$$\varphi(A) \geq \varphi(A \cap \{f \leq a\}) + \varphi(A \cap \{f > a\}) \geq \varphi(A \cap \{f \leq a\}) + \varphi(A \cap \{f \geq b\})$$

To show the converse, we need to prove that for any $r \in \mathbb{R}$, the set

$$E := \{f \leq r\}$$

satisfies

$$\varphi(A) \geq \varphi(A \cap E) + \varphi(A - E)$$

for each $A \subseteq X$. We define

$$B_j := A \cap \left\{r + \frac{1}{j+1} \leq f \leq r + \frac{1}{j}\right\}$$

We shall show

$$\infty > \varphi(A) \geq \varphi\left(\bigcup_{j=1}^{\infty} B_{2j}\right) = \sum_{j=1}^{\infty} \varphi\left(B_{2j}\right)$$

by induction. We fix k and assume

$$\varphi\left(\bigcup_{j=1}^{k} B_{2j}\right) = \sum_{j=1}^{k} \varphi\left(B_{2j}\right)$$

We set

$$A_k := \bigcup_{j=1}^{k} B_{2j}$$

Then

$$\left(A_k \cup B_{2(k+1)}\right) \cap \left\{f \leq r + \frac{1}{2(k+1)}\right\} = B_{2(k+1)}$$

$$\left(A_k \cup B_{2(k+1)}\right) \cap \left\{f \geq r + \frac{1}{2k+1}\right\} = A_k$$

We have

$$\varphi\left(\bigcup_{j=1}^{k+1} B_{2j}\right) = \varphi\left(A_k \cup B_{2(k+1)}\right) \geq \varphi\left(B_{2(k+1)}\right) + \varphi(A_k) = \sum_{j=1}^{k+1} \varphi\left(B_{2j}\right)$$

Thus we inductively show that

$$\varphi(A) \geq \varphi\left(\bigcup_{j=1}^{\infty} B_{2j}\right) = \sum_{j=1}^{\infty} \varphi\left(B_{2j}\right)$$

Similarly, one can show that

$$\varphi(A) \geq \varphi\left(\bigcup_{j=1}^{\infty} B_{2j-1}\right) = \sum_{j=1}^{\infty} \varphi\left(B_{2j-1}\right)$$

Then the series

$$\sum_{j=1}^{\infty} \varphi\left(B_j\right) \leq 2\varphi(A) < \infty$$

convergents. The tail of the series can be arbitrarily small. For each $\varepsilon > 0$, there exists $m \in \mathbb{N}$ such that

$$\varepsilon > \sum_{j=m}^{\infty} \varphi\left(B_j\right) \geq \varphi\left(\bigcup_{j=m}^{\infty} B_j\right)$$

Where

$$\bigcup_{j=m}^{\infty} B_j = A \cap \left\{r < f \leq r + \frac{1}{m}\right\}$$

We define a measure ψ

$$\psi(F) \coloneqq \varphi(A \cap F), \quad \forall F \subseteq X$$

Then

$$\varepsilon > \psi\left(\left\{r < f \leq r + \frac{1}{m}\right\}\right)$$

Thus

$$\begin{aligned}
\varphi(A \cap E) + \varphi(A - E) &= \psi(E) + \psi(E^c) \\
&= \psi(\{f \leq r\}) + \psi(\{f > r\}) \\
&\leq \psi(\{f \leq r\}) + \psi\left(\left\{f > r + \frac{1}{m}\right\}\right) + \psi\left(\left\{r < f \leq r + \frac{1}{m}\right\}\right) \\
&\leq \psi(\{f \leq r\}) + \psi\left(\left\{f \geq r + \frac{1}{m}\right\}\right) + \varepsilon \\
&= \varphi(A \cap \{f \leq r\}) + \varphi\left(A \cap \left\{f \geq r + \frac{1}{m}\right\}\right) + \varepsilon \\
&\leq \varphi(A) + \varepsilon
\end{aligned}$$

Since ε is arbitrary, we have

$$\varphi(A \cap E) + \varphi(A - E) \leq \varphi(A)$$

Thus E is measurable. □

Exercises

Exercise 4.1.17

Prove that a function defined on $\mathbb{R}^n$ that is continuous everywhere except for a set of Lebesgue measure zero is a Lebesgue measurable function. In particular, conclude that a nondecreasing function defined on $[0, 1]$ is Lebesgue measurable.

Proof. If f is such function. Let N be the measure-zero set where f is not continuous. f is continuous on the topological subspace $\mathbb{R}^n \setminus N$, thus is measurable on $(\mathbb{R}^n \setminus N, \mathcal{B}(\mathbb{R}^n \setminus N))$. Then for each open sets $U \subseteq \mathbb{R}$, we have

$$\begin{aligned} f^{-1}(U) &= \left(f^{-1}(U) \cap (\mathbb{R}^n \setminus N)\right) \cup \left(f^{-1}(U) \cap N\right) \\ &= \left(f|_{\mathbb{R}^n \setminus N}^{-1}(U)\right) \cap \left(f^{-1}(U) \cup N\right) \end{aligned}$$

Where $f|_{\mathbb{R}^n \setminus N}^{-1}(U) \in \mathcal{B}(\mathbb{R}^n \setminus N)$. Note that

$$\mathcal{B}(\mathbb{R}^n \setminus N) = \{B \cap (\mathbb{R}^n \setminus N) : B \in \mathcal{B}(\mathbb{R}^n)\} \subseteq \mathcal{L}(n)$$

Since $f^{-1}(U) \cap N \in \mathcal{L}(n)$, we have $f^{-1}(U) \in \mathcal{L}(n)$. f is Lebesuge-measurable.

Since a nondecreasing function is continuous except for a countable set, which is Lebesgue-measura-zero. It is Lebesgue measurable from our argument above. □

4.2 Limits of Measurable Functions

Definition 4.2.1

Let $(X, \mathcal{M}, \mu)$ be a measure space, and let $\{f_i\}$ be a sequence of measurable functions defined on X. The ***upper*** and ***lower envelopes*** of $\{f_i\}$ are defined respectively as

$$\sup_i f_i(x) = \sup\{f_i(x) : i = 1, 2, \dots\}$$

and

$$\inf_i f_i(x) = \inf\{f_i(x) : i = 1, 2, \dots\}.$$

Also, the ***upper*** and ***lower limits*** of $\{f_i\}$ are defined as

$$\limsup_{i\to\infty} f_i(x) = \inf_{j\geq 1}\left(\sup_{i\geq j} f_i(x)\right)$$

and

$$\liminf_{i\to\infty} f_i(x) = \sup_{j\geq 1}\left(\inf_{i\geq j} f_i(x)\right).$$

Theorem 4.2.2

Let $\{f_i\}$ be a sequence of measurable functions defined on the measure space $(X,\mathcal{M},\mu)$. Then $\sup_i f_i$, $\inf_i f_i$, $\limsup_{i\to\infty} f_i$, and $\liminf_{i\to\infty} f_i$ are all measurable functions.

Proof. For each $a\in\mathbb{R}$ the identity

$$X\cap\{x : \sup_i f_i(x) > a\} = \bigcup_{i=1}^{\infty}(X\cap\{f_i(x) > a\})$$

implies that $\sup_i f_i$ is measurable. The measurability of the lower envelope follows from

$$\inf_i f_i(x) = -\sup_i(-f_i(x)).$$

Now that it has been shown that the upper and lower envelopes are measurable, it is immediate that the upper and lower limits of $\{f_i\}$ are also measurable. □

Theorem 4.2.3 ▶ Egorov

Let $(X,\mathcal{M},\mu)$ be a finite measure space and suppose $\{f_i\}$ and f are measurable functions that are finite almost everywhere on X. Also, suppose that $\{f_i\}$ converges pointwise a.e. to f. Then for each $\varepsilon > 0$ there exists a set $A\in\mathcal{M}$ such that $\mu(A^c) < \varepsilon$ and $\{f_i\}\to f$ uniformly on A.

Remark.
定义一个依赖于 $x\in X$ 和两个自然数索引 $n,k\in\mathbb{Z}_+$ 的逻辑陈述 $P_{n,k}(x)$, 它的值为 1 或 0, 代表真与假.

- k 表示精度, k 越大条件越严格.

- n 表示阶段或步骤

我们可以定义

- 逐点条件:

$$\forall k \in \mathbb{Z}_+, \forall x \in E, \exists N \in \mathbb{Z}_+, \forall n \geq N, P_{n,k}(x) = 1$$

即对于任意的精度 k , 在每个点上可以最终保持的.
设 $E_{n,k} = P_{n,k}^{-1}(1)$, 即该精度在某一阶段成立的点. 那么逐点条件用集合, 就是

$$E \subseteq \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} E_{n,k} = \liminf_{n \to \infty} E_{n,k}, \quad \forall k \in \mathbb{Z}_+$$

- 一致条件:

$$\forall k \in \mathbb{Z}_{>0}, \exists N_k \in \mathbb{Z}_{>0}, \forall n \geq N_k, \forall x \in E, P_{n,k}(x) = 1$$

即对于任意的精度 k, 最终总是全局保持的.

Egorov 定理就是说, 如果 $P_{n,k}(x)$ 在每个点总是最终精确的, 那么它差不多总是最终整体精确的.

Proof. Let E be the set on which the function f_i, $i = 1, 2, \cdots$ are defined and finite. Let F be the set on which f convergent pointwise to f. Let $A_0 = E \cap F$. We have by hypothesis $\mu(A_0^c) = 0$. For each $\varepsilon > 0$, define

$$E_{i,k} = \left\{x \in A_0 : |f_i - f| < \frac{1}{k}\right\}$$

Define

$$A_{i,k} = \bigcap_{j=i}^{\infty} E_{j,k} = \left\{x \in A_0 : \left|f_j(x) - f(x)\right| < \frac{1}{k}, \forall j \geq i\right\}$$

Then $A_{1,k} \subseteq A_{2,k} \subseteq \cdots$, $A_0 = \bigcup_{i=1}^{\infty} A_{i,k}, \forall k$. $A_0 \setminus A_{1,k} \supseteq A_0 \setminus A_{2,k}, \cdots$. $\bigcap_{i=1}^{\infty} \left(A_0 \setminus A_{i,k}\right) = \varnothing$.

$$0 = \mu(\varnothing) = \lim_{i \to \infty} \mu\left(A_0 \setminus A_{i,k}\right)$$

Then for each k , and for each $\varepsilon_k > 0$, there exists i_k such that

$$\mu\left(A_0 \setminus A_{i_k,k}\right) < \varepsilon_k$$

Remark.
这里就是在说，处处最终成立的条件，是差不多最终一直成立的.

$$E \subseteq \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} P_n \implies E_{\text{大测度, 舍弃了不多于}\varepsilon} \subseteq \bigcap_{n=N_\varepsilon}^{\infty} P_n$$

如果 P_n 都是可测条件

For each $\varepsilon > 0$, we take $\varepsilon_k = \frac{\varepsilon}{2^k}$, let $A = \bigcap_{k=1}^{\infty} A_{i_k,k}$. Then

$$\mu(A_0 \setminus A) = \mu\left(\bigcup_{k=1}^{\infty} A_0 \setminus A_{i_k,k}\right) \leq \sum_{k=1}^{\infty} \mu(A_0 \setminus A_{i_k,k}) < \sum_{k=1}^{\infty} \varepsilon_k = \varepsilon$$

$$\mu(X \setminus A) \leq \mu(X \setminus A_0) + \mu(A_0 \setminus A) = 0 + \mu(A_0 \setminus A) < \varepsilon$$

And note that for each k , since $A \subseteq A_{i_k,k}$ we have

$$|f_j(x) - f(x)| < \frac{1}{k}, \quad \forall x \in A, j \geq i_k$$

Then

$$\lim_{i\to\infty} \sup_{x\in A} |f_i - f| = 0$$

That is $\{f_i\}$ convergents uniformly to f on A. □

Corollary 4.2.4

若存在收敛于零的正项数列 $\{a_k\}$ 使得对于

$$A_k = \{x : |f_k(x) - f(x)| \geq a_k\}$$

有级数 $\sum_{k=1}^{\infty} m(A_k)$ 收敛. 则 f_k 是近一致收敛于 f 的.

Corollary 4.2.5

Let $\{E_n\}$ be a sequence of measurable functions. If

$$E \subseteq \liminf_{n\to\infty} E_n$$

Then for each $\varepsilon > 0$, there exists measurable set E_ε, and $N_\varepsilon \in \mathbb{Z}_{>0}$, such that $\mu\left(E_\varepsilon^c\right) < \varepsilon$ and

$$E_\varepsilon \subseteq \bigcap_{n=N_\varepsilon} E_n$$

Remark.
也就是说，如果一列可测的条件 P_n，对于每个点 $x \in E$ 总是最终保持的. 相当于说每个 x 总会在每一个阶段加入并保持在 P 中，那么我们可以让还没有加入的那些成员任意地少，然后把它们抛弃掉，使得剩下的成员在一个共同的阶段后总是全都保持在里面的.

Definition 4.2.6

A sequence of measurable functions $\{f_i\}$ defined relative to the measure space $(X, \mathcal{M}, \mu)$ is said to ***converge in measure*** to a measurable function f if for every $\varepsilon > 0$, we have

$$\lim_{i\to\infty} \mu(X \cap \{x : |f_i(x) - f(x)| \geq \varepsilon\}) = 0.$$

Theorem 4.2.7

Let $(X, \mathcal{M}, \mu)$ be a finite measure space, and suppose $\{f_i\}$ and f are measurable functions that are finite a.e. on X. If $\{f_i\}$ converges to f a.e. on X, then $\{f_i\}$ converges to f in measure.

Remark.
测度的有限性是为了保证 Egorov 的使用，以及从一致逼近过渡到测度逼近的可行性.

Theorem 4.2.8

Let $(X, \mathcal{M}, \mu)$ be a measure space and let $\{f_i\}$ and f be measurable functions such that $f_i \to f$ in measure. Then there exists a subsequence $\{f_{i_j}\}$ such that

$$\lim_{j\to\infty} f_{i_j}(x) = f(x)$$

for μ-a.e. $x \in X$.

Remark.
依测度收敛导致子列的 a.e. 收敛.

Proof sketch.
将收敛描述成某种“逃逸难度”，依测度收敛就是在任意逃逸难度（函数值差大于 $\frac{1}{k}$）下，逃逸的可能性随着进程的推进最终会趋于“不可能逃逸”（逃逸出去的点最终是零测的）. 于是我们选取某些进程 i_k，使得在这个进程之后，即便逃逸难度很小，逃逸可能性也会很微弱. 依测度收敛保证了几乎所有的点都被这些进程给捕获. 于是最终几乎所有点在任意简单的难度下都不能逃逸，也就代表了几乎处处收敛.

Proof. Let i_1 be an integer such that

$$\mu\left(\left\{x \in X : \left|f_{i_1}(x) - f(x)\right| \geq 1\right\}\right) < \frac{1}{2}$$

Assume that we take $i_1, \cdots, i_k$, then we take $i_{k+1} > i_k$ such that

$$\mu\left(\left\{x \in X : \left|f_{i_{k+1}}(x) - f(x)\right| \geq \frac{1}{k+1}\right\}\right) < \frac{1}{2^{k+1}}$$

Let

$$A_j = \bigcup_{k=j}^{\infty}\left\{x \in X : \left|f_{i_k} - f\right| \geq \frac{1}{k}\right\}$$

Then

$$A_1 \supseteq A_2 \supseteq \cdots$$

$$\mu\left(A_j\right) \leq \sum_{k=j}^{\infty}\mu\left(\left\{x \in X : \left|f_{i_k} - f\right| \geq \frac{1}{k}\right\}\right) < \sum_{k=j}^{\infty}\frac{1}{2^k} = \frac{1}{2^{j-1}}$$

Specifically, $\mu(A_1) < \infty$, define $A = \bigcap_{j=1}^{\infty} A_j$ we have

$$\mu(A) = \lim_{j\to\infty}\mu\left(A_j\right) = 0$$

For each $x \in A^c = \bigcup_{j=1}^{\infty} A_j^c$, there exists k_0 such that $x \in A_{k_0}^c$,

$$A_{k_0}^c = \bigcap_{k=k_0}^{\infty} \left\{ x \in X : |f_{i_k} - f| < \frac{1}{k} \right\}$$

Which mean that

$$|f_{i_k}(x) - f(x)| < \frac{1}{k}, \quad \forall k \geq k_0$$

Thus $\lim_{k\to\infty} f_{i_k}(x) = f(x)$. Since x is arbitrary, $\{f_{i_k}\}_k$ converges to f on A^c. □

Theorem 4.2.9

- A sequence $\{f_i\}$ of a.e. finite-valued measurable functions on a measure space $(X, \mathcal{M}, \mu)$ is ***fundamental in measure*** if for every $\varepsilon > 0$,

$$\mu(\{x : |f_i(x) - f_j(x)| \geq \varepsilon\}) \to 0$$

- Prove that if $\{f_i\}$ is fundamental in measure, then there is a measurable function f to which the sequence $\{f_i\}$ converges in measure. as i and $j \to \infty$.

Theorem 4.2.10

Let $(X, \mathcal{M}, \mu)$ be a finite measure space. Suppose that $\{f_i\}_{i=1}^{\infty}$ and f are measurable functions. Prove that $f_i \to f$ in measure if and only if each subsequence of f_i has a subsequence that converges to f μ-a.e.

4.3 Approximation of Measurable Functions

Theorem 4.3.1

Let $f : X \to \overline{\mathbb{R}}$ be an arbitrary (possibly nonmeasurable) function. Then the following hold:

(i) There exists a sequence of simple functions, $\{f_i\}$, such that $f_i(x) \to f(x)$ for each $x \in X$.

(ii) If f is nonnegative, the sequence can be chosen such that $f_i \uparrow f$.

(iii) If f is bounded, the sequence can be chosen such that $f_i \to f$ uniformly on X.
(iv) If f is measurable, the f_i can be chosen to be measurable.

Proof. Step1. All for $f \geq 0$:

Assume that $f \geq 0$. For $i \in \mathbb{Z}_{>0}$, partition $[0, i)$ in to $i \cdot 2^i$ half-open intervals of the form $[\frac{k-1}{2^i}, \frac{k}{2^i}), k = 1, 2, \cdots, i \cdot 2^i$. Label these intervals $H_{i,k}$ and let

$$A_{i,k} = f^{-1}\left(H_{i,k}\right), \quad A_i = f^{-1}\left([i, \infty]\right)$$

Define f_i on X as

$$f_i(x) = \begin{cases} \frac{k-1}{2^i}, & x \in A_{i,k} \\ i, & x \in A_i \end{cases}$$

- If f is measurable, then the sets $A_{i,k}$ and A_i are measurable, and thus so are the functions f_i.

It is easy to verify that

$$f_1 \leq f_2 \leq \cdots \leq f$$

If $f(x) < \infty$, then for every $i > f(x)$, we have

$$|f_i(x) - f(x)| < \frac{1}{2^i}$$

and hence $f_i(x) \to f(x)$. If $f(x) = \infty$, then $f_i(x) = i \to f(x)$. In any case, we obtains

$$\lim_{i \to \infty} f_i(x) = f(x), \quad x \in X$$

- If f is bounded by M. Then $A_i = \varnothing$ for all $i > M$, and therefore $|f_i(x) - f(x)| < \frac{1}{2^i}$ for all $x \in X$, thus showing that $f_i \to f$ uniformly .

Step 2. The general case:

Applying to f^+ and f^-, (i), (iii), (iv) holds .

□

Theorem 4.3.2 ▸ Lusin's theorem

Suppose $(X, \mathcal{M}, \mu)$ is a measure space where X is a metric space and μ is a finite Borel measure. Let $f : X \to \mathbb{R}$ be a measurable function that is finite almost everywhere. Then for every $\varepsilon > 0$ there is a closed set $F \subset X$ with $\mu(X \setminus F) < \varepsilon$ such that f is continuous on F in the relative topology.

Corollary 4.3.3

Let $E \subseteq \mathbb{R}^n$ be a Lebesgue measurable set. f is a a.e. finite measurable function on E. Then for any $\delta > 0$, there exists a continuous g on $\mathbb{R}^n$ such that

$$m(\{x \in E : f(x) \neq g(x)\}) < \delta$$

If E is bounded, g can chosen to be compactly supported.

Remark.

- If f is bounded. Then from Tiesze Extension Theorem, g can chosen to satisfy $\sup_{x \in \mathbb{R}^n} |g| \leq \sup_{x \in F} |f|$

Proof. From Lusin's Theorem, for each $\delta > 0$, there exists a closed set F in E, with $m(E \setminus F) < \delta$ such that f is continuous on F with the relative topology. The function $f|_F$ can be extended to a continuous functin g on $\mathbb{R}^n$ such that

$$f(x) = g(x), \quad \forall x \in F$$

Then it follows that

$$m(\{x \in E : f(x) \neq g(x)\}) \leq m(E \setminus F) < \delta$$

If E is bounded, assume that $E \subseteq B(0, k)$. Then from Urysohn Lemma , there is a continuous function φ with $0 \leq \varphi \leq 1$ such that

$$\varphi(F) = \{1\}, \quad \varphi\left((B(0,k))^c\right) = \{0\}$$

We substitute g by $g\varphi$, it is compactly supported. $\square$

5

Integration

5.1 Integration Respect to a Measure

Theorem 5.1.1 ▸ Monotone Convergence Theorem

Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $0 \leq f_1 \leq f_2 \leq \cdots$ is an increasing sequence of $\mathcal{S}$-measurable functions. Define $f : X \to [0, \infty)$ by

$$f(x) = \lim_{k\to\infty} f_k(x).$$

Then

$$\lim_{k\to\infty} \int f_k d\mu = \int f d\mu.$$

Proof. f is $\mathcal{S}$-measurable as the pointwise limitation of $\{f_k\}$. Since $f_k \leq f$ for all k, we have $\int f_k \, \mathrm{d}\mu \leq \int f \, \mathrm{d}\mu$. Thus

$$\lim_{k\to\infty} \int f_k \, \mathrm{d}\mu \leq \int f \, \mathrm{d}\mu$$

To show the converse, take arbitrary simple function $s = \sum_{j=1}^{n} c_j \chi_{A_j}$ with standard representation such that $f \geq s$. We fix a $t \in (0, 1)$, and define

$$E_k := \left\{ x \in X : f_k(x) \geq t \sum_{j=1}^{n} c_j \chi_{A_j}(x) \right\}$$

Then E_k s are measurable and

$$E_1 \subseteq E_2 \subseteq \cdots, \quad \bigcup_{k=1}^{\infty} E_k = X$$

Then it follows that $\mu\left(A_j\right) = \lim_{k\to\infty} \mu\left(A_j \cap E_k\right)$ for each j. We have

$$\int f_k \,\mathrm{d}\mu \geq \int t \sum_{j=1}^{n} c_j \chi_{A_j \cap E_k} \,\mathrm{d}\mu \geq t \sum_{j=1}^{n} c_j \mu\left(A_j \cap E_k\right)$$

By taking $k \to \infty$, we get

$$\lim_{k\to\infty} \int f_k \,\mathrm{d}\mu \geq t \sum_{j=1}^{n} c_j \mu\left(A_j\right)$$

Taking the suprerum of the inequality above over all simple functions s , we get

$$\lim_{k\to\infty} \int f_k \,\mathrm{d}\mu \geq t \int f \,\mathrm{d}\mu$$

Finally , taking the limit of t increasing to 1 shows that

$$\lim_{k\to\infty} \int f_k \,\mathrm{d}\mu \geq \int f \,\mathrm{d}\mu$$

□

Lemma 5.1.2 ▸ Fatou's lemma

If $\{f_k\}_{k=1}^{\infty}$ is a sequence of nonnegative μ-measurable functions, then

$$\int_X \liminf_{k\to\infty} f_k d\mu \leq \liminf_{k\to\infty} \int_X f_k d\mu.$$

Proof. Let $g_k(x) = \inf_{n\geq k} f_n(x)$. Then $0 \leq g_1 \leq g_2 \leq \cdots$ is an increasing suquence of $\mathcal{S}$-measurable functions.Then

$$g(x) = \liminf_{k\to\infty} f_k$$

It follows from Monotone Convergence Theorem that

$$\int_X \liminf_{k\to\infty} f_k \,\mathrm{d}\mu = \lim_{k\to\infty} \int_X g_k \,\mathrm{d}\mu = \lim_{k\to\infty} \int_X g_k \,\mathrm{d}\mu$$

Since for each g_k, we have

$$g_k(x) = \inf_{n \geq k} f_n(x) \leq f_k(x)$$

then

$$\int_X g_k \, \mathrm{d}\mu \leq \int_X f_k \, \mathrm{d}\mu, \quad \forall k \geq 1$$

By taking the lim inf relative to k both sides, we have

$$\lim_{k \to \infty} \int_X g_k \, \mathrm{d}\mu = \liminf_{k \to \infty} \int_X g_k \, \mathrm{d}\mu \leq \liminf_{k \to \infty} f_k \, \mathrm{d}\mu.$$

Thus

$$\int_X \liminf_{k \to \infty} f_k \, \mathrm{d}\mu \leq \liminf_{k \to \infty} \int_X f_k \, \mathrm{d}\mu$$

□

5.2 Limits of Integrations

5.2.1 Bounded Convergence Theorem

Proposition 5.2.1 ▸ bounding an integral

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $E \in \mathcal{S}$, and $f : X \to [-\infty, \infty]$ is a function such that $\int_E f \, d\mu$ is defined. Then

$$\left| \int_E f \, d\mu \right| \leq \mu(E) \sup_E |f|.$$

Proof. Let $c = \sup_E |f|$, then

$$\begin{aligned}\left|\int_E f \, d\mu\right| &= \left|\int \chi_E f \, d\mu\right| \\ &\le \int |\chi_E f| \, d\mu \\ &\le \int \chi_E c \, d\mu \\ &= \mu(E) c = \mu(E) \sup_E |f|\end{aligned}$$

□

5.2.2 Dominated Convergence Theorem

Proposition 5.2.2 ▸ integrals on small sets are small

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $g : X \to [0, \infty]$ is $\mathcal{S}$-measurable, and $\int g d\mu < \infty$. Then for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\int_B g d\mu < \varepsilon$$

for every set $B \in \mathcal{S}$ such that $\mu(B) < \delta$.

Remark.

- $g \in L^1(\mu)$ induces a finite measure

$$\nu(B) = \int_B g \, d\mu, \quad B \in \mathcal{S}$$

- For each $\varepsilon > 0$, there exists $\delta > 0$ such that $\nu(B) < \varepsilon$ whenever $\mu(B) < \delta$.

Proof. Since $\int g \, d\mu < \infty$, for each $\varepsilon > 0$, there exists a simple function $h : X \to [0, \infty]$ such that $h \le g$,

$$0 \le \int g \, d\mu - \int h \, d\mu < \frac{\varepsilon}{2}$$

Suppose that $c = \sup_X h$, which is finite. Then for $\delta = \frac{\varepsilon}{2c}$, we have for each $B \in \mathcal{S}$ such that $\mu(B) < \delta$, there is

$$\int_B h \, \mathrm{d}\mu < \mu(B) c < \frac{\varepsilon}{2}$$

$$\int_B g \, \mathrm{d}\mu \leq \int_B h \, \mathrm{d}\mu + \frac{\varepsilon}{2} < \varepsilon$$

□

Proposition 5.2.3 ▸ integrable functions live mostly on sets of finite measure

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $g : X \to [0, \infty]$ is $\mathcal{S}$-measurable, and $\int g d\mu < \infty$. Then for every $\varepsilon > 0$, there exists $E \in \mathcal{S}$ such that $\mu(E) < \infty$ and

$$\int_{X \setminus E} g d\mu < \varepsilon.$$

Proof. Let $h : X \to [0, \infty]$ be a simple function such that $0 \leq h \leq g$,

$$\int g \, \mathrm{d}\mu - \int h \, \mathrm{d}\mu < \frac{\varepsilon}{2}$$

Define

$$E = h^{-1}((0, \infty))$$

It follows from the definition of the simple function and its integrability that $\mu(E) < \infty$. Furthermore,

$$\int_{X \setminus E} h \, \mathrm{d}\mu = \int_{X \setminus E} 0 \cdot \, \mathrm{d}\mu = 0$$

Thus

$$\int_{X \setminus E} g \, \mathrm{d}\mu \leq \frac{\varepsilon}{2} + \int_{X \setminus E} h \, \mathrm{d}\mu < \varepsilon$$

□

Theorem 5.2.4 ▸ Dominated Convergence Theorem

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $f : X \to [-\infty, \infty]$ is $\mathcal{S}$-measurable, and $f_1, f_2, \ldots$ are $\mathcal{S}$-measurable functions from X to $[-\infty, \infty]$ such that

$$\lim_{k\to\infty} f_k(x) = f(x)$$

for almost every $x \in X$. If there exists an $\mathcal{S}$-measurable function $g : X \to [0, \infty]$ such that

$$\int g d\mu < \infty \quad \text{and} \quad |f_k(x)| \le g(x)$$

for every $k \in \mathbb{Z}^+$ and almost every $x \in X$, then

$$\lim_{k\to\infty} \int f_k d\mu = \int f d\mu.$$

Proof sketch.
将积分差先控制在 g 几乎落在其上的有限测度集 E, 在将 E 上的积分分别控制在一个小测度集 B_ε 和使得 f_k 一致收敛的大测度集 $E \setminus B_\varepsilon$ 上.

Proof. For each $E \in \mathcal{S}$, and $k \in \mathbb{Z}^+$,

$$\begin{aligned}\left|\int f_k \,\mathrm{d}\mu - \int f \,\mathrm{d}\mu\right| &\le \int_E |f_k - f| \,\mathrm{d}\mu + \left|\int_{X\setminus E} f_k \,\mathrm{d}\mu\right| + \left|\int_{X\setminus E} f \,\mathrm{d}\mu\right| \\ &\le \int_E |f_k - f| \,\mathrm{d}\mu + 2\int_{X\setminus E} g \,\mathrm{d}\mu\end{aligned}$$

From proposition 5.2.3, for every $\varepsilon > 0$, there exists $E_\varepsilon \in \mathcal{S}$ such that $\mu(E_\varepsilon) < \infty$ and

$$\int_{X\setminus E_\varepsilon} g \,\mathrm{d}\mu < \frac{\varepsilon}{4}$$

From proposition 5.2.2, there exists $\delta > 0$ such that for each $F \in \mathcal{S}, \mu(F) < \delta$, we have

$$\int_F g \,\mathrm{d}\mu < \frac{\varepsilon}{8}$$

The Egorov's theorem implise that there exists measurable set $B_\varepsilon \in \mathcal{S}$ with

$\mu(E_\varepsilon \setminus B_\varepsilon) < \delta$ such that f_k converges uniformly to f on B_ε. Then we have

$$\lim_{k\to\infty} \int_{B_\varepsilon} |f_k - f| \, \mathrm{d}\mu \leq \lim_{k\to\infty} \mu(B_\varepsilon) \sup |f_k - f| = 0$$

Then there exists N such that

$$\int_{B_\varepsilon} |f_k - f| \, \mathrm{d}\mu < \frac{\varepsilon}{4}, \quad k \geq N$$

Furthermore , for each k we have

$$\int_{E_\varepsilon \setminus B_\varepsilon} |f_k - f| \, \mathrm{d}\mu \leq 2 \int_{E_\varepsilon \setminus B_\varepsilon} g \, \mathrm{d}\mu < \frac{\varepsilon}{4}$$

Therefore

$$\int_{E_\varepsilon} |f_k - f| \leq \int_{E_\varepsilon \setminus B_\varepsilon} |f_k - f| \, \mathrm{d}\mu + \int_{B_\varepsilon} |f_k - f| \, \mathrm{d}\mu < \frac{\varepsilon}{2}, \quad k \geq N$$

then it follows that

$$\left| \int f_k \, \mathrm{d}\mu - \int f \, \mathrm{d}\mu \right| \leq \int_{E_\varepsilon} |f_k - f| \, \mathrm{d}\mu + 2 \int_{X \setminus E_\varepsilon} g \, \mathrm{d}\mu < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \quad k \geq N$$

That is

$$\lim_{k\to\infty} \int f_k \, \mathrm{d}\mu = \int f \, \mathrm{d}\mu$$

□

5.3 Integrable Functions and Continuous Functions

Definition 5.3.1

Denote $C_c(\mathbb{R}^n)$ by the collection of compactly supported continuous functions.

Theorem 5.3.2

Let E be a measurable set. $C_c(\mathbb{R}^n)$ is dense in $L(E)$. That is if $f \in L(E)$, then for each $\varepsilon > 0$, there exists $g \in C_c(\mathbb{R}^n)$ such that

$$\int_E |f - g| \, \mathrm{d}\mu < \varepsilon$$

Remark.

- We can choose g such that $\sup_{x \in \mathbb{R}^n} |g| \leq \sup_{x \in E} |f|$

Proof. Let $f \in L(E)$, then for each $\varepsilon > 0$, there exists a simple function φ with compact support on E such that

$$\int_E |f - \varphi| \, \mathrm{d}\mu < \frac{\varepsilon}{2}$$

Let $F = \operatorname{supp} \varphi$, then F is bounded . Set $|\varphi(x)| \leq M, \forall x \in F$. From corollary 4.3.3, there is a $g \in C_c(\mathbb{R}^n)$ with such that

$$m(\{x \in F : |g - \varphi| > 0\}) < \frac{\varepsilon}{4M}, \quad \sup_{x \in \mathbb{R}^n} |g| \leq M$$

Then

$$\int_E |g - \varphi| \, \mathrm{d}\mu = \int_{\{|\varphi - g| > 0\}} |g - \varphi| \, \mathrm{d}\mu \leq 2Mm(\{|\varphi - g| > 0\}) < \frac{\varepsilon}{2}$$

We have

$$\int_E |f - g| \, \mathrm{d}\mu \leq \int_E |f - \varphi| \, \mathrm{d}\mu + \int_E |g - \varphi| \, \mathrm{d}\mu < \varepsilon$$

□

Corollary 5.3.3

Let $f \in L(E)$, then there exists a sequence $\{g_k(x)\}$ consists of compactly supported continous functions on $\mathbb{R}^n$, such that

- $$\lim_{k \to \infty} \int_E |f - g_k| \, \mathrm{d}\mu = 0$$

- $$\lim_{k\to\infty} g_k(x) = f(x), \quad a.e. x \in E$$

Remark.
- g_k can choosen to satisfy $\sup_{x\in E}|g_k| \leq \sup_{x\in E}|f|$

Proposition 5.3.4

Let $f \in L^1(\mathbb{R}^n)$. If for any $\varphi \in C_c(\mathbb{R}^n)$, there is

$$\int_{\mathbb{R}^n} f \cdot \varphi \, \mathrm{d}\mu = 0$$

Then $f = 0$, a.e.$x \in \mathbb{R}^n$.

Proof. Let $E = \{f > 0\}$. There is a sequence of compactly supported continuous functions $\{\varphi_k\}$ such that

$$\int_{\mathbb{R}^n} |\chi_E - \varphi_k| \, \mathrm{d}\mu = 0$$

$$\lim_{k\to\infty} \varphi_k(x) = \chi_E(x), \quad a.e. x \in \mathbb{R}^n,$$

$$|\varphi_k| \leq 1, \forall k$$

Since $|\varphi_k f| \leq |f|$, we have from dominated convergence theorem that

$$\begin{aligned} 0 < \int_E f \, \mathrm{d}\mu &= \int_{\mathbb{R}^n} \chi_E \cdot f \, \mathrm{d}\mu \\ &= \lim_{k\to\infty} \int_{\mathbb{R}^n} f \cdot \varphi_k \, \mathrm{d}\mu \end{aligned} \tag{5.1}$$

But $\lim_{\mathbb{R}^n} f \cdot \varphi_k \, \mathrm{d}\mu = 0$ for each k, which is a contradiction. □

Proposition 5.3.5

Let $f \in L^1([a, b])$, if for each differentiable function φ with $\operatorname{supp} \varphi \subseteq$

(a, b), there is

$$\int_{[a,b]} f(x)\varphi'(x)\,\mathrm{d}\mu = 0$$

Then $f(x) = c$, a.e.$x \in [a, b]$.

Theorem 5.3.6

If $f \in L^1(\mathbb{R}^n)$, then

$$\lim_{h\to 0}\int_{\mathbb{R}^n} |f(x+h) - f(x)|\,\mathrm{d}x = 0$$

Proof sketch.
紧支连续函数是一致连续的, 因此 f 和紧支的一致连续函数仅仅相差一个积分很小的函数. 而紧支的一致连续函数的积分自动是关于平移连续的.

Proof. For each $\varepsilon > 0$, we can decompose f into $f = f_1 + f_2$, where $f_1 \in C_c(\mathbb{R}^n)$, and

$$\int_{\mathbb{R}^n} |f_2|\,\mathrm{d}\mu < \frac{\varepsilon}{4}$$

f_1 is uniformly continuous , there exists $\delta > 0$ such that for each h with $|h| < \varepsilon$

$$\int_{\mathbb{R}^n} |f_1(x+h) - f_1(x)|\,\mathrm{d}\mu = \int_{\operatorname{supp} f_1} |f_1(x+h) - f(x)|\,\mathrm{d}\mu < \frac{\varepsilon}{2}$$

Then we have

$$\begin{aligned}\int_{\mathbb{R}^n} |f(x+h) - f(x)|\,\mathrm{d}\mu &\le \int_{\mathbb{R}^n} |f_1(x+h) - f_1(x)|\,\mathrm{d}\mu + \int_{\mathbb{R}^n} |f_2(x+h) - f_2(x)|\,\mathrm{d}\mu \\ &\le \int_{\mathbb{R}^n} |f_1(x+h) - f_1(x)|\,\mathrm{d}\mu + 2\int_{\mathbb{R}^n} |f_2(x)|\,\mathrm{d}\mu \\ &< \varepsilon\end{aligned} \tag{5.2}$$

Thus

$$\lim_{h\to\infty}\int_{\mathbb{R}^n} |f(x+h) - f(x)|\,\mathrm{d}x = 0$$

□

Proposition 5.3.7

Let $E \subseteq \mathbb{R}^n$ be a measurable set. If $f \in L(E)$, then there exists a sequence of measurable set functions with compact support , say $\{\varphi_k\}$ such that

- $$\lim_{k\to\infty} \varphi_k(x) = f(x), \quad a.e. x \in E$$
- $$\lim_{k\to\infty} \int_E |f - \varphi_k| \, \mathrm{d}\mu = 0$$

Proof. For each $\varepsilon > 0$, there exists $g \in C_c(\mathbb{R}^n)$ such that

$$\int_E |f - g| \, \mathrm{d}\mu < \frac{\varepsilon}{2}$$

We suppose that supp $\subseteq I$, where I is a closed cube. Since g is uniformly continuous, we can seperate I into 2^m small cubes I_k such that

$$|g(x) - g(\xi_k)| < \frac{\varepsilon}{2\mu(I)}, \quad k = 1, 2, \cdots, 2^m$$

where ξ_k is a point in I_k. Define $\varphi(x) = \sum_{k=1}^{2^m} g(\xi_k) \chi_{I_k}(x)$, then note that $\sum_{k=1}^{2^m} \mu(I_k) = \mu(I)$, we have

$$\int_I |g - \varphi| \, \mathrm{d}x < \sum_{k=1}^{2^m} \mu(I_k) \frac{\varepsilon}{2\mu(I)} = \frac{\varepsilon}{2}$$

Take $\varphi_k = \frac{1}{k}$, $k = 1, 2, \cdots$, we get a sequence $\{\varphi_k\}$ such that

$$\lim_{k\to\infty} \int_E |f - \varphi_k| \, \mathrm{d}x = 0$$

Then $\{\varphi_k\}$ convergent to f in measure, the proposition follows from Riesz Theorem. □

5.4 Riemann and Lebesgue Integration

Definition 5.4.1

By a ***partition*** $\mathcal{P}$ of $[a,b]$, we mean a finite set of points $\{x_i\}_{i=0}^{m}$ such that $a = x_0 < x_1 < \cdots < x_m = b$. Let

$$\|\mathcal{P}\| := \max\{x_i = x_{i=1} : 1 \leq i \leq m\}$$

For each $i \in \{1, \cdots, m\}$ let x_i^* be an arbitrary point of the inteval $[x_{i-1}, x_i]$. A bounded function $f : [a,b] \to \mathbb{R}$ is ***Riemann integrable*** if

$$\lim_{\|\mathcal{P}\|\to 0} \sum_{i=1}^{m} f\left(x_i^*\right)(x_i - x_{i-1})$$

exists, in which case the value is the Riemann integral of f over $[a,b]$, which we will denote by

$$(R) \int_a^b f(x)\, \mathrm{d}x$$

Lemma 5.4.2

Given a partition $\mathcal{P} = \{x_i\}_{i=0}^{m}$ of $[a,b]$, set

$$U(\mathcal{P}) = \sum_{i=1}^{m} \left[\sup_{x\in[x_{i-1},x_i]} f(x)\right](x_i - x_{i-1})$$

$$L(\mathcal{P}) = \sum_{i=1}^{m} \left[\inf_{x\in[x_{i-1},x_i]} f(x)\right](x_i = x_{i-1})$$

Then

$$L(\mathcal{P}) \leq \sum_{i=1}^{m} f\left(x_i^*\right)(x_i - x_{i-1}) \leq U(\mathcal{P})$$

for every choice of the x_i^*.

We see that a bounded function f is Riemann integrable if and only if

$$\lim_{\|\mathcal{P}\|\to 0} (U(\mathcal{P}) - L(\mathcal{P})) = 0$$

in which case

$$(R)\int_a^b f\,\mathrm{d}x = \lim_{\|\mathcal{P}\|\to 0} U(\mathcal{P}) = \lim_{\|\mathcal{P}\|\to 0} L(\mathcal{P})$$

Lemma 5.4.3

Soppose that $\mathcal{P} = \{x_i\}_{i=1}^m$ is a partition of $[a,b]$, $z \in [a,b] \setminus \mathcal{P}$, and $\mathcal{Q} = \mathcal{P} \cup \{z\}$. Thus $\mathcal{P} \subseteq \mathcal{Q}$, $\mathcal{Q}$ is called a refinement of $\mathcal{P}$. We have

$$L(\mathcal{P}) \leq L(\mathcal{Q}) \leq U(\mathcal{Q}) \leq U(\mathcal{P})$$

Theorem 5.4.4

If $f : [a,b] \to \mathbb{R}$ is a bounded Riemann integrable function, then f is Lebesgue integrable and

$$(R)\int_a^b f(x)\,\mathrm{d}x = \int_{[a,b]} f\,\mathrm{d}\lambda$$

Proof. Let $\{\mathcal{P}_k\}_{k=1}^{\infty}$ be a sequence of partitions of $[a,b]$ such that $\mathcal{P}_k \subseteq \mathcal{P}_{k+1}$ and $\|\mathcal{P}_k\| \to \infty$ as $k \to \infty$. Write $\mathcal{P}_k = \left\{x_j^k\right\}_{j=0}^{m_k}$. For each k, define l_k, u_k by

$$l_k(x) = \sum_{j=1}^{m_k} \chi_{\left[x_{j-1},x_j\right)} \inf_{t\in\left[x_{j-1},x_j\right]} f(t)$$

$$u_k(x) = \sum_{j=1}^{m_k} \chi_{\left[x_{j-1},x_j\right)} \sup_{t\in\left[x_{j-1},x_j\right]} f(t)$$

Then

$$\int_{[a,b]} l_k\,\mathrm{d}\lambda = L(\mathcal{P}_k) \leq U(\mathcal{P}_k) = \int_{[a,b]} u_k\,\mathrm{d}\lambda$$

Since l_k is increasing, u_k is decresing , the limit

$$l(x) := \lim_{k\to\infty} l_k(x), \quad u(x) := \lim_{k\to\infty} u_k(x)$$

exsists and are measurable and $l \leq f \leq u$. Since f is Riemann measurable, it follows from Dominated Convergence Theorem (($u_1 - l_1$) as a dominating function)that

$$\int_{[a,b]} (u-l)\, \mathrm{d}\lambda = \lim_{k\to\infty} \int_{[a,b]} (u_k - l_k)\, \mathrm{d}\lambda = \lim_{k\to\infty} \left(U(\mathcal{P}_k) - L(\mathcal{P}_k)\right) = 0$$

Thus $u = f = l$, a.e. $x \in [a,b]$. Then

$$\int_{[a,b]} f\, \mathrm{d}\lambda = \lim_{k\to\infty} \int_{[a,b]} u_k\, \mathrm{d}\lambda = \lim_{k\to\infty} U(\mathcal{P}_k) = (R)\int_a^b f(x)\, \mathrm{d}x$$

□

Theorem 5.4.5

A bounded function $f : [a,b] \to \mathbb{R}$ is Riemann integrable if and only if f is continuous λ-a.e. on $[a,b]$.

Proof. Suppose f is Riemann integrable and let $\{\mathcal{P}_k\}$ be a sequence of partitions of $[a,b]$ such that $\mathcal{P}_k \subseteq \mathcal{P}_{k+1}$ and $\lim_{k\to\infty} \|\mathcal{P}_k\| = 0$. Let l_k, u_k as in the proof of the above theorem. If $x \in [a,b] \setminus N$, $l(x) = u(x)$ and $\varepsilon > 0$, then there is an integer k such that

$$u_k(x) - l_k(x) < \varepsilon$$

Let $\mathcal{P}_k = \left\{x_j^k\right\}_{j=0}^{m_k}$. Then $x \in \left[x_{j-1}^k, x_j^k\right)$ for some $j \in \{1, \cdots, m_k\}$. For every $y \in \left(x_{j-1}^k, x_j^k\right)$

$$|f(y) - f(x)| \leq u_k(x) - l_k(x) < \varepsilon$$

Thus f is continuous at x. Since $l(x) = u(x)$ for λ-a.e. $x \in [a,b]$ and $\lambda(N) = 0$, we see that f is continuous at λ-a.e. point of $[a,b]$.

Now suppose f is bounded, $N \subseteq [a,b]$ with $\lambda(N) = 0$, and f is continuous at each point of $[a,b] - N$. Let $\{\mathcal{P}_k\}$ be a sequence of partitions of $[a,b]$ such that $\lim_{k\to\infty} \|\mathcal{P}_k\| = 0$. For each k, define l_k, u_k as above. If $x \in [a,b] - N$

and $\varepsilon > 0$, then there exists a $\delta > 0$ such that

$$|f(x) - f(y)| < \frac{\varepsilon}{2}$$

whenever $|y - x| < \delta$. There is a k_0 such that $\|\mathcal{P}_k\| < \frac{\delta}{2}$ whenever $k > k_0$, thus

$$u_k(x) - l_k(x) \leq 2\sup\{|f(x) - f(y)| : |y - x| < \delta\} < \varepsilon$$

whenever $k > k_0$. Thus

$$\lim_{k\to\infty}(u_k(x) - l_k(x)) = 0$$

for each $x \in [a, b] \setminus N$. By the dominated convergence theorem

$$\lim_{k\to\infty}(U(\mathcal{P}_k) - L(\mathcal{P}_k)) = \lim_{k\to\infty}\int_{[a,b]}(u_k - l_k)\, d\lambda = 0$$

$\square$

5.5 Improper Integrals

Definition 5.5.1

Let $a \in \mathbb{R}$ and let $f : [a, \infty) \to \mathbb{R}$ be a function that is Riemann integrable on each subinterval of $[a, \infty)$. The improper integral of f is defined as

$$(I)\int_a^\infty f(x)\, dx := \lim_{b\to\infty}(R)\int_a^b f(x)\, dx. \tag{5.3}$$

If the limit is finite, we say that the improper integral of f exist. We have the following result.

Theorem 5.5.2

et $f : [a, \infty) \to \mathbb{R}$ be a nonnegative function that is Riemann integrable on each subinterval of $[a, \infty]$. Then

$$\int_{[a,\infty)} f\, d\lambda = \lim_{b\to\infty}(R)\int_a^b f\, dx. \tag{5.4}$$

Thus, f is Lebesgue integrable on $[a,\infty)$ if and only if the improper integral $(I)\int_a^\infty f(x)\,dx$ exists. Moreover, in this case, $\int_{[a,\infty)} f(x)\,d\lambda = (I)\int_a^\infty f(x)\,dx$.

Proof. Let $b_n, n = 1,2,3,\cdots$ be any sequence with $b_n \to \infty$, $b_n > a$. We define $f_n = f\chi_{[a,b_n]}$. The monotone Convergence theorem yields

$$\int_{[a,\infty)} f\,\mathrm{d}\lambda = \lim_{n\to\infty} \int_{[a,\infty)} f_n\,\mathrm{d}\lambda = \lim_{n\to\infty} \int_{[a,b_n]} f\,\mathrm{d}\lambda$$

Since f is Riemann integrable on each interval $[a,b_n]$, we have

$$\int_{[a,b_n]} = (R)\int_a^{b_n} f\,\mathrm{d}x$$

and we conclude that

$$\int_{[a,\infty)} f\,\mathrm{d}\lambda = \lim_{n\to\infty}(R)\int_a^{b_n} f\,\mathrm{d}x = \lim_{b\to\infty}(R)\int_a^{b} f\,\mathrm{d}x$$

The last equality follows from the nonnegativity of f. □

Theorem 5.5.3

Let $f : [a,\infty) \to \mathbb{R}$ be Riemann integrable on every subinterval of $[a,\infty)$. Then f is Lebesgue integrable if and only if the improper integral $(I)\int_a^\infty |f(x)|dx$ exists. Moreover, in this case,

$$\int_{[a,\infty)} f\,d\lambda = (I)\int_a^\infty f(x)\,dx. \tag{5.5}$$

Proof. Note that $f^+ = \frac{|f|+f}{2}$, $f^- = \frac{|f|-f}{2}$ is Riemann integrable, if f is Riemann integrable. And we have

$$(R)\int_a^{b_n} |f|\,\mathrm{d}x = (R)\int_a^{b_n} f^+\,\mathrm{d}x + (R)\int_a^{b_n} f^-\,\mathrm{d}x$$

The remaining is straightforward to check. $\square$

6 Iterated Integrals

6.1 Products of Measure Spaces

Definition 6.1.1 ▸ product of two σ-algebras; $\mathcal{S} \otimes \mathcal{T}$

Suppose $(X, \mathcal{S})$ and $(Y, \mathcal{T})$ are measurable spaces. Then

- the ***product*** $\mathcal{S} \otimes \mathcal{T}$ is defined to be the smallest σ-algebra on $X \times Y$ that contains
$$\{A \times B : A \in \mathcal{S}, B \in \mathcal{T}\};$$
- a ***measurable rectangle*** in $\mathcal{S} \otimes \mathcal{T}$ is a set of the form $A \times B$, where $A \in \mathcal{S}$ and $B \in \mathcal{T}$.

Definition 6.1.2 ▸ cross sections of sets; $[E]_a$ and $[E]^b$

Suppose X and Y are sets and $E \subset X \times Y$. Then for $a \in X$ and $b \in Y$, the cross sections $[E]_a$ and $[E]^b$ are defined by

$$[E]_a = \{y \in Y : (a, y) \in E\} \quad \text{and} \quad [E]^b = \{x \in X : (x, b) \in E\}.$$

Proposition 6.1.3 ▸ cross sections of measurable sets are measurable

Suppose $\mathcal{S}$ is a σ-algebra on X and $\mathcal{T}$ is a σ-algebra on Y. If $E \in \mathcal{S} \otimes \mathcal{T}$, then

$$[E]_a \in \mathcal{T} \text{ for every } a \in X \quad \text{and} \quad [E]^b \in \mathcal{S} \text{ for every } b \in Y.$$

Proof sketch.
将截面可测的集合的全体记作 $\mathcal{E}$. 则 $\mathcal{E}$ 是一个 σ-algebra. 可测方块的截面都是可测的, 故 $\mathcal{E}$ 包含了 $\mathcal{S} \otimes \mathcal{T}$.

Definition 6.1.4 ▸ cross sections of functions; $[f]_a$ and $[f]^b$

Suppose X and Y are sets and $f : X \times Y \to \mathbf{R}$ is a function. Then for $a \in X$ and $b \in Y$, the cross section functions $[f]_a : Y \to \mathbf{R}$ and

$[f]^b : X \to \mathbf{R}$ are defined by

$$[f]_a(y) = f(a, y) \text{ for } y \in Y \quad \text{and} \quad [f]^b(x) = f(x, b) \text{ for } x \in X.$$

Proposition 6.1.5

Suppose $\mathcal{S}$ is a σ-algebra on X and $\mathcal{T}$ is a σ-algebra on Y. Suppose $f : X \times Y \to \mathbb{R}$ is an $\mathcal{S} \otimes \mathcal{T}$-measurable function. Then

$$[f]_a \text{ is a } \mathcal{T}\text{-measurable function on } Y \text{ for every } a \in X$$

and

$$[f]^b \text{ is an } \mathcal{S}\text{-measurable function on } X \text{ for every } b \in Y.$$

Proof sketch.
截面函数的原像集等于原像集的截面:

$$\left([f]_a\right)^{-1}(D) = [f^{-1}(D)]_a$$

6.2 Monotone Class Theorem

Definition 6.2.1 ▸ algebra

Suppose W is a set and $\mathcal{A}$ is a set of subsets of W. Then $\mathcal{A}$ is called an ***algebra*** on W if the following three conditions are satisfied:

- $\varnothing \in \mathcal{A}$;
- if $E \in \mathcal{A}$, then $W \setminus E \in \mathcal{A}$;
- if E and F are elements of $\mathcal{A}$, then $E \cup F \in \mathcal{A}$.

Proposition 6.2.2 ▸ the set of finite unions of measurable rectangles is an algebra

Suppose $(X, \mathcal{S})$ and $(Y, \mathcal{T})$ are measurable spaces. Then
(a) the set of finite unions of measurable rectangles in $\mathcal{S} \otimes \mathcal{T}$ is an algebra on $X \times Y$;
(b) every finite union of measurable rectangles in $\mathcal{S} \otimes \mathcal{T}$ can be written as a finite union of disjoint measurable rectangles in $\mathcal{S} \otimes \mathcal{T}$.

Definition 6.2.3 ▶ monotone class

Suppose W is a set and $\mathcal{M}$ is a set of subsets of W. Then $\mathcal{M}$ is called a ***monotone class*** on W if the following two conditions are satisfied:

- If $E_1 \subset E_2 \subset \cdots$ is an increasing sequence of sets in $\mathcal{M}$, then
$$\bigcup_{k=1}^{\infty} E_k \in \mathcal{M};$$
- If $E_1 \supset E_2 \supset \cdots$ is a decreasing sequence of sets in $\mathcal{M}$, then
$$\bigcap_{k=1}^{\infty} E_k \in \mathcal{M}.$$

Theorem 6.2.4 ▶ Monotone Class Theorem

Suppose $\mathcal{A}$ is an algebra on a set W. Then the smallest σ-algebra containing $\mathcal{A}$ is the smallest monotone class containing $\mathcal{A}$.

Proof sketch.

- 设 $\mathcal{M}$ 是包含了 $\mathcal{A}$ 的最小的单调类, $\mathcal{M}$ 包含于 $\sigma(\mathcal{A})$ 是容易的. 主要说明另一边.
- 取 $A \in \mathcal{A}$, 令 $\mathcal{E}$ 是 $\mathcal{M}$ 中在“并 A ”操作下封闭的集合的全体. 则 $\mathcal{E}$ 可以继承 $\mathcal{M}$ 的单调类属性. 从而 $\mathcal{M} \subseteq \mathcal{E}, A \cup E \in \mathcal{M}$ 对于所有的 E 成立.
- 令 $\mathcal{D}$ 是 $\mathcal{M}$ 中在“与 $\mathcal{M}$ 中任意元素做并”操作下封闭的集合全体, 则 $\mathcal{D}$ 可以继承 $\mathcal{M}$ 的单调类属性. 则上面的结果就是在说 $A \in \mathcal{D}$, 从而 $\mathcal{A} \subseteq \mathcal{D}$. 得到 $\mathcal{M} \subseteq \mathcal{D}$. 这就是说 $\mathcal{M}$ 在有限并下封闭.
- 有限并加上对递增列封闭, 就是对可数并封闭.
- 令 $\mathcal{M}'$ 是 $\mathcal{M}$ 中在“补集”操作下封闭的全体, 则 $\mathcal{A} \subseteq \mathcal{M}'$. $\mathcal{M}'$ 可以继承 $\mathcal{M}$ 的单调类属性 (可以认为单调类的定义是关于补集对偶的). 故 $\mathcal{M} \subseteq \mathcal{M}'$, $\mathcal{M}$ 在补集下封闭.

6.3 Products of Measures

Theorem 6.3.1 ▶ measure of cross section is a measurable function

Suppose $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ are σ-***finite*** measure spaces. If $E \in \mathcal{S} \otimes \mathcal{T}$, then

1. $x \mapsto \nu([E]_x)$ is an $\mathcal{S}$-measurable function on X;
2. $y \mapsto \mu([E]^y)$ is a $\mathcal{T}$-measurable function on Y.

Proof sketch.
考虑 (a)
先考虑有限测度的情况下.

- 令 $\mathcal{M}$ 是具有性质 (a) 的 E 的全体.
- 所有可测方块具有性质 (a).
- 令 $\mathcal{A}$ 表示可测方块生成的 algebra. $[\cdot]_x$ 是保持集合运算的, 可以由此将 $[E]_x$ 按无交并的可测方块的并分解 $[E_1]_x \cup [E_2]_x \cup \cdots \cup [E_n]_x$.
- 由此, $E \in \mathcal{A}$ 的函数 $x \mapsto \nu([E]_x)$ 是 $x \mapsto \nu([E_i]_x)$ 的和, 是可测的. 故 $\mathcal{A} \subseteq \mathcal{M}$.
- ν 的自上自下连续性, 可测函数对逐点极限的封闭性 (需要测度是有限的), 给出 $\mathcal{M}$ 是单调类.
- 由单调类定理, $\mathcal{M}$ 包含 $\mathcal{S} \otimes \mathcal{T}$.

推广到 σ-有限的情况, 只需要取出 Y 的一列递增的有限测度分解 $Y_1 \subseteq Y_2 \subseteq \cdots$,$\bigcup_{k=1}^{\infty} Y_k = Y$. 做逼近 $\nu([E]_x) = \lim_{k\to\infty} \nu([E \cap (X \times Y_k)]_x)$ 即可.

Definition 6.3.2 ▶ product of two measures; $\mu \times \nu$

Suppose $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ are σ-finite measure spaces. For $E \in \mathcal{S} \otimes \mathcal{T}$, define $(\mu \times \nu)(E)$ by

$$(\mu \times \nu)(E) = \int_X \int_Y \chi_E(x, y)\, d\nu(y)\, d\mu(x).$$

Theorem 6.3.3 ▶ product of two measures is a measure

Suppose $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ are σ-finite measure spaces. Then $\mu \times \nu$ is a measure on $(X \times Y, \mathcal{S} \otimes \mathcal{T})$.

Proof sketch.
利用单调收敛定理即可说明可数可加性.

6.4 Iterated Integrals

Theorem 6.4.1 ▶ Tonelli's Theorem

Suppose $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ are σ-***finite*** measure spaces. Suppose $f : X \times Y \to [0, \infty]$ is $\mathcal{S} \otimes \mathcal{T}$-measurable. Then

(a) $x \mapsto \displaystyle\int_Y f(x, y)\, d\nu(y)$ is an $\mathcal{S}$-measurable function on X,

(b) $y \mapsto \displaystyle\int_X f(x, y)\, d\mu(x)$ is a $\mathcal{T}$-measurable function on Y,

and

$$\int_{X\times Y} f\, d(\mu\times\nu) = \int_X \int_Y f(x, y)\, d\nu(y)\, d\mu(x) = \int_Y \int_X f(x, y)\, d\mu(x)\, d\nu(y).$$

Proof sketch.
先说明 $f = \chi_E$ 的情况.

- 这个情况下 (a) (b) 就是 Theorem 6.3.1. 令 $\mathcal{M}$ 是使得 χ_E 满足积分交换性的集合.
- 可测方块是属于 $\mathcal{M}$ 的, 进而可测方块生成的代数 $\mathcal{A}$ 含于 $\mathcal{M}$.
- 先考虑测度 μ, ν 有限的情况. 由单调收敛定理, $\mathcal{M}$ 对递增集合列封闭. 当 μ, ν 测度有限时, 由控制收敛定理 $\mathcal{M}$ 对递减列封闭. 故此时 $\mathcal{M}$ 是单调类.
- 由单调类定理, 当 μ, ν 有限时, $\mathcal{M}$ 包含 $\mathcal{S} \otimes \mathcal{T}$.
- 推广到 σ-有限的情况, 做分解 $X_1 \subseteq X_2 \cdots \subseteq X$, $Y_1 \subseteq Y_2 \cdots \subseteq Y$. 则对于 $E \in \mathcal{S} \otimes \mathcal{T}$, $\left(E \cap \left(X_j \times Y_k\right)\right) \in \mathcal{M}$. 单调收敛定理可以给出 $E \cap (X \times Y_k) \in \mathcal{M}$, 再用一次单调收敛定理给出 $E = E \cap (X \times Y) \in \mathcal{M}$.

对于 f 是一般的非负可测函数的情况, 找一列趋于 f 的递增的简单函数 $\{f_k\}$. 题述性质对 f_k 是成立的. 单调收敛定理给出 (a) 和 (b). 再用单调收敛定理穿过两层积分 $\int_X \int_Y$ 给出积分的交换性.

Theorem 6.4.2 ▶ Fubini's Theorem

Suppose $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ are σ-finite measure spaces. Suppose

$$f : X \times Y \to [-\infty, \infty]$$

is $\mathcal{S} \otimes \mathcal{T}$-measurable and

$$\int_{X\times Y} |f|\, d(\mu \times \nu) < \infty.$$

Then

$$\int_Y |f(x,y)|\, d\nu(y) < \infty \text{ for almost every } x \in X$$

and

$$\int_X |f(x,y)|\, d\mu(x) < \infty \text{ for almost every } y \in Y.$$

Furthermore,

(a) $x \mapsto \displaystyle\int_Y f(x,y)\, d\nu(y)$ is an $\mathcal{S}$-measurable function on X,

(b) $y \mapsto \displaystyle\int_X f(x,y)\, d\mu(x)$ is a $\mathcal{T}$-measurable function on Y,

and

$$\int_{X\times Y} f\, d(\mu\times\nu) = \int_X \int_Y f(x,y)\, d\nu(y)\, d\mu(x) = \int_Y \int_X f(x,y)\, d\mu(x)\, d\nu(y).$$

Proof sketch.

- 对 $|f|$ 使用 Tonelli 定理中的积分等式, 得到累次绝对积分的可积性, 同时得到单边绝对积分的几乎处处有限.
- 对 f^+, f^- 使用 Tonelli 定理中的积分函数的可测性结论, 并由可积性可知未定义的部分是零测集, 得到 $x \mapsto \int_Y f(x,y)\, \mathrm{d}\nu(y)$ 的可测性.
- 将 f 做正负部分解, 对正负部积分分别使用 Tonelli 定理得到积分等式.

6.5 Area Under the Graph

Definition 6.5.1 ▸ region under the graph

Suppose X is a set and $f : X \to [0,\infty]$ is a function. Then the ***region under the graph*** of f, denoted U_f, is defined by

$$U_f = \{(x,t) \in X \times (0,\infty) : 0 < t < f(x)\}.$$

Proposition 6.5.2 ▸ area under the graph of a function equals the integral

Suppose $(X, \mathcal{S}, \mu)$ is a σ-finite measure space and $f : X \to [0, \infty]$ is an $\mathcal{S}$-measurable function. Let $\mathcal{B}$ denote the σ-algebra of Borel subsets of $(0, \infty)$, and let λ denote Lebesgue measure on $((0, \infty), \mathcal{B})$. Then $U_f \in \mathcal{S} \otimes \mathcal{B}$ and

$$(\mu \times \lambda)(U_f) = \int_X f \, d\mu = \int_{(0,\infty)} \mu(\{x \in X : t < f(x)\}) \, d\lambda(t).$$

Proof sketch.
令 $F(x,t) = f(x) - t$, 则 U_f 是 $F^{-1}((0,\infty))$ 是可测的. 积分等式由 Tonelli 定理得到.

7

Lp Spaces

7.1 $\mathcal{L}^p$

Hölder's Inequality

Definition 7.1.1 ▶ $\|f\|_p$

Suppose that $(X, \mathcal{S}, \mu)$ is a measure space, $0 < p < \infty$, and $f : X \to \mathbb{F}$ is $\mathcal{S}$-measurable. Then the ***p-norm*** of f is denoted by $\|f\|_p$ and is defined by

$$\|f\|_p = \left(\int |f|^p \, d\mu\right)^{1/p}.$$

Also, $\|f\|_\infty$, which is called the ***essential supremum*** of f, is defined by

$$\|f\|_\infty = \inf\{t > 0 : \mu(\{x \in X : |f(x)| > t\}) = 0\}.$$

Definition 7.1.2 ▶ $L^p(\mu)$

Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $0 < p \le \infty$. The ***Lebesgue space*** $L^p(\mu)$, sometimes denoted $L^p(X, \mathcal{S}, \mu)$, is defined to be the set of $\mathcal{S}$-measurable functions $f : X \to \mathbb{F}$ such that $\|f\|_p < \infty$.

Theorem 7.1.3 ▶ $L^p(\mu)$ is a vector space

Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $0 < p < \infty$. Then
(a)

$$\|f + g\|_p^p \le 2^p\big(\|f\|_p^p + \|g\|_p^p\big)$$

and
(b)

$$\|\alpha f\|_p = |\alpha| \, \|f\|_p$$

for all $f, g \in L^p(\mu)$ and all $\alpha \in \mathbb{F}$. Furthermore, with the usual operations of addition and scalar multiplication of functions, $L^p(\mu)$ is a vector space.

Definition 7.1.4 ▶ dual exponent; p'

For $1 \leq p \leq \infty$, the ***dual exponent*** of p is denoted by p' and is the element of $[1,\infty]$ such that

$$\frac{1}{p} + \frac{1}{p'} = 1.$$

Lemma 7.1.5 ▶ Young's inequality

Suppose $1 < p < \infty$. Then

$$ab \leq \frac{a^p}{p} + \frac{b^{p'}}{p'}$$

for all $a \geq 0$ and $b \geq 0$.

Theorem 7.1.6 ▶ Hölder's Inequality

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $1 \leq p \leq \infty$, and $f, h : X \to \mathbb{F}$ are $\mathcal{S}$-measurable. Then

$$\|fh\|_1 \leq \|f\|_p \, \|h\|_{p'}.$$

Theorem 7.1.7 ▶ $L^q(\mu) \subset L^p(\mu)$ if $p < q$ and $\mu(X) < \infty$

Suppose $(X, \mathcal{S}, \mu)$ is a finite measure space and $0 < p < q < \infty$. Then

$$\|f\|_p \leq \mu(X)^{(q-p)/(pq)} \|f\|_q$$

for all $f \in L^q(\mu)$. Furthermore, $L^q(\mu) \subset L^p(\mu)$.

Minkowski's Inequality

Theorem 7.1.8 ▶ *formula for $\|f\|_p$*

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $1 \leq p < \infty$, and $f \in L^p(\mu)$. Then

$$\|f\|_p = \sup\left\{ \left| \int f h \, d\mu \right| : h \in L^{p'}(\mu) \text{ and } \|h\|_{p'} \leq 1 \right\}.$$

Theorem 7.1.9 ▶ Minkowski's inequality

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $1 \leq p \leq \infty$, and $f, g \in \mathcal{L}^p(\mu)$. Then
$$\|f+g\|_p \leq \|f\|_p + \|g\|_p.$$

7.2 L^p

Definition of $L^p(\mu)$

Definition 7.2.1 ▶ $\mathcal{Z}(\mu)$; $\tilde{f}$

Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $0 < p \leq \infty$.

- $\mathcal{Z}(\mu)$ denotes the set of $\mathcal{S}$-measurable functions from X to $\mathbf{F}$ that equal 0 almost everywhere.
- For $f \in \mathcal{L}^p(\mu)$, let $\tilde{f}$ be the subset of $\mathcal{L}^p(\mu)$ defined by
$$\tilde{f} = \{f + z : z \in \mathcal{Z}(\mu)\}.$$

Definition 7.2.2 ▶ $L^p(\mu)$

Suppose μ is a measure and $0 < p \leq \infty$.

- Let $L^p(\mu)$ denote the collection of subsets of $\mathcal{L}^p(\mu)$ defined by
$$L^p(\mu) = \{\tilde{f} : f \in \mathcal{L}^p(\mu)\}.$$
- For $\tilde{f}, \tilde{g} \in L^p(\mu)$ and $\alpha \in \mathbf{F}$, define $\tilde{f} + \tilde{g}$ and $\alpha\tilde{f}$ by
$$\tilde{f} + \tilde{g} = (f+g)^{\sim} \quad \text{and} \quad \alpha\tilde{f} = (\alpha f)^{\sim}.$$

Definition 7.2.3 ▶ $\|\cdot\|_p$ *on* $L^p(\mu)$

Suppose μ is a measure and $0 < p \leq \infty$. Define $\|\cdot\|_p$ on $L^p(\mu)$ by
$$\|\tilde{f}\|_p = \|f\|_p$$
for $f \in \mathcal{L}^p(\mu)$.

Theorem 7.2.4 ▶ $L^p(\mu)$ *is a normed vector space*

Suppose μ is a measure and $1 \le p \le \infty$. Then $L^p(\mu)$ is a vector space and $\|\cdot\|_p$ is a norm on $L^p(\mu)$.

$L^p(\mu)$ is a Banach Space

Theorem 7.2.5

Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $1 \le p \le \infty$. Suppose $f_1, f_2, \ldots$ is a sequence of functions in $L^p(\mu)$ such that for every $\varepsilon > 0$, there exists $n \in \mathbb{Z}^+$ such that

$$\|f_j - f_k\|_p < \varepsilon$$

for all $j \ge n$ and $k \ge n$. Then there exists $f \in L^p(\mu)$ such that

$$\lim_{k\to\infty} \|f_k - f\|_p = 0.$$

Proof sketch.
对于 $1 \le p < \infty$:

- 取 $\sum_{k=1}^{\infty} \|f_k - f_{k-1}\|_p < \infty$ 的子列.
- 使用 Minkowski 说明绝对 $\sum_{k=1}^{\infty} |f_k - f_{k-1}|$ 的几乎处处收敛. 原级数几乎处处收敛, 等于极限 $\lim_{m\to\infty} f_m(x)$.
- 使用 Fatou's Lemma, 把 $\|f_k - f\|_p$ 中的极限函数 f 拆解, 控制成 $\liminf_{j\to\infty} \left\|f_k - f_j\right\|_p$ 以带入 Cauchy 条件.

8 Hilbert Space

8.1 Orthonormal Bases

Definition 8.1.1 ▸ orthonormal family

A family $\{e_k\}_{k\in\Gamma}$ in an inner product space is called an ***orthonormal family*** if

$$\langle e_j, e_k\rangle = \begin{cases} 0 & \text{if } j \neq k, \\ 1 & \text{if } j = k \end{cases}$$

for all $j, k \in \Gamma$.

Proposition 8.1.2 ▸ finite orthonormal families

Suppose Ω is a finite set and $\{e_j\}_{j\in\Omega}$ is an orthonormal family in an inner product space. Then

$$\left\| \sum_{j\in\Omega} \alpha_j e_j \right\|^2 = \sum_{j\in\Omega} |\alpha_j|^2$$

for every family $\{\alpha_j\}_{j\in\Omega}$ in $\mathbf{F}$.

Definition 8.1.3 ▸ unordered sum; $\sum_{k\in\Gamma} f_k$

Suppose $\{f_k\}_{k\in\Gamma}$ is a family in a normed vector space V. The ***unordered sum*** $\sum_{k\in\Gamma} f_k$ is said to ***converge*** if there exists $g \in V$ such that for every $\varepsilon > 0$, there exists a finite subset Ω of Γ such that

$$\left\| g - \sum_{j\in\Omega'} f_j \right\| < \varepsilon$$

for all finite sets Ω' with $\Omega \subset \Omega' \subset \Gamma$. If this happens, we set $\sum_{k\in\Gamma} f_k = g$. If there is no such $g \in V$, then $\sum_{k\in\Gamma} f_k$ is left undefined.

Theorem 8.1.4 ▶ linear combinations of an orthonormal family

Suppose $\{e_k\}_{k\in\Gamma}$ is an orthonormal family in a Hilbert space V. Suppose $\{\alpha_k\}_{k\in\Gamma}$ is a family in $\mathbf{F}$. Then

(a) the unordered sum $\sum\limits_{k\in\Gamma} \alpha_k e_k$ converges $\iff$ $\sum\limits_{k\in\Gamma} |\alpha_k|^2 < \infty$.

(b) Furthermore, if $\sum_{k\in\Gamma} \alpha_k e_k$ converges, then

$$\left\|\sum_{k\in\Gamma} \alpha_k e_k\right\|^2 = \sum_{k\in\Gamma} |\alpha_k|^2.$$

Theorem 8.1.5

Suppose $\{e_k\}_{k\in\Gamma}$ is an orthonormal family in an inner product space V and $f \in V$. Then

$$\sum_{k\in\Gamma} |\langle f, e_k\rangle|^2 \leq \|f\|^2.$$

Theorem 8.1.6 ▶ closure of the span of an orthonormal family

Suppose $\{e_k\}_{k\in\Gamma}$ is an orthonormal family in a Hilbert space V. Then

(a)

$$\overline{\operatorname{span}\{e_k\}_{k\in\Gamma}} = \left\{\sum_{k\in\Gamma} \alpha_k e_k \,:\, \{\alpha_k\}_{k\in\Gamma} \text{ is a family in } \mathbf{F} \text{ and } \sum_{k\in\Gamma} |\alpha_k|^2 < \infty\right\}.$$

(b) Furthermore,

$$f = \sum_{k\in\Gamma} \langle f, e_k\rangle e_k$$

for every $f \in \overline{\operatorname{span}\{e_k\}_{k\in\Gamma}}$.

8.2 Parseval's Identity

Definition 8.2.1 ▶ orthonormal basis

An orthonormal family $\{e_k\}_{k\in\Gamma}$ in a Hilbert space V is called an ***orthonormal basis*** of V if

$$\overline{\operatorname{span}\{e_k\}_{k\in\Gamma}} = V.$$

Theorem 8.2.2 ▶ Parseval's identity

Suppose $\{e_k\}_{k\in\Gamma}$ is an orthonormal basis of a Hilbert space V and $f, g \in V$. Then

1. $f = \sum\limits_{k\in\Gamma} \langle f, e_k\rangle e_k$;
2. $\langle f, g\rangle = \sum\limits_{k\in\Gamma} \langle f, e_k\rangle \overline{\langle g, e_k\rangle}$;
3. $\|f\|^2 = \sum\limits_{k\in\Gamma} |\langle f, e_k\rangle|^2$.

Definition 8.2.3 ▶ separable

A normed vector space is called ***separable*** if it has a countable subset whose closure equals the whole space.

Theorem 8.2.4 ▶ existence of orthonormal bases for separable Hilbert spaces

Every separable Hilbert space has an orthonormal basis.

Theorem 8.2.5 ▶ orthonormal bases as maximal elements

Suppose V is a Hilbert space, $\mathcal{A}$ is the collection of all orthonormal subsets of V, and Γ is an orthonormal subset of V. Then Γ is an orthonormal basis of V if and only if Γ is a maximal element of $\mathcal{A}$.

Theorem 8.2.6 ▶ existence of orthonormal bases for all Hilbert spaces

Every Hilbert space has an orthonormal basis.

Theorem 8.2.7 ▸ Riesz Representation Theorem

Suppose φ is a bounded linear functional on a Hilbert space V and $\{e_k\}_{k\in\Gamma}$ is an orthonormal basis of V. Let

$$h = \sum_{k\in\Gamma} \overline{\varphi(e_k)}e_k.$$

Then

$$\varphi(f) = \langle f, h\rangle$$

for all $f \in V$. Furthermore, $\|\varphi\| = \left(\sum_{k\in\Gamma} |\varphi(e_k)|^2\right)^{1/2}$.

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